INCH-POUND
MIL-M-38510/306E
17 June 2003
SUPERSEDING
MIL-M-38510/306D
16 NOVEMBER 1987

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW-POWER SCHOTTKY TTL, SHIFT REGISTERS, CASCADABLE, MONOLITHIC SILICON

Inactive for new design after 18 April 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, low power, shift register microcircuits. Two product assurance classes and a choice of case outlines and lead finishes and are reflected in the complete part number. For this product, the requirements of MIL-M-38510 have been superseded by MIL-PRF-38535, (see 6.3).

1.2 Part number. The part number should be in accordance with MIL-PRF-38535, and as specified herein.

1.2.1 Device types. The device types should be as follows:

Device type	Circuit
01	4 bit bi-directional shift register
02	4 bit parallel-access shift register
03	4 bit parallel-access shift register
04	5 bit shift register
05	8 bit parallel-out shift register
06	4 bit right-shift, left-shift register, 3-state outputs
07	4 bit cascadable shift register, 3-state outputs
08	8 bit parallel-in shift register with clock inhibit
09	8 bit parallel-in shift register with clear

1.2.2 Device class. The device class should be the product assurance level as defined in MIL-PRF-38535.

Beneficial comments (recommendations, additions deletions) and any pertinent data which may be used in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAS, P.O. Box 3990, Columbus OH 43216-5000, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.3 Case outlines. The case outlines should be as designated in MIL-STD-1835 and as follows:

Outline letter	Descriptive designator	Terminals	Package style
A	GDFP5-F14 or CDFP6-F14	14	Flat pack
B	GDFP4-14	14	Flat pack
C	GDIP1-T14 or CDIP2-T14	14	Dual-in-line
D	GDFP1-F14 or CDFP2-F14	14	Flat pack
E	GDIP1-T16 or CDIP2-T16	16	Dual-in-line
F	GDFP2-F16 or CDFP3-F16	16	Flat pack
X	CQCC2-N20	20	Square leadless chip carrier
2	CQCC1-N20	20	Square leadless chip carrier

1.3 Absolute maximum ratings.

Supply voltage range ..... -0.5 V dc to 7.0 V dc
Input voltage range ..... -1.5 V dc at -18 mA to 5.5 V dc
Storage temperature range ..... -65° to +150°C
Maximum power dissipation per register, (P_D) 1/ :
- Device type 01 ..... 127 mW dc
- Device type 02, 03 ..... 116 mW dc
- Device type 04 ..... 110 mW dc
- Device type 05 ..... 149 mW dc
- Device type 06, 07 ..... 160 mW dc
- Device type 08 ..... 198 mW dc
- Device type 09 ..... 209 mW dc

Lead temperature (soldering, 10 seconds) ..... 300°C
Thermal resistance, junction to case (θ_{JC}):
Cases A, B, C, D, E, F, 2, and X (See MIL-STD-1835)
Junction temperature (T_J) 2/..... 175°C

1.4 Recommended operating conditions.

Supply voltage (V_{CC}) ..... 4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage (V_{IH}) ..... 2.0 V dc
Maximum low level input voltage (V_{IL}) ..... 0.7 V dc
Case operating temperature range (T_C) ..... -55° to +125°C
Minimum clock pulse width:
- Device type 01, 03, 05, 07, 09..... 20 ns
- Device type 02 ..... 18 ns
- Device type 04, 06, 08..... 25 ns

Minimum clear pulse width:
- Device type 01, 09 ..... 20 ns
- Device type 02 ..... 15 ns
- Device type 04 ..... 30 ns
- Device type 05, 07 ..... 25 ns

Minimum load pulse width:
- Device type 08 ..... 30 ns

Minimum setup time at mode control:
- Device type 01 ..... 30 ns
- Device type 03, 06 ..... 20 ns

1/ Must withstand the added P_D due to short-circuit test (e.g., I_{OS}).
2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with MIL-PRF-38535.

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Minimum setup time at shift/load:

Device type 02	25 ns
Device type 07	20 ns
Device type 08	42 ns
Device type 09	30 ns
Minimum setup time at serial data:	
Device type 08	10 ns
Minimum setup time at serial or parallel data:	
Device type 01, 02, 03, 05, 06, 07	20 ns
Device type 04	30 ns
Device type 09	18 ns
Minimum setup time at preset:	
Device type 04	30 ns
Minimum setup time at inhibit:	
Device type 08	30 ns
Minimum hold time:	
Device type 01, 02, 03, 04, 05, 07	10 ns
Device type 06	20 ns
Device type 08	3 ns
Device type 09	2 ns
Minimum enable or inhibit time of clock:	
Device type 03	20 ns
Maximum release time shift/load:	
Device type 02	10 ns

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Departments of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-38535 - Integrated Circuits (Microcircuits) Manufacturing, General Specification for.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-883 - Test Method Standard for Microelectronics.
MIL-STD-1835 - Interface Standard Electronic Component Case Outlines

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Microcircuits furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.3 and 6.4).

3.2 Item requirements. The individual item requirements shall be in accordance with MIL-PRF-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-38535 and herein.

3.3.1 Terminal connections and logic diagrams. The terminal connections and logic diagrams shall be as specified on figure 1.

3.3.2 Truth tables. The truth tables and timing diagrams shall be as specified on figure 2.

3.3.3 Logic diagrams. The logic diagrams shall be as specified on figure 3.

3.3.4 Schematic circuits. The schematic circuits shall be maintained by the manufacturer and made available to the qualifying activity and the preparing activity (DSCC-VAS) upon request.

3.3.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.4 Lead material and finish. The lead material and finish shall be in accordance with MIL-PRF-38535 (see 6.6).

3.5 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.6 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.7 Marking. Marking shall be in accordance with MIL-PRF-38535.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 12 (see MIL-PRF-38535, appendix A).

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-PRF-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein.

4.2 Screening. Screening shall be in accordance with MIL-PRF-38535 and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. The burn-in test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1015 of MIL-STD-883.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. Additional screening for space level product shall be as specified in MIL-PRF-38535, appendix B.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ -55°C ≤ T_C ≤ +125°C unless otherwise specified		Device types	Limits		Unit
					Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5 V V_{IN} = 2.0 V	I_{OH} = -1.0 mA	06,07	2.4		V
			I_{OH} = -400 µA	01,02,03 04,05, 07 (QD'), 08,09	2.5		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IN} = 0.7 V	I_{OL} = 4 mA	01,02,03 04,05, 07 (QD'), 08,09		0.4	V
			I_{OL} = 12 mA	06,07			
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA, T_C = 25°C		All		-1.5	V
High-level input current for all inputs except S/L for type 08	I_{IH1}	V_{CC} = 5.5 V, I_{IN} = 2.7 V		01,02,05, 06,07,08, 09		20	µA
	I_{IH2}	V_{CC} = 5.5 V, I_{IN} = 5.5 V				100	
High-level input current at any input except mode	I_{IH3}	V_{CC} = 5.5 V, I_{IN} = 2.7 V		03		20	µA
	I_{IH4}	V_{CC} = 5.5 V, I_{IN} = 5.5 V				100	
High-level input current at any input except preset enable	I_{IH5}	V_{CC} = 5.5 V, I_{IN} = 2.7 V		04		20	µA
	I_{IH6}	V_{CC} = 5.5 V, I_{IN} = 5.5 V				100	
High-level input current at mode	I_{IH7}	V_{CC} = 5.5 V, I_{IN} = 2.7 V		03		40	µA
	I_{IH8}	V_{CC} = 5.5 V, I_{IN} = 5.5 V				200	
High-level input current at preset enable	I_{IH9}	V_{CC} = 5.5 V, I_{IN} = 2.7 V		04		100	µA
	I_{IH10}	V_{CC} = 5.5 V, I_{IN} = 5.5 V				500	
High-level input current at S/L	I_{IH11}	V_{CC} = 5.5 V, I_{IN} = 2.7 V		08		60	µA
	I_{IH12}	V_{CC} = 5.5 V, I_{IN} = 5.5 V				300	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \leq T_C \leq +125°C$ unless otherwise specified	Device types	Limits		Unit
				Min	Max	
Off-state output current, high level voltage applied	I_{OZH}	V_{CC} = 5.5 V, V_O = 2.7 V	06,07		20	µA
Off-state output current, low level voltage applied	I_{OZL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	06,07		-20	µA
Low-level input current (for all inputs except S/L, serial in & data for types 08 and 09)	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	01,02,06	-.03	-.44	mA
			05	-.10	-.44	
			07	-.03	-.40	
			08,09	-.001	-.72	
Low-level input current at any input except clock	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	03	-.06	-.76	mA
Low-level input current at any input except preset enable	I_{IL3}		04	-.16	-.4	mA
Low-level input current at clock	I_{IL4}		03	-.03	-.44	mA
Low-level input current at preset enable	I_{IL5}		04	-.6	-2.0	mA
Low-level input current at data and serial in	I_{IL6}		08	-.100	-.380	mA
			09	-.100	-.340	mA
Low-level input current at S/L	I_{IL7}		08	-.001	-1.14	mA
			09	-.001	-.380	mA
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V 2/	01,02,03, 04,05,08, 09	-15	-100	mA
			06,07	-15	-130	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \leq T_C \leq +125°C$ unless otherwise specified	Device types	Limits		Unit
				Min	Max	
Supply current	I_{CC}	V_{CC} = 5.5 V	04		20	mA
			02,03		21	
			01		23	
			05		27	
			06		29	
			07		34	
			08		36	
			09		38	
Maximum shift frequency	f_{MAX}	V_{CC} = 5.0 V	04	17		MHz
			06	18		
			01,03, 05,07	20		
			02	25		
			08	20		
			09	20		
Propagation delay time, low-to-high level from clock	t_{PLH1}	V_{CC} = 5.0 V, C_L = 50 pF ±10% R_L = 2 kΩ for types 01 thru 05, 08 and 09. See figures 9 and 10 for R_L for types 06 and 07	01,02	5	41	ns
			03,05		48	
			07		56	
			04		68	
			06		46	
			08		58	
			09		40	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \leq T_C \leq +125°C$ unless otherwise specified	Device types	Limits		Unit
				Min	Max	
Propagation delay time, low-to-high level form preset or preset enable	t_{PLH2}	V_{CC} = 5.0, C_L = 50 pF ±10% R_L = 2 kΩ for types 01 thru 05, 08 and 09. See figures 9 and 10 for R_L types 06 and 07	02	5	53	ns
			04		60	
Propagation delay time, high-to-low level from clock	t_{PLH1}		01,02	5	47	ns
			03,05,07		56	
			04		68	
			06		52	
			08		58	
			09		46	
Propagation delay time, high-to-low level from clear	t_{PLH2}		01,02	05	53	ns
			07		56	
			05		62	
			04		90	
Propagation delay time, low to high level from S/L	t_{PLH5}		08,09	5	52	ns
Propagation delay time, high to low level from S/L or clear	t_{PHL5}		08,09	5	52	ns
Propagation delay time, high to low level from data	t_{PHL3}		08	5	46	ns
Propagation delay time, low to high level from data	t_{PLH3}		08	5	39	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \leq T_C \leq +125°C$ unless otherwise specified	Device types	Limits		Unit
				Min	Max	
Propagation delay time, low to high level from data	t_{PLH4}	V_{CC} = 5.0 V, C_L = 50 pF ±10% R_L = 2kΩ for types 01 thru 05, 08 and 09. See figures 9 and 10 for R_L types 06 and 07	08	5	46	ns
Propagation delay time, high to low level from data	t_{PHL4}		08	5	39	ns
Output enable time to low level	t_{ZL}	See figures 9 and 10 for conditions	06	5	45	ns
			07	5	53	ns
Output enable time to high level	t_{ZH}		06	5	39	ns
			07		53	
Output disable time from low level	t_{LZ}		07	5	53	ns
			06		71	
Output disable time from high level	t_{HZ}		07	5	53	ns
			06		84	

1/ Complete terminal condition shall be as specified in table III.
2/ Not more than one output should be shorted at a time.

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TABLE II. Electrical test requirements.

MIL-PRF-38535 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters	1	1
Final electrical test parameters	1*, 2, 3, 7, 9, 10, 11	1*, 2, 3, 9
Group A test requirements	1, 2, 3, 7, 8, 9, 10, 11	1, 2, 3, 7, 8, 9, 10, 11
Group B test when using the method 5005 QCI option.	1, 2, 3 9, 10, 11	N/A
Group C end-point electrical parameters	1, 2, 3, 9, 10, 11	1, 2, 3
Group D end-point electrical parameters	1, 2, 3	1, 2, 3

*PDA applies to subgroup 1.

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4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-38535 .

4.4 Technology Conformance inspection (TCI). Technology conformance inspection shall be in accordance with MIL-PRF-38535 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table III of MIL-PRF-38535 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of MIL-PRF-38535.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table IV of MIL-PRF-38535 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. The steady-state life test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table V of MIL-PRF-38535. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

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Pin number	Device type 01		Device type 02		Device type 03		Device type 04	
	CASES							
	2, X	E,F	2, X	E,F	2, X	A,B,C, and D	2, X	E,F
1	NC	CLEAR	NC	CLR	NC	SER INP	NC	CLK
2	CLEAR	SHF RHT SER INP	CLR	J	SER INP	A	CLK	A
3	SHF RHT SER INP	A	J	$\overline{K}$	A	B	A	B
4	A	B	$\overline{K}$	A	B	C	B	C
5	B	C	A	B	NC	D	C	V_{CC}
6	NC	D	NC	C	C	MODE CONT	NC	D
7	C	SHF LEFT SER INP	B	D	NC	GND	V_{CC}	E
8	D	GND	C	GND	D	CLK 2 L SHF LOAD	D	PRESET ENABLE
9	SHF LEFT SER INP	SO	D	SHF/ LOAD	MODE CONT	CLK1 R SHF	E	SER INP
10	GND	S1	GND	CLK	GND	QD	PRESET ENABLE	QE
11	NC	CLOCK	NC	$\overline{Q}$ D	NC	QC	NC	QD
12	SO	QD	SHF/LOAD	Q D	CLK 2 L SHF/LOAD	QB	SER INP	GND
13	S1	QC	CLK	QC	CLK1 R SHF	QA	QE	QC
14	CLK	QB	$\overline{Q}$ D	QB	QD	V_{CC}	QD	QB
15	QD	QA	QD	QA	NC		GND	QA
16	NC	V_{CC}	NC	V_{CC}	QC		NC	CLR
17	QC		QC		NC		QC	
18	QB		QB		QB		QB	
19	QA		QA		QA		QA	
20	V_{CC}		V_{CC}		V_{CC}		CLR	

FIGURE 1. Terminal connections.

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Pin number	Device type 05		Device type 06		Device type 07		Device type 08	
	2, X	A,B,C, and D	2, X	A,B,C and D	2, X	E,F	2, X	E, F
1	NC	A	NC	SER INP	NC	CLR	NC	SHF LOAD
2	A	B	SER INP	A	CLR	SER INP	SHF LOAD	CLOCK
3	B	QA	A	B	SER INP	A	CLOCK	E
4	QA	QB	B	C	A	B	E	F
5	NC	QC	NC	D	B	C	F	G
6	QB	QD	C	MODE CONT	NC	D	NC	H
7	NC	GND	NC	GND	C	LOAD SHF	G	$\overline{Q}$ H
8	QC	CLK	D	OUTPUT CONT	D	GND	H	GND
9	QD	CLR	MODE CONT	CLK	LOAD SHF	OUTPUT CONT	$\overline{Q}$ H	QH
10	GND	QE	GND	QD	GND	CLK	GND	SER INP
11	NC	QF	NC	QC	NC	QD'	NC	A
12	CLK	QG	OUTPUT CONT	QB	OUTPUT CONT	QD	QH	B
13	CLR	QH	CLK	QA	CLK	QC	SER INP	C
14	QE	V_{CC}	QD	V_{CC}	QD'	QB	A	D
15	NC		NC		QD	QA	B	CLOCK INHIBIT
16	QF		QC		NC	CC	NC	V_{CC}
17	NC		NC		QC		C	
18	QG		QB		QB		D	
19	QH		QA		QA		CLOCK INHIBIT	
20	V_{CC}		V_{CC}		V_{CC}		V_{CC}	

FIGURE 1. Terminal connections - Continued.

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Pin number	Device type 09	
	2, X	E,F
1	NC	SERIAL INPUT
2	SERIAL INPUT	A
3	A	B
4	B	C
5	C	D
6	NC	CLOCK INHIBIT
7	D	CLOCK
8	CLOCK INHIBIT	GND
9	CLK	CLEAR
10	GND	E
11	NC	F
12	CLR	G
13	E	OUTPUT QH
14	F	INPUT H
15	G	SHIFT LOAD
16	NC	V_{CC}
17	QH	
18	INPUT H	
19	SHIFT LOAD	
20	V_{CC}	

FIGURE 1. Terminal connections - Continued.

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Device type 01

INPUTS										OUTPUTS			
CLEAR	MODE		CLOCK	SERIAL		PARALLEL				Q_A	Q_B	Q_C	Q_D
	S1	S0		LEFT	RIGHT	A	B	C	D				
L	X	X	X	X	X	X	X	X	X	L	L	L	L
H	X	X	L'	X	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}
H	H	H	↑	X	X	a	b	c	d	a	b	c	d
H	L	H	↑	X	H	X	X	X	X	H	Q_{An}	Q_{Bn}	Q_{Cn}
H	L	H	↑	X	L	X	X	X	X	L	Q_{An}	Q_{Bn}	Q_{Cn}
H	H	L	↑	H	X	X	X	X	X	Q_{Bn}	Q_{Cn}	Q_{Dn}	H
H	H	L	↑	L	X	X	X	X	X	Q_{Bn}	Q_{Cn}	Q_{Dn}	L
H	L	L	X	X	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}

H = high level (steady state)
L = low level (steady state)
X = irrelevant (any input, including transitions)
↑ = transition from low to high level

a, b, c, d = the level of steady state input at inputs A, B, C, or D, respectively.
Q_{A0}, Q_{B0}, Q_{C0}, Q_{D0} = level of Q_A, Q_B, Q_C, or Q_D, respectively, before the indicated steady state input conditions were established.
Q_{An}, Q_{Bn}, Q_{Cn}, Q_{Dn} = the level of Q_A, Q_B, Q_C or Q_D, respectively, before the most recent ↑ transition of the clock.

Typical clear, load, right-shift, left shift, inhibit, and clear sequences.

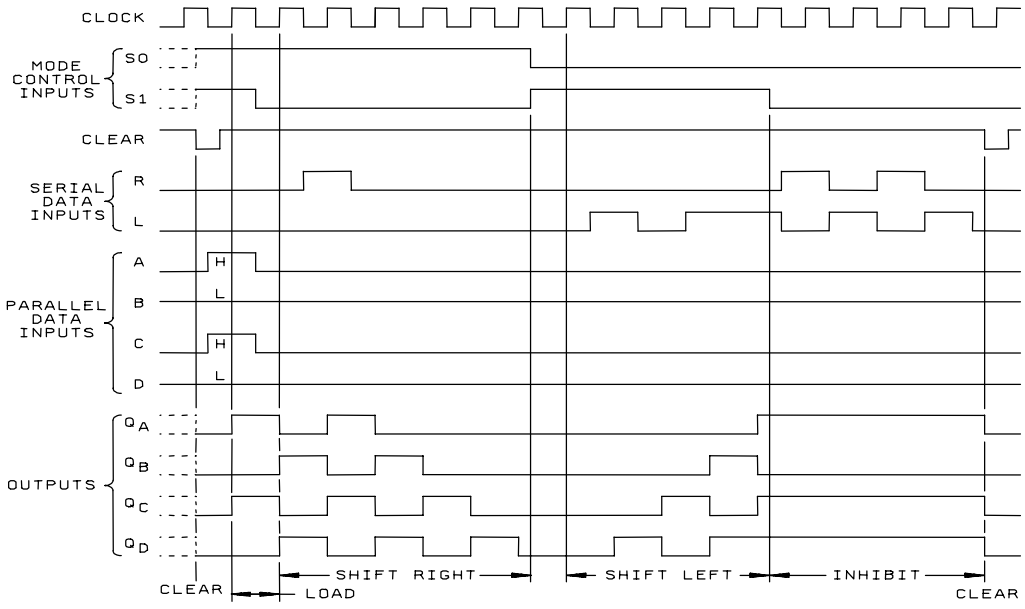


FIGURE 2. Truth tables and timing diagrams.

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Device type 02

INPUTS									OUTPUTS				
CLEAR	SHIFT/ LOAD	CLOCK	SERIAL		PARALLEL				Q_A	Q_B	Q_C	Q_D	$\overline{Q}_D$
			J	$\overline{K}$	A	B	C	D					
L	X	X	X	X	X	X	X	X	L	L	L	L	H
H	L	↑	X	X	a	b	c	d	a	b	c	d	$\overline{d}$
H	H	L	X	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}	$\overline{Q}_{D0}$
H	H	↑	L	H	X	X	X	X	Q_{A0}	Q_{A0}	Q_{Bn}	Q_{Cn}	$\overline{Q}_{Cn}$
H	H	↑	L	L	X	X	X	X	L	Q_{An}	Q_{Bn}	Q_{Cn}	$\overline{Q}_{Cn}$
H	H	↑	H	H	X	X	X	X	H	Q_{An}	Q_{Bn}	Q_{Cn}	$\overline{Q}_{Cn}$
H	H	↑	H	L	X	X	X	X	$\overline{Q}_{An}$	Q_{An}	Q_{Bn}	Q_{Cn}	$\overline{Q}_{Cn}$

H = high level (steady state)
L = low level (steady state)
X = irrelevant (any input, including transitions)
↑ = transition from low to high level
a, b, c, d = the level of steady state input at inputs A, B, C, or D, respectively.
Q_{A0}, Q_{B0}, Q_{C0}, Q_{D0} = level of Q_A, Q_B, Q_C, or Q_D, respectively, before the indicated steady state input conditions were established.
Q_{An}, Q_{Bn}, Q_{Cn}, = the level of Q_A, Q_B, or Q_C, respectively, before the most recent transition of the clock.

Typical clear, shift, and load sequences.

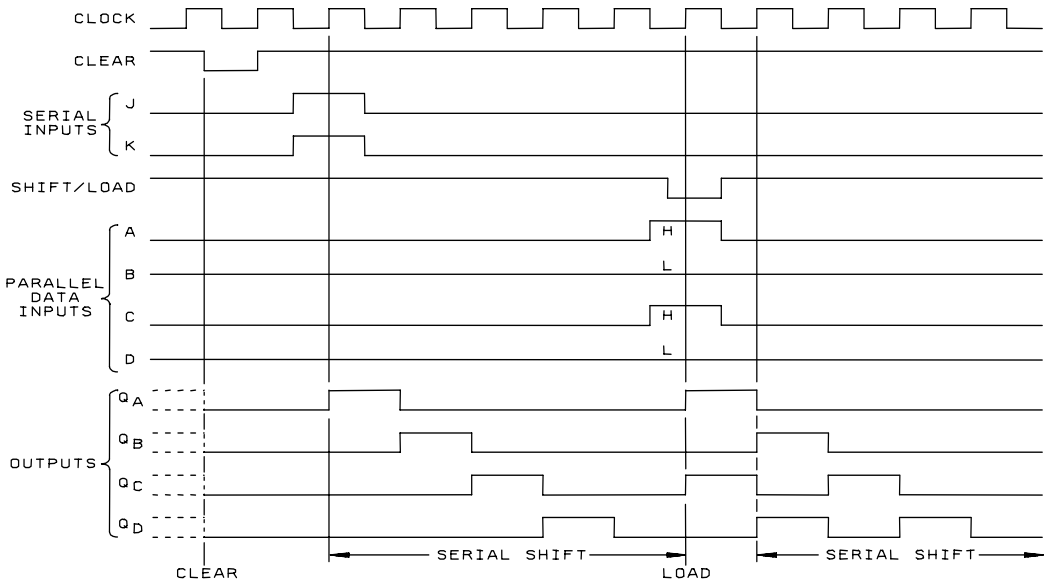


FIGURE 2. Truth tables and timing diagrams - Continued.

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Device type 03

MODE CONTROL	CLOCKS		SERIAL	PARALLEL				Q_A	Q_B	Q_C	Q_D
	2 (L)	1 (R)		A	B	C	D				
H	H	X	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}
H	↓	X	X	a	b	c	d	a	b	c	d
H	↓	X	X	Q_B ↑	Q_C ↑	Q_D ↑	d	Q_{Bn}	Q_{Cn}	Q_{Dn}	d
L	L	H	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}
L	X	↓	H	X	X	X	X	H	Q_{An}	Q_{Bn}	Q_{Cn}
L	X	↓	L	X	X	X	X	L	Q_{An}	Q_{Bn}	Q_{Cn}
↑	L	L	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}
↓	L	L	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}
↓	L	H	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}
↑	H	L	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}
↑	H	H	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}

[+]Shifting left requires external connection of Q_B to A, Q_C to B, and Q_D to C.
Serial data is entered to input D.

H = High level (steady state), L = Low level (steady state),
X = Irrelevant (any input, including transitions)
↓= Transition from high to low level, ↑= Transition from low to high level a, b, c,
d = the level of steady state input at inputs A, B, C, or D, respectively.
Q_{A0}, Q_{B0}, Q_{C0}, Q_{D0} = level of Q_A, Q_B, Q_C, or Q_D, respectively,
before the indicated steady state input conditions were established.
Q_{An}, Q_{Bn}, Q_{Cn}, Q_{Dn} = the level of Q_A, Q_B, Q_C, or Q_D, respectively,
most recent ↓ transition of the clock.

FIGURE 2. Truth tables and timing diagrams - Continued.

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Device type 04

INPUTS													
CLEAR	PRESET ENABLE	PRESET					CLOCK	SERIAL	Q_A	Q_B	Q_C	Q_D	Q_E
		A	B	C	D	E							
L	L	X	X	X	X	X	X	X	L	L	L	L	L
L	X	L	L	L	L	L	X	X	L	L	L	L	L
H	H	H	H	H	H	H	X	X	H	H	H	H	H
H	H	L	L	L	L	L	L	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}	Q_{E0}
H	H	H	L	H	L	H	L	X	H	Q_{B0}	H	Q_{D0}	H
H	L	X	X	X	X	X	L	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}	Q_{E0}
H	L	X	X	X	X	X	↑	H	H	Q_{An}	Q_{Bn}	Q_{Cn}	Q_{Dn}
H	L	X	X	X	X	X	↑	L	L	Q_{An}	Q_{Bn}	Q_{Cn}	Q_{Dn}

H = high level (steady state), L = low level (steady state)
X = irrelevant (any input, including transitions)
↑ = transition from low to high level
Q_{A0}, $Q_{B0,}$ etc. = the level of Q_A, Q_B, etc., respectively before the indicated steady state input conditions were established.
Q_{An}, Q_{Bn}, etc. = the level of Q_A, Q_B, etc., respectively before the most recent ↑ transition of the clock.

Typical clear, shift, preset and shift sequences

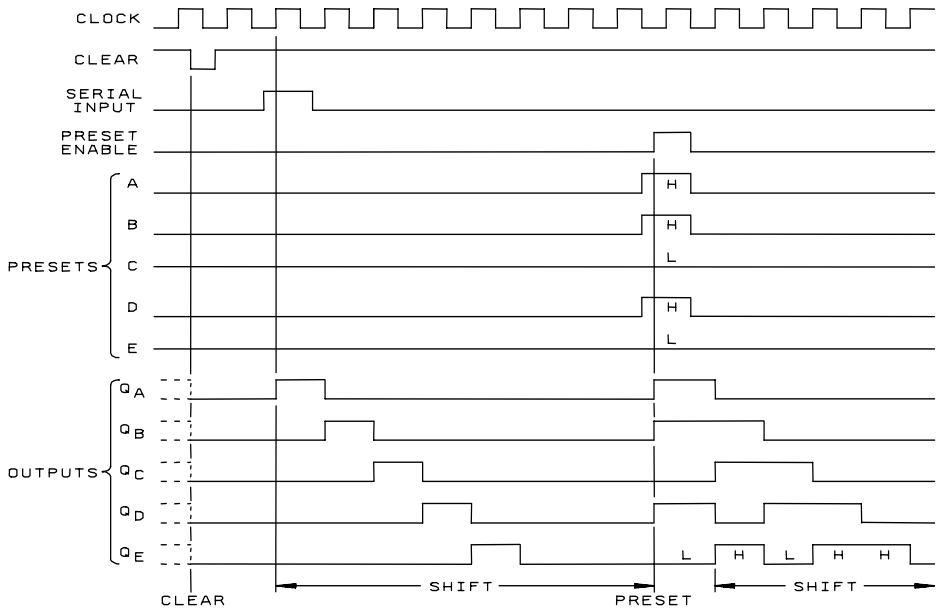


FIGURE 2. Truth tables and timing diagrams - Continued.

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Device type 05

INPUTS				OUTPUTS		
CLEAR	CLOCK	A	B	Q_A	Q_B .... Q_H	
L	X	X	X	L	L	L
H	L	X	X	Q_{A0}	Q_{B0}	Q_{H0}
H	↑	H	H	H	Q_{An}	Q_{Gn}
H	↑	L	X	L	Q_{An}	Q_{Gn}
H	↑	X	L	L	Q_{An}	Q_{Gn}

H = high level (steady state), L = low level (steady state)
X = irrelevant (any input, including transitions)
↑ = transition from low to high level
Q_{A0}, Q_{B0}, Q_{H0} = the level of Q_A, Q_B, or Q_H, respectively, before the indicated steady state input conditions were established.
Q_{An}, Q_{Gn} = the level of Q_A, or Q_G before the most recent ↑ transition of the clock; indicates a one-bit shift.

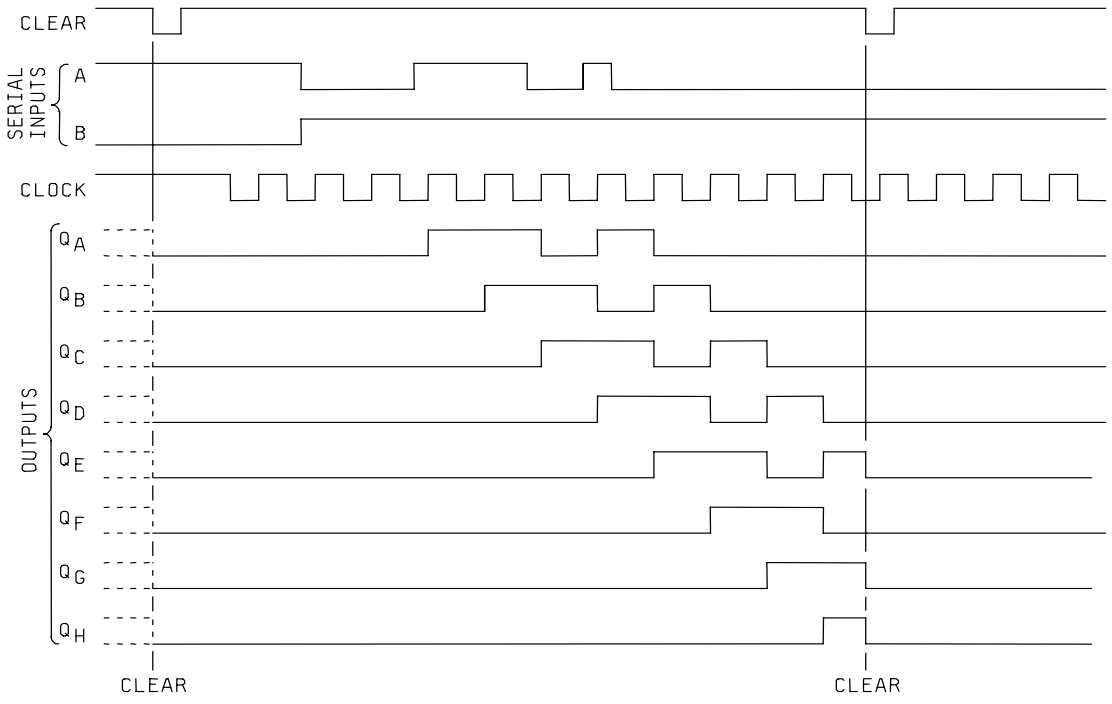


FIGURE 2. Truth tables and timing diagrams - Continued.

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Device type 06

INPUTS							OUTPUTS			
MODE CONTROL	CLOCK	SERIAL	PARALLEL				Q_A	Q_B	Q_C	Q_D
			A	B	C	D				
H	H	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}
H	↓	X	a	b	c	d	a	b	c	d
H	↓	X	Q_B ↑	Q_C ↑	Q_D ↑	d	Q_{Bn}	Q_{Cn}	Q_{Dn}	d
L	H	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}
L	↓	H	X	X	X	X	H	Q_{An}	Q_{Bn}	Q_{Cn}
L	↓	L	X	X	X	X	L	Q_{An}	Q_{Bn}	Q_{Cn}
When the output control is low, the outputs are disabled to high impedance state. however, sequential operation of the registers is not affected.										

[+]Shifting left requires external connection of Q_B to A, Q_C to B, and Q_D to C. Serial data is entered to input D.

H = high level (steady state), L = low level (steady state)
X = irrelevant (any input, including transitions)
↓= transition from high to low level.
a, b, c,d = the level of steady state input at inputs A, B, C, or D, respectively.
Q_{A0}, Q_{B0}, Q_{C0}, Q_{D0} = the level of Q_A, Q_B, Q_C, or Q_D, respectively, before the indicated steady state input conditions were established.
Q_{An}, Q_{Bn}, Q_{Cn}, Q_{Dn} = the level of Q_A, Q_B, Q_C, or Q_D, respectively, before the most recent ↓ transition of the clock.

Device type 07

INPUTS								3 STATE OUTPUTS				CASCADE OUTPUT
CLEAR	LOAD/SHIFT CONTROL	CLOCK	SERIAL	PARALLEL				Q_A	Q_B	Q_C	Q_D	$Q_{D'}$
				A	B	C	D					
L	X	X	X	X	X	X	X	L	L	L	L	L
H	H	H	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}	Q_{D0}
H	H	↓	X	a	b	c	d	a	b	c	d	d
H	L	H	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}	Q_{D0}
H	L	↓	H	X	X	X	X	H	Q_{An}	Q_{Bn}	Q_{Cn}	Q_{Cn}
H	L	↓	L	X	X	X	X	L	Q_{An}	Q_{Bn}	Q_{Cn}	Q_{Cn}
When the output control is low, the outputs are disabled to high impedance state. however, sequential operation of the registers is not affected.												

H = high level (steady state), L = low level (steady state),
X = irrelevant (any input, including transitions)
↓= transition from high to low level.
Q_{A0}, Q_{B0}, Q_{C0}, Q_{D0} = the level of Q_A, Q_B, Q_C, or Q_D, respectively, before the indicated steady state input conditions were established.
Q_{An}, Q_{Bn}, Q_{Cn}, Q_{Dn} = the level of Q_A, Q_B, Q_C, or Q_D, respectively, before the most recent ↓ transition of the clock.

FIGURE 2. Truth tables and timing diagrams - Continued.

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Device type 08

Inputs					Internal Outputs		Output Q_H
Shift/ load	Clock inhibit	Clock	Serial	Parallel A....H	Q_A	Q_B	
L	X	X	X	a.....h	a	b	h
H	L	L	X	X	Q_{A0}	Q_{B0}	Q_{H0}
H	L	↑	H	X	H	Q_{An}	Q_{Gn}
H	L	↑	L	X	L	Q_{An}	Q_{Gn}
H	H	X	X	X	Q_{A0}	Q_{B0}	Q_{H0}

H = High level (steady state), L = Low level (steady state)
X = Irrelevant (any input, including transitions)
↑= Transition from low to high level
Q_{A0}, Q_{B0}, Q_{H0} = the level of Q_A, Q_B, or Q_H, respectively, before the indicated steady state input conditions were established.
Q_{An}, Q_{Gn} = The level of Q_A or Q_G before the most recent transition of the clock; indicates a one-bit shift.

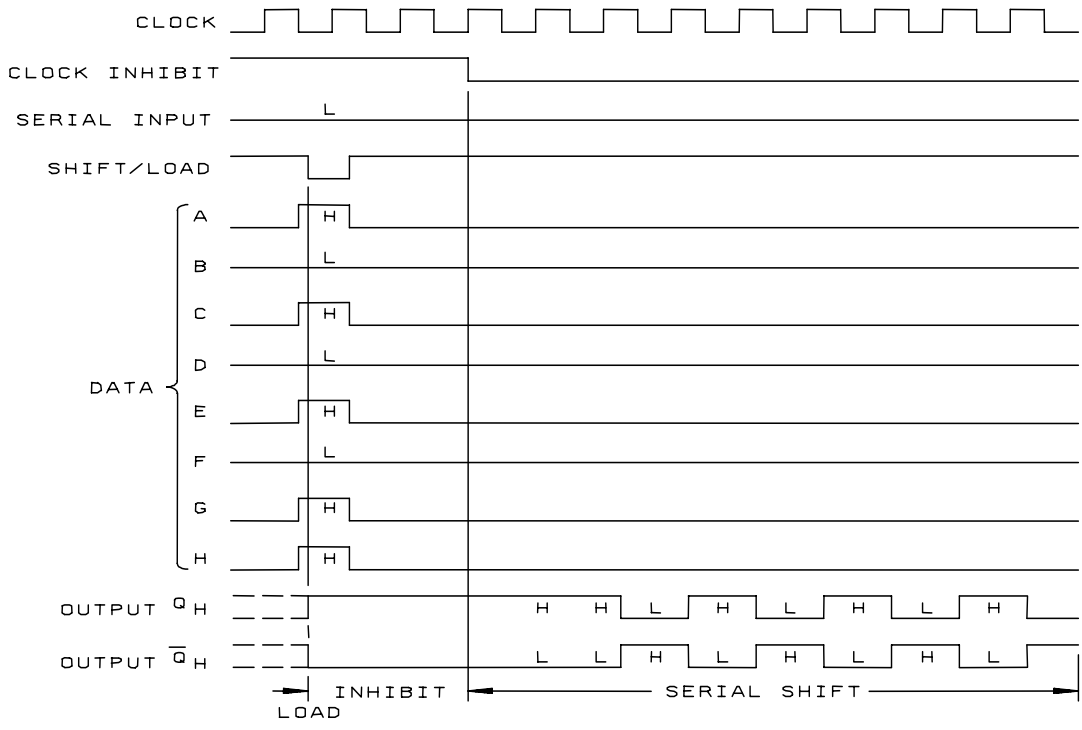


FIGURE 2. Truth tables and timing diagrams - Continued.

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Device type 09

Clear	Shift/ load	Clock inhibit	Clock	Serial	Parallel A....H	Internal Outputs Q_A	Internal Outputs Q_B	Output Q_H
L	X	X	X	X	X	L	L	L
H	X	L	L	X	X	Q_{A0}	Q_{B0}	Q_{H0}
H	L	L	↑	X	a....h	a	b	h
H	H	L	↑	H	X	H	Q_{An}	Q_{Gn}
H	H	L	↑	L	X	L	Q_{An}	Q_{Gn}
H	X	H	↑	X	X	Q_{A0}	Q_{B0}	Q_{H0}

H = High level (steady state), L = Low level (steady state)
X = Irrelevant (any input, including transitions)
↑ = Transition from low to high level
Q_{A0}, Q_{B0}, Q_{H0} = the level of Q_A, Q_B, or Q_H respectively, before the indicated steady state input conditions were established.
Q_{An}, Q_{Gn} = The level of Q_A or Q_G before the most recent ↑ transition of the clock; indicates a one-bit shift.

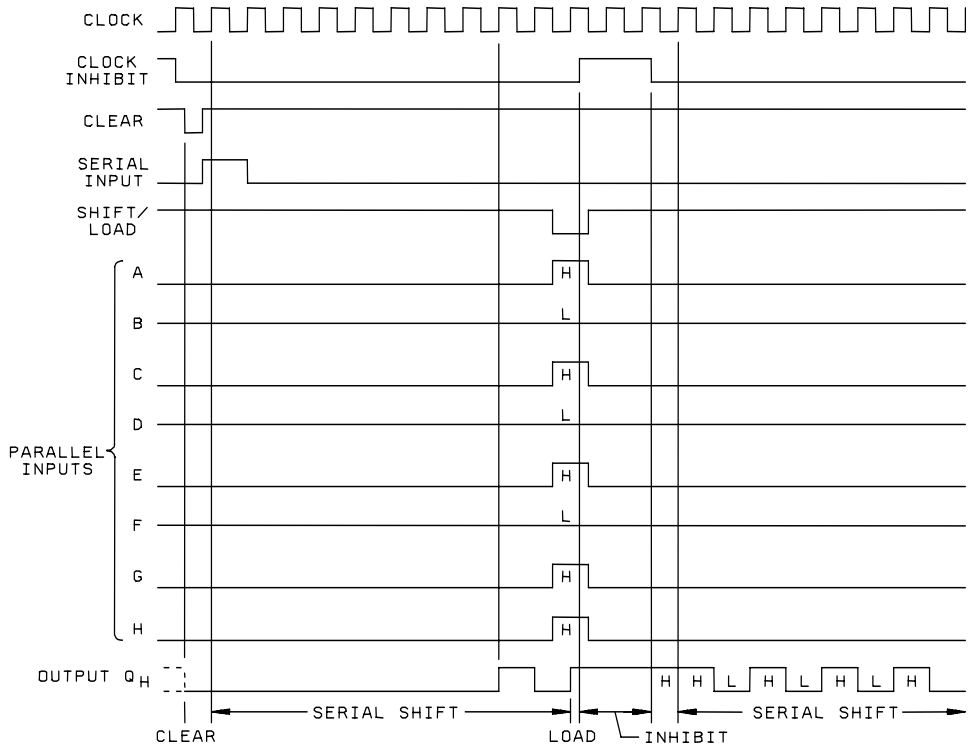


FIGURE 2. Truth tables and timing diagrams - Continued.

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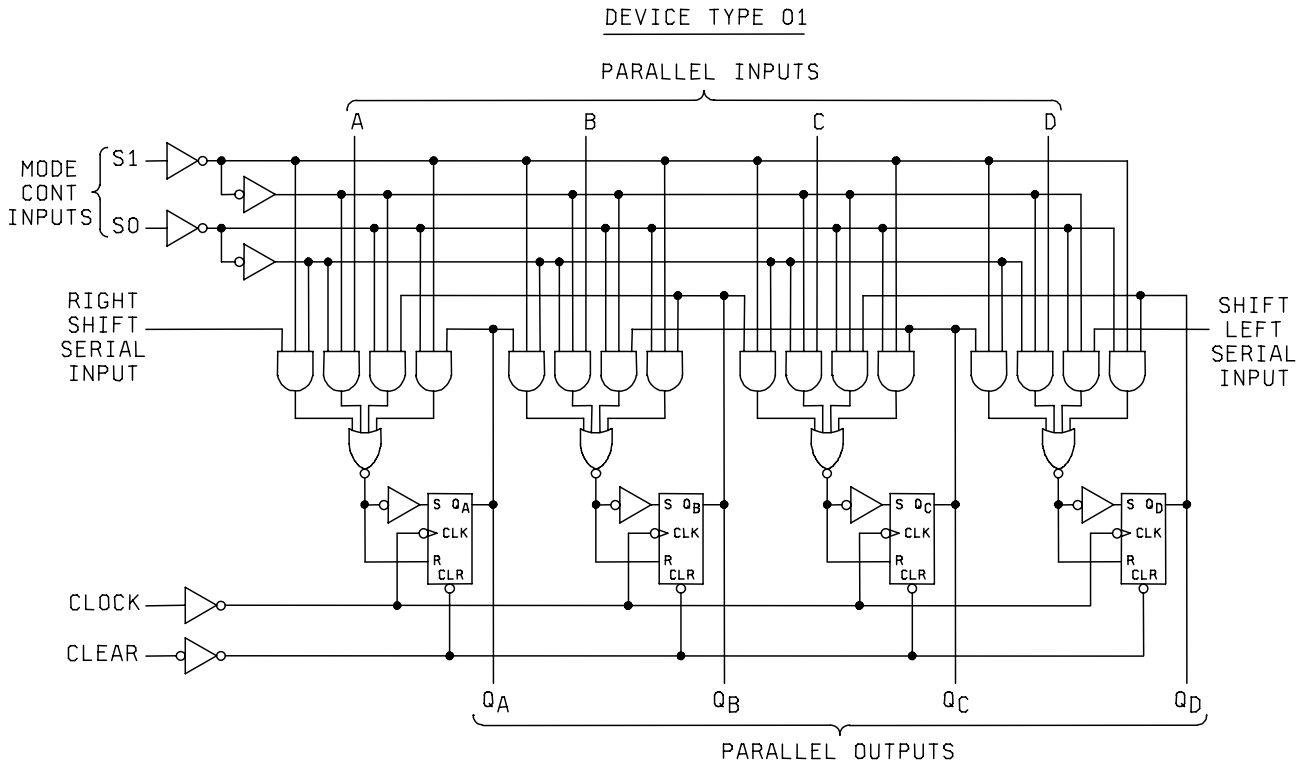
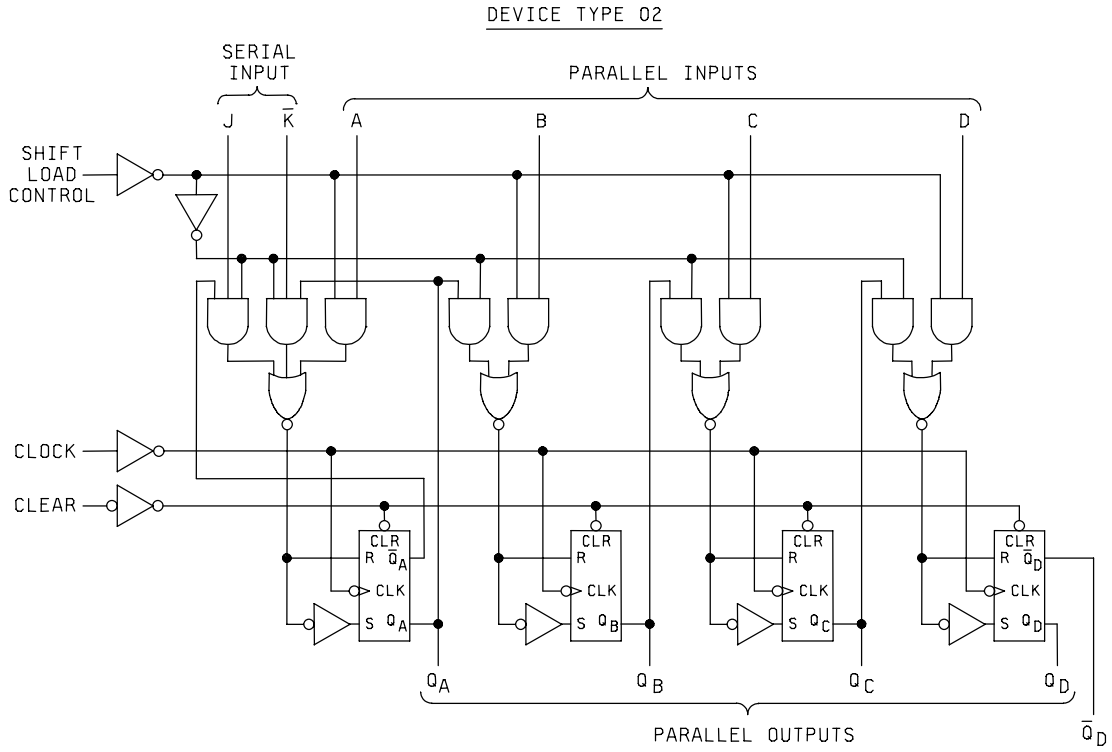


FIGURE 3. Logic diagrams.

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... dynamic input activated from a high level to a low level

FIGURE 3. Logic diagrams - Continued.

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DEVICE TYPE 03

CIRCUITS A,B,C,D, AND E

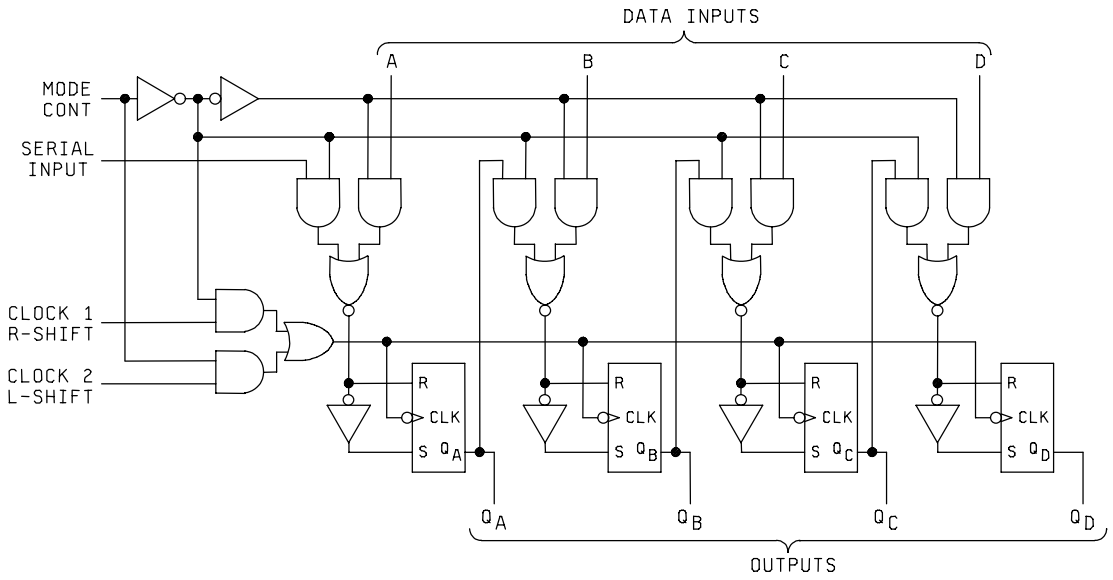


FIGURE 3. Logic diagrams - Continued.

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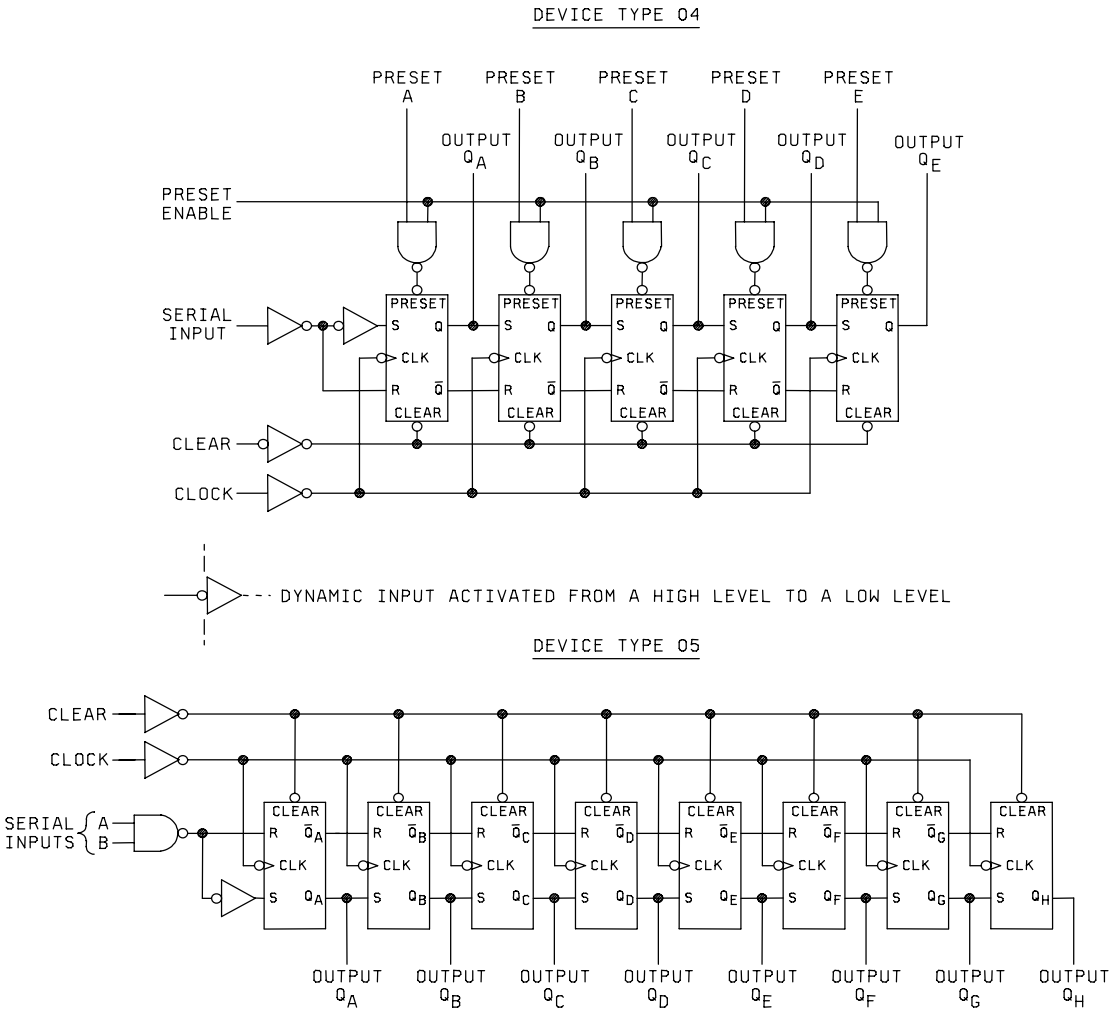


FIGURE 3. Logic diagram - Continued.

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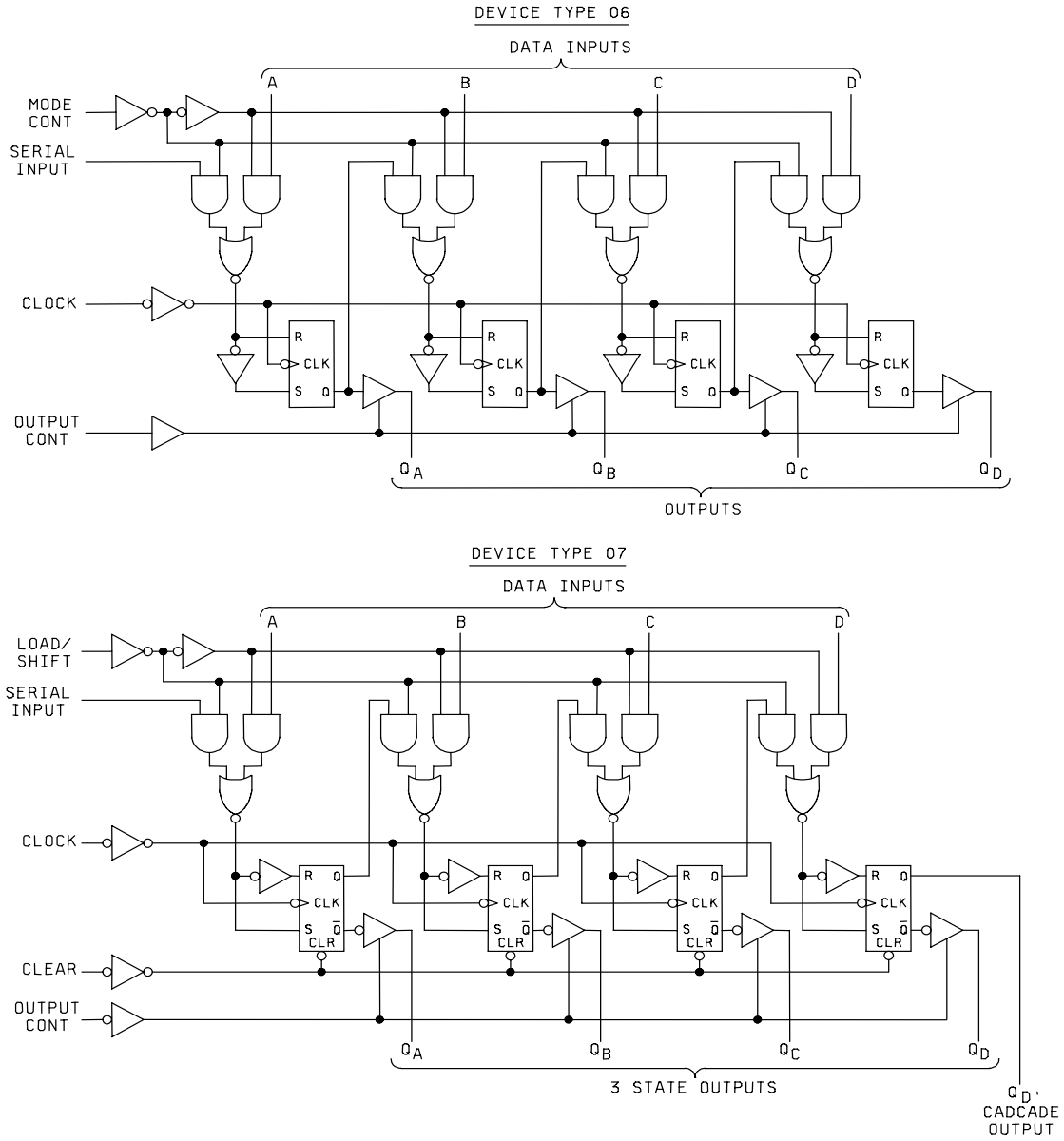
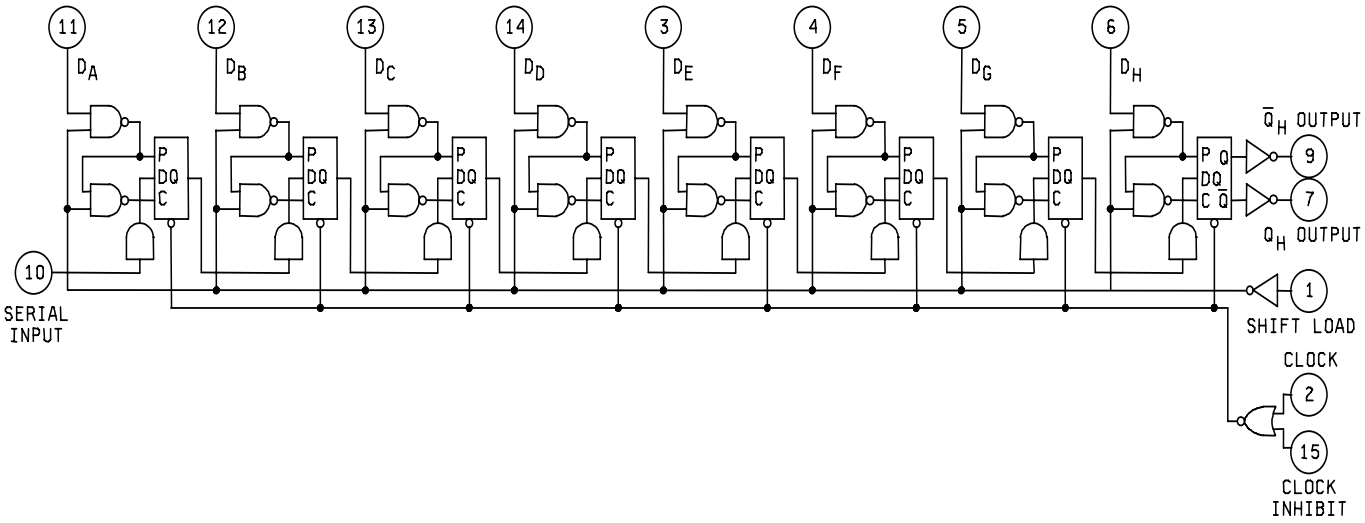


FIGURE 3. Logic diagrams - Continued.

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DEVICE TYPE 08

CIRCUIT A



Pin numbers are for cases E and F only.

FIGURE 3. Logic diagrams - Continued.

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DEVICE TYPE 08

CIRCUIT C AND F

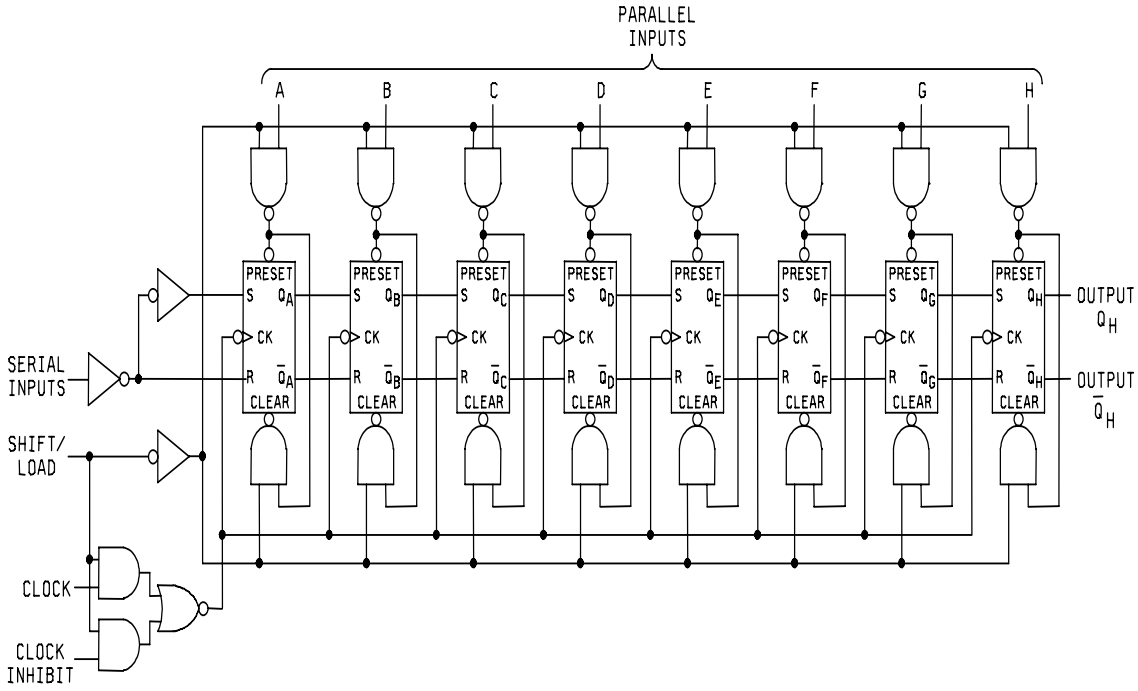
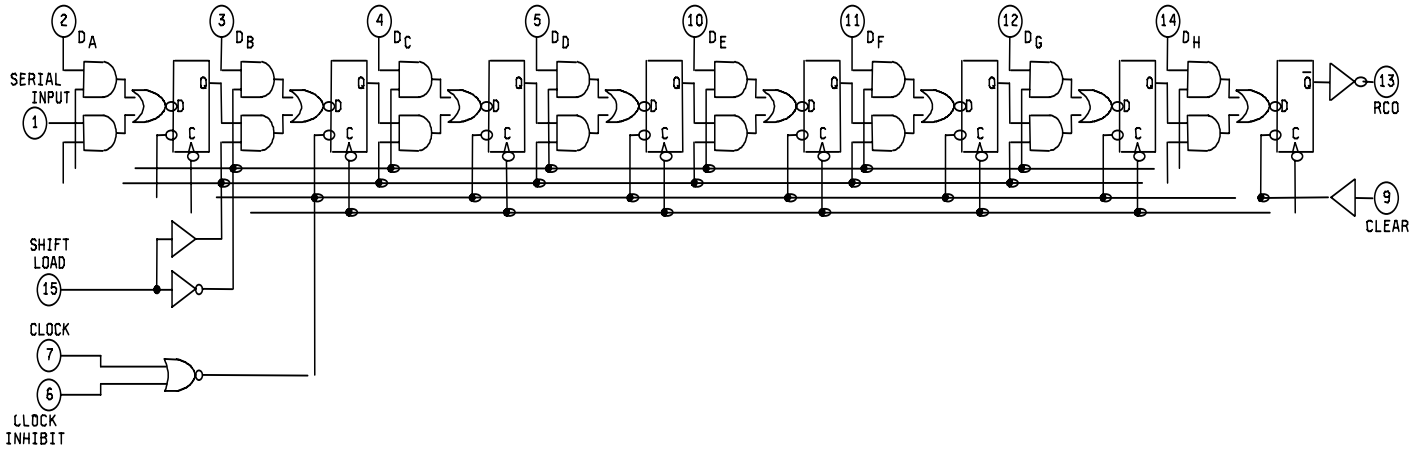


FIGURE 3. Logic diagrams - Continued.

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Device Type 09



Pin numbers are for cases E and F only.

FIGURE 3. Logic diagrams - Continued.

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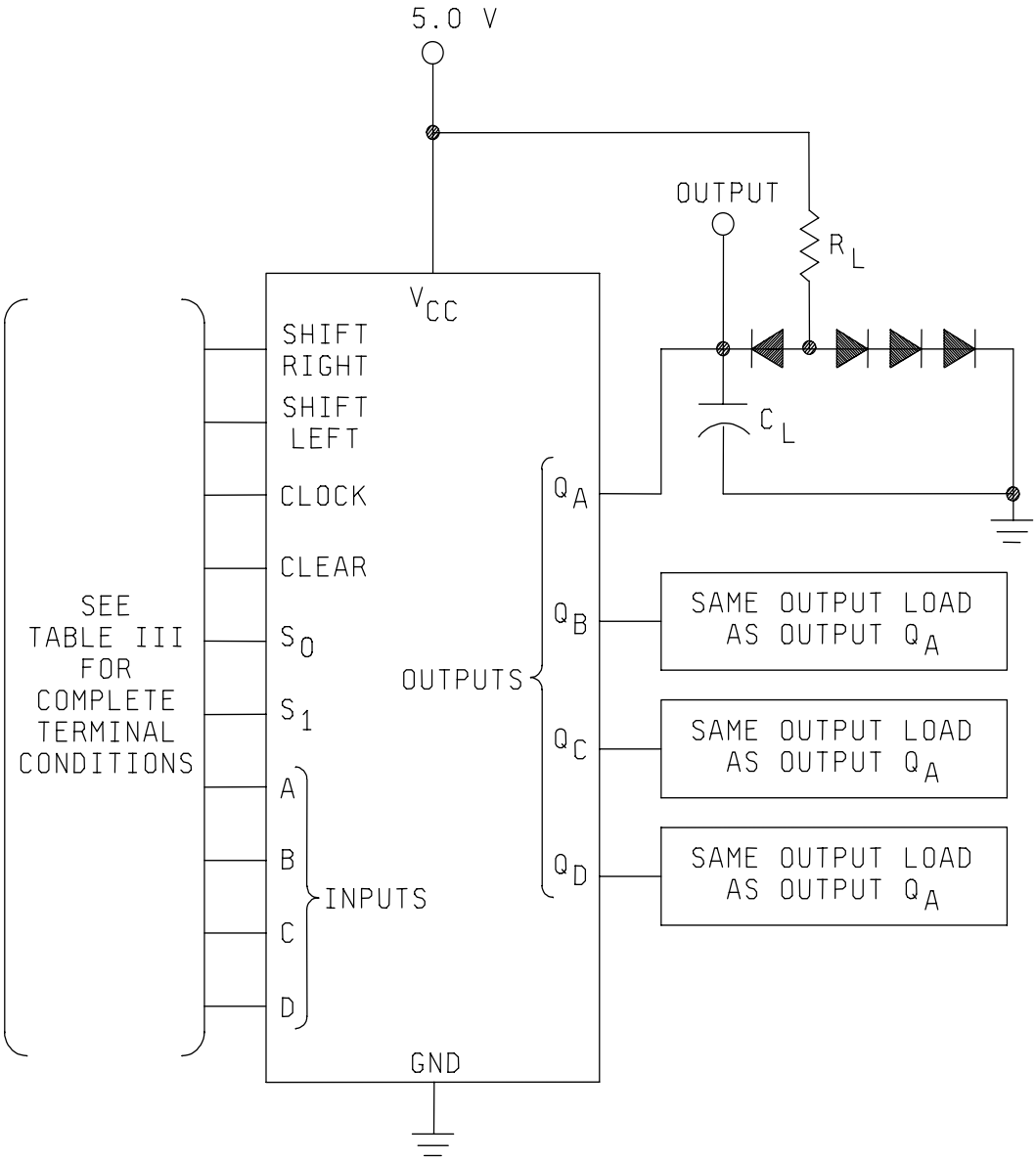
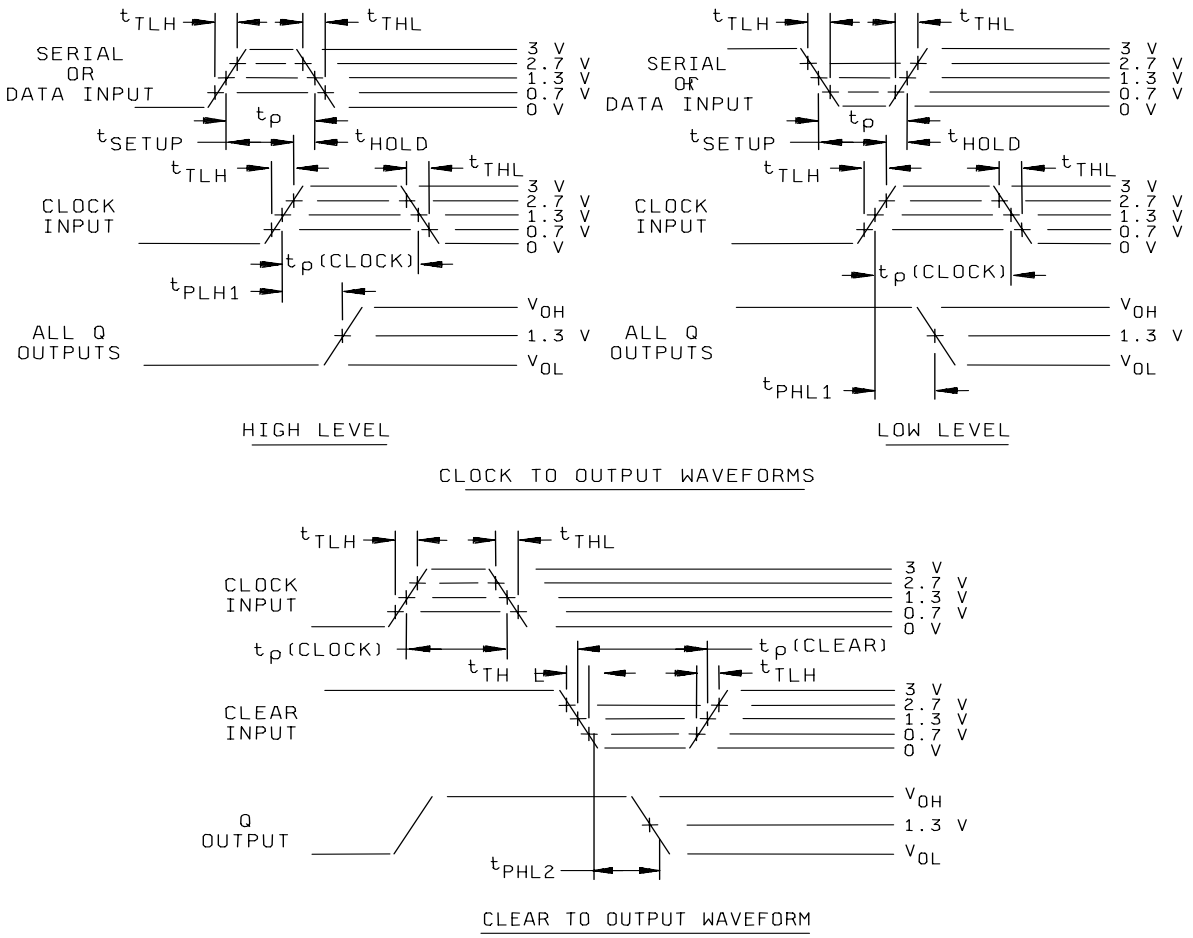


FIGURE 4. Switching test circuit and waveforms for device type 01.

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NOTES:

1. Clock pulse characteristics: PRR ≤ 1.0 Mhz, t_{TLH} ≤ 15 ns, t_{THL} ≤ 6 ns, t_p (clock) ≥ 20 ns.

2. Serial or data pulse characteristics: t_{THL} ≤ 15 ns, t_{THL} ≤ 6 ns, t_{SETUP} = 20 ns, t_{HOLD} = 10 ns, t_p (serial) or t_p (data) = 30 ns.

3. Clear pulse characteristics: t_{THL} ≤ 15 ns, t_{THL} ≤ 6 ns; t_p (clear) = 20 ns.

4. C_L = 50 pF ±10 percent incliding scope, probe, wiring and stray capacitance without package in test fixture.

5. All diodes are 1N3064, 1N916 or equivalent.

6. R_L = 2.0 kΩ ±5percent.

7. Prior to initiating tests, the output shall be placed in the proper state.

FIGURE 4. Switching test circuit and waveforms for device type 01 - Continued.

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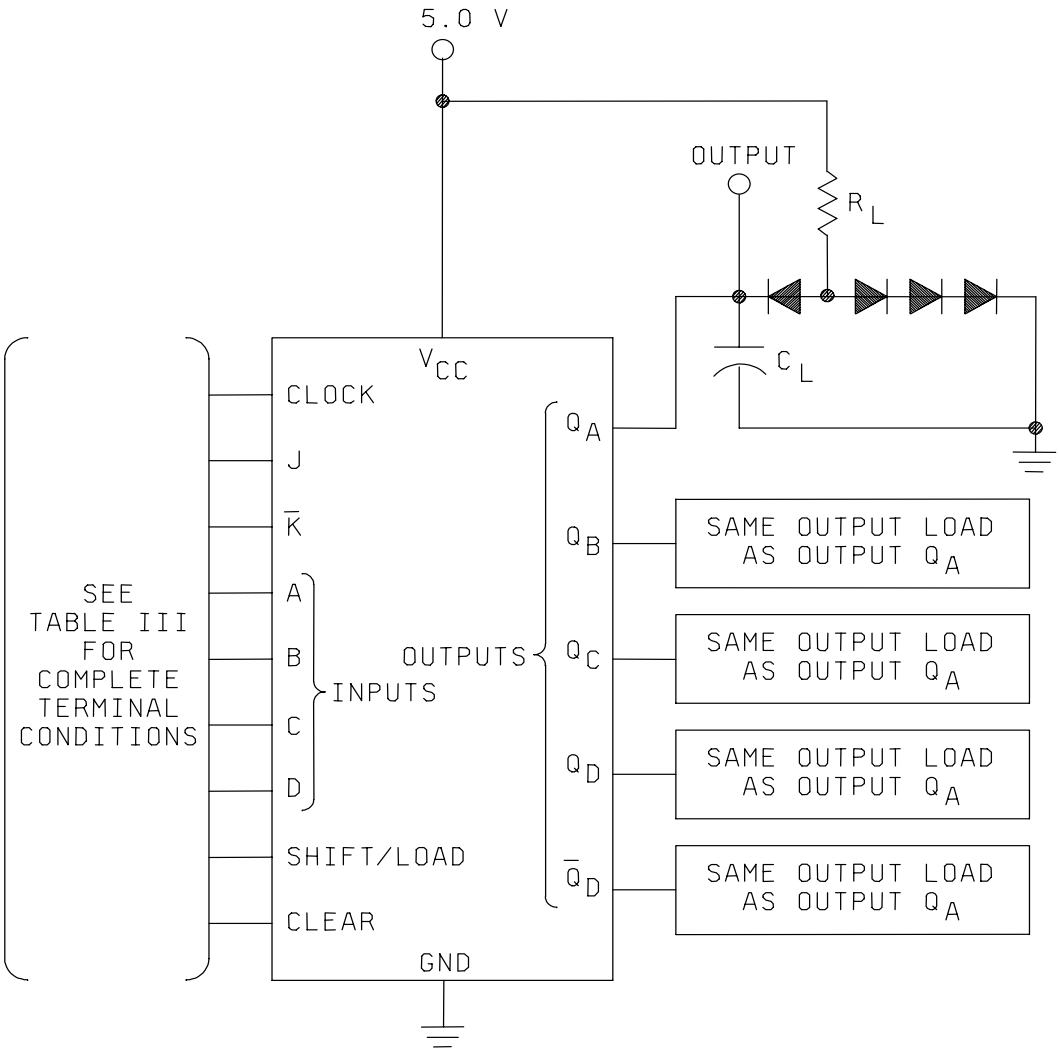


FIGURE 5. Switching test circuit and waveforms for device type 02.

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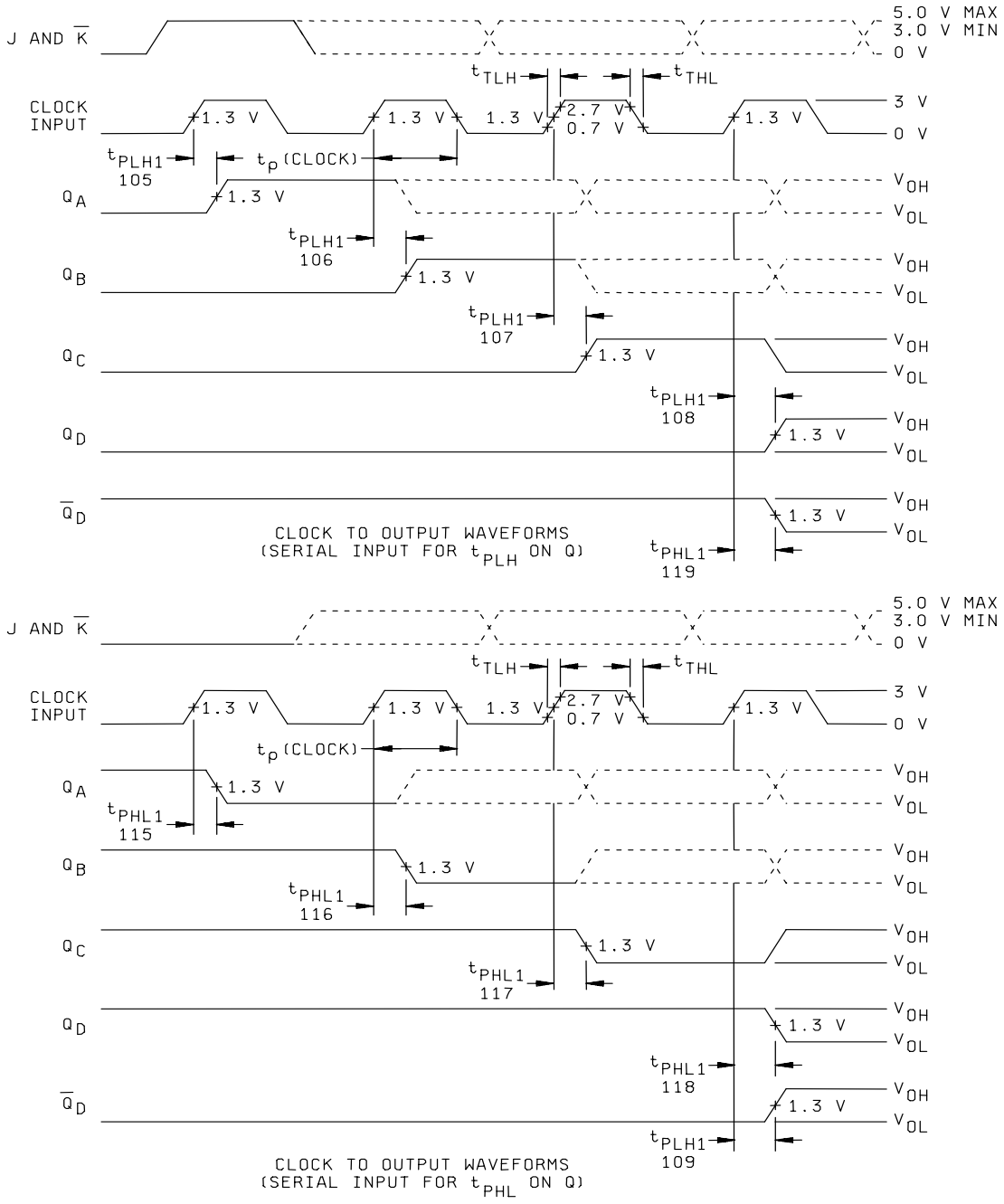
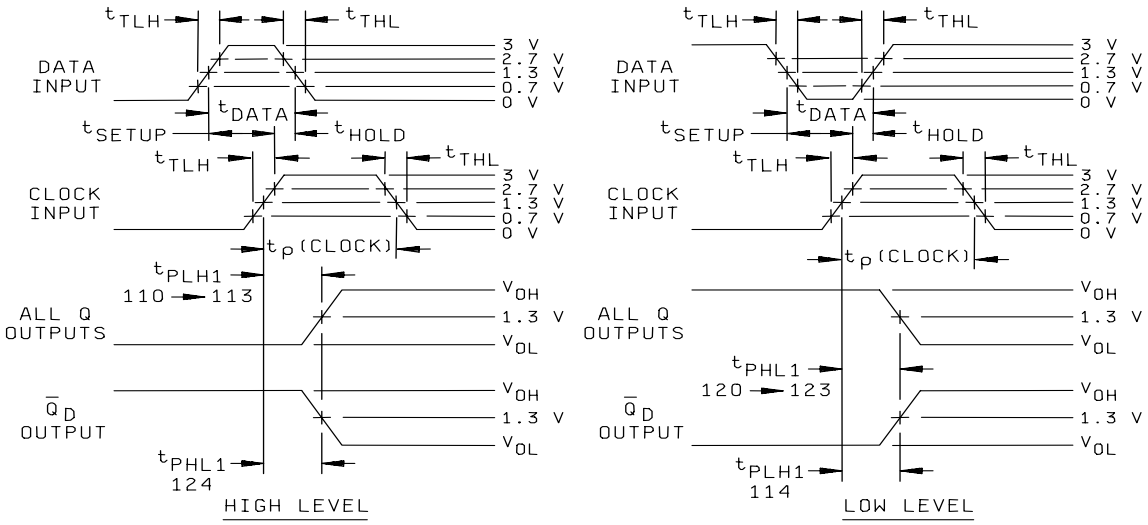
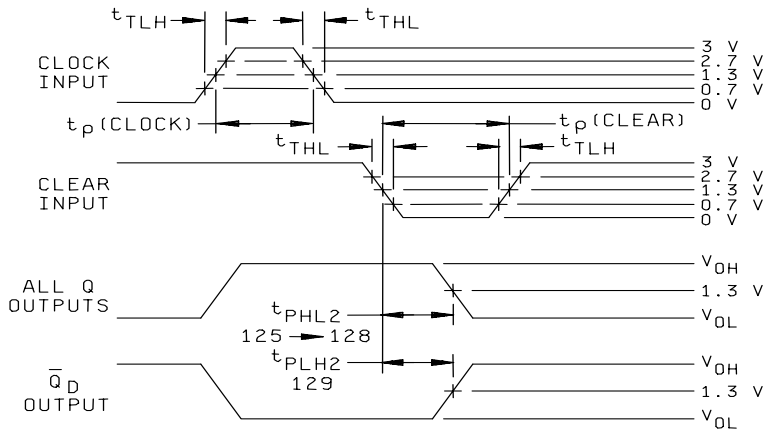


FIGURE 5. Switching test circuit and waveforms for device type 02 - Continued.

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CLOCK TO OUTPUT WAVEFORMS (PARALLEL INPUT)



CLEAR TO OUTPUT WAVEFORMS

NOTES:
1. Clock pulse characteristics: PRR ≤ 1.0 MHz, t_{TLH} ≤ 15 ns, t_{TLH} ≤ 6 ns, t_p (clock) ≥ 18 ns.
2. Data pulse characteristics: t_{TLH} ≤ 20 ns, t_{THL} ≤ 6 ns, t_{SETUP} = 20 ns, t_{HOLD} = 10 ns, t_{DATA} = 30 ns.
3. Clear pulse characteristics: t_{TLH} ≤ 15 ns, t_{THL} ≤ 6 ns; t_p (clear) = 15 ns.
4. C_L = 50 pF ±10 percent including scope, probe, wiring and stray capacitance without package in test fixture.
5. All diodes are 1N3064, 1N916 or equivalent.
6. R_L = 2.0 kΩ ± 5 percent.
7. Prior to initiating tests, the output shall be placed in the proper state.

FIGURE 5. Switching test circuit and waveforms for device type 02 - Continued.

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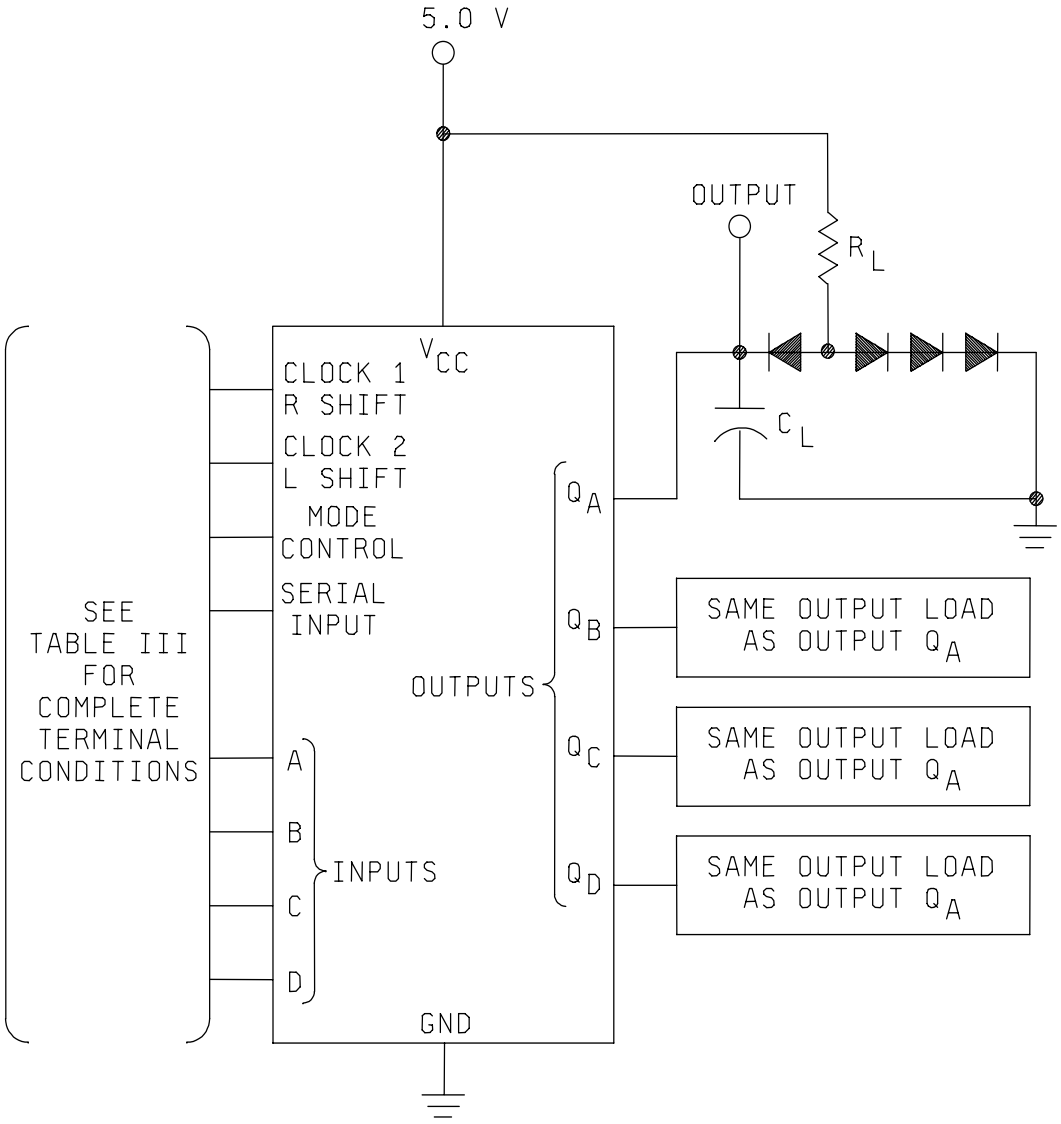


FIGURE 6. Switching test circuit and waveforms for device type 03.

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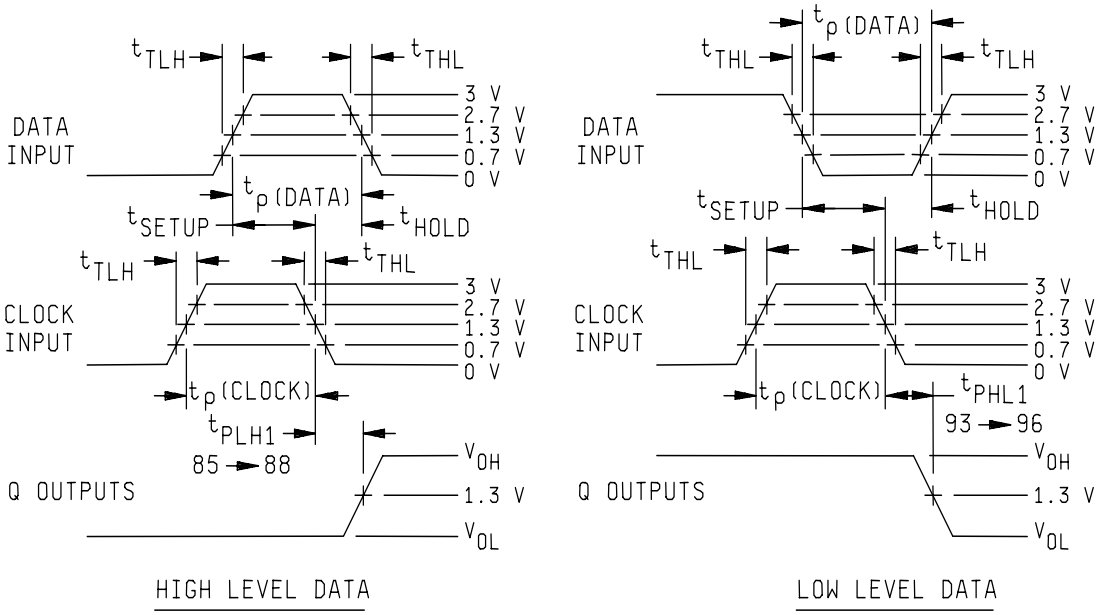
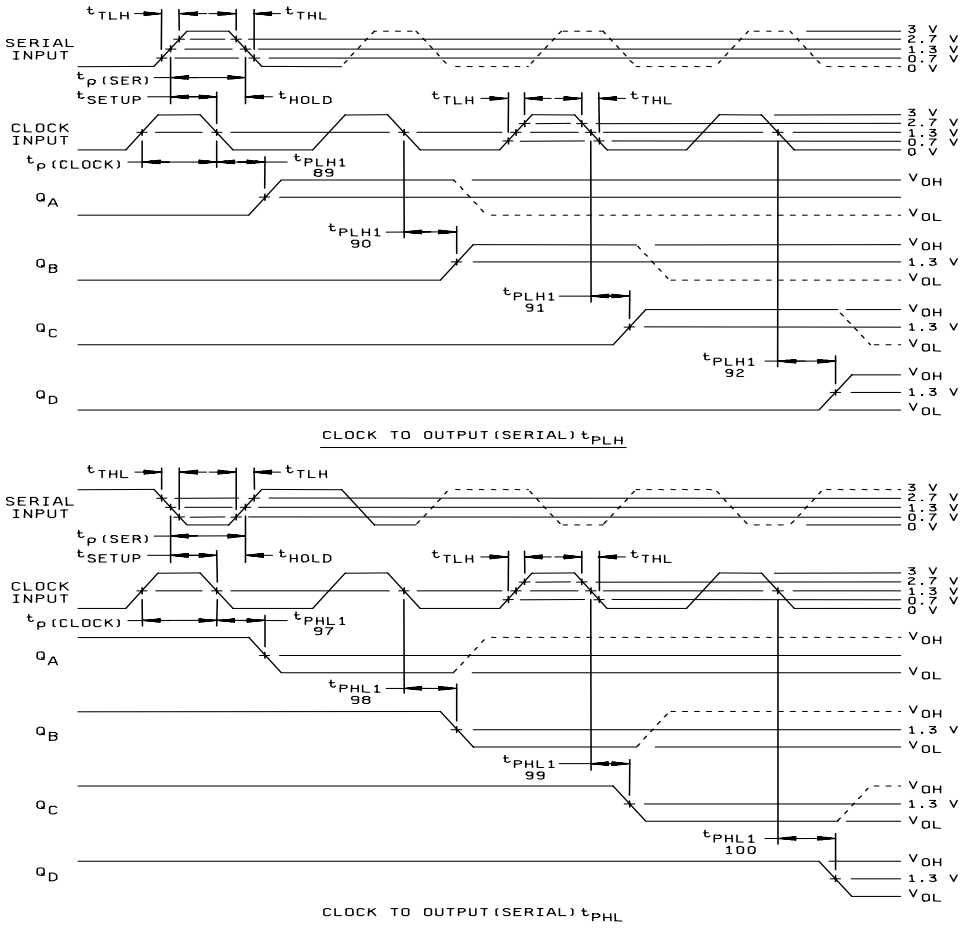


FIGURE 6. Switching test circuit and waveforms for device type 03 - Continued.

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NOTES:

1. Clock pulse characteristics: PRR ≤ 1.0 MHz, t_{TLH} ≤ 15 ns, t_{THL} ≤ 6 ns, t_p (clock) ≥ 20 ns.

2. Serial data pulse characteristics: t_{TLH} ≤ 15 ns, t_{THL} ≤ 6 ns, t_p (SER) or t_p (DATA) = 30 ns, t_{SETUP} = 20 ns, t_{HOLD} = 10 ns.

3. C_L = 50 pF ±10 percent including scope, probe, wiring and stray capacitance without package in test fixture.

4. R_L = 2.0 kΩ ±5%.

5. All diodes are 1N3064, 1N916 or equivalent.

6. Prior to initiating tests, the output shall be placed in the proper state.

FIGURE 6. Switching test circuit and waveforms for device type 03 - Continued.

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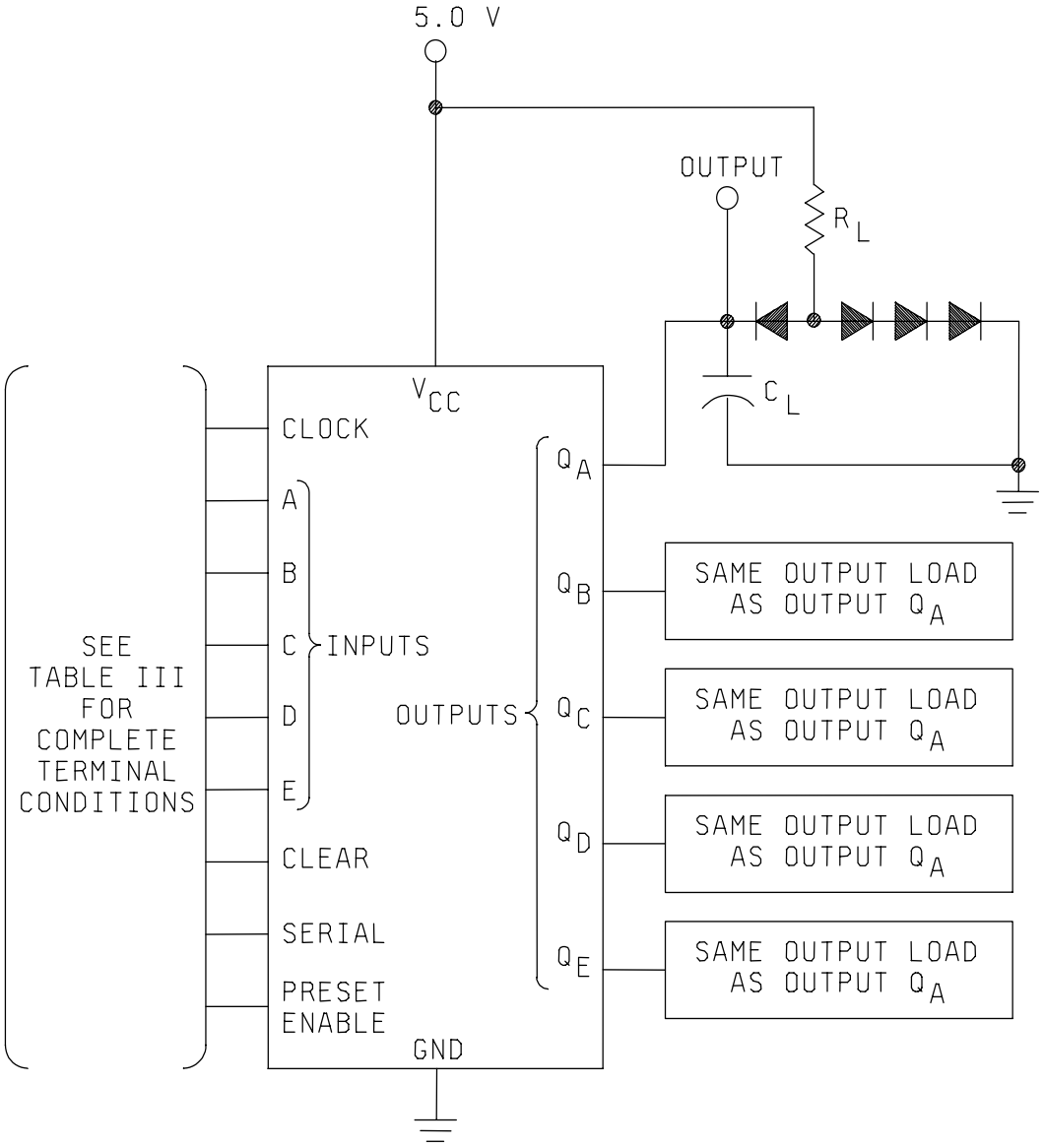


FIGURE 7. Switching test circuit and waveforms for device type 04.

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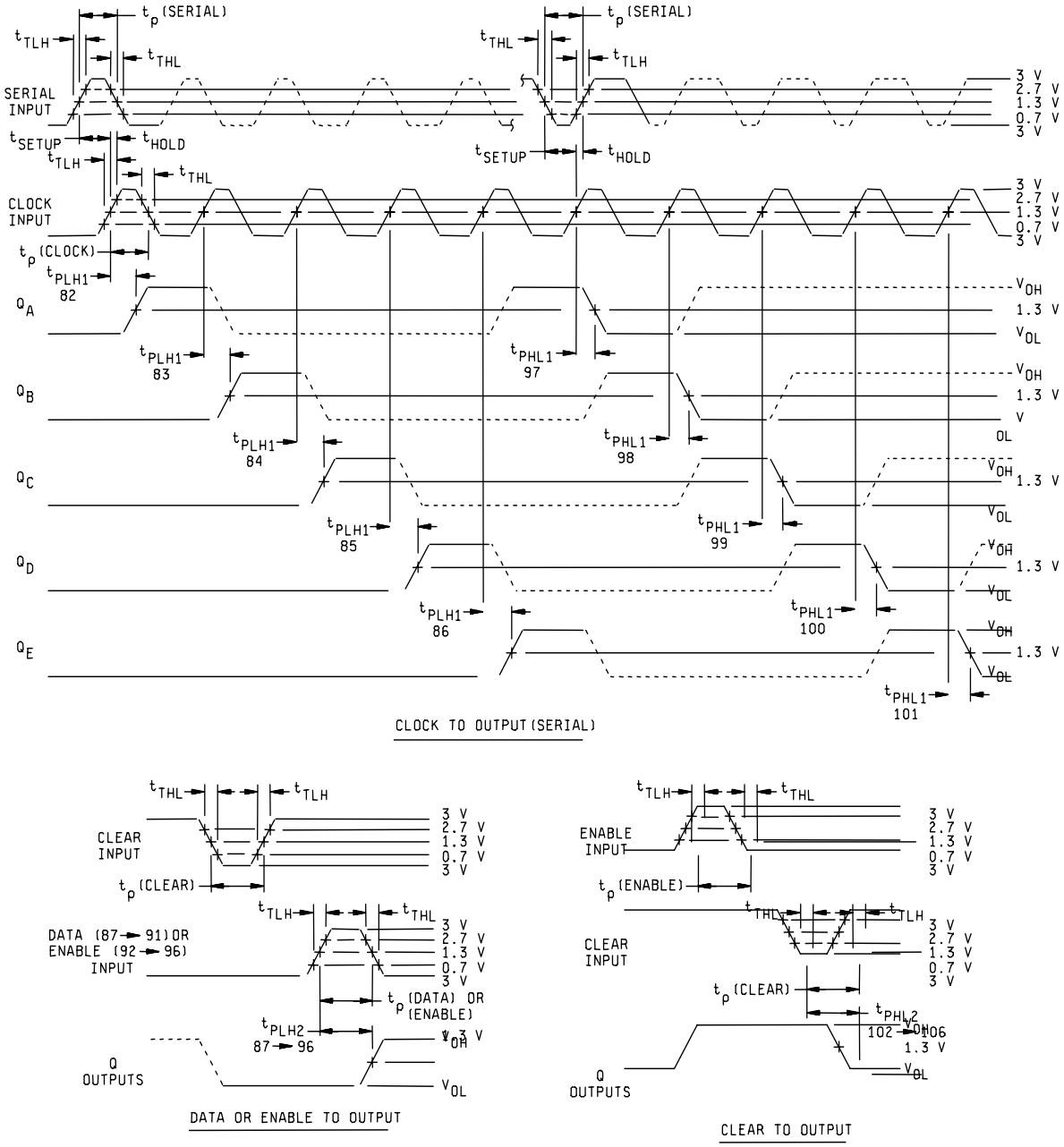


FIGURE 7. Switching test circuit and waveforms for device type 04 - Continued.

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NOTES:
1. Clock pulse characteristics: PRR ≤ 1.0 MHz, t_{TLH} ≤ 15 ns, t_{THL} ≤ 6 ns, t_p (clock) ≥ 25 ns.

2. Serial data pulse characteristics: t_{TLH} ≤ 15 ns, t_{THL} ≤ 6 ns, t_p = 30 ns.

3. Clear, data, and enable pulse characteristics: t_{TLH} ≤ 15 ns, t_{THL} ≤ 6 ns, t_p = 30 ns.

4. C_L = 50 pF ±10 percent including scope, probe, wiring and stray capacitance without package in test fixture.

4. R_L = 2.0 kΩ ±5%.

5. All diodes are 1N3064, 1N916 or equivalent.

6. Prior to initiating tests, the output shall be placed in the proper state.

FIGURE 7. Switching test circuit and waveforms for device type 04 - Continued.

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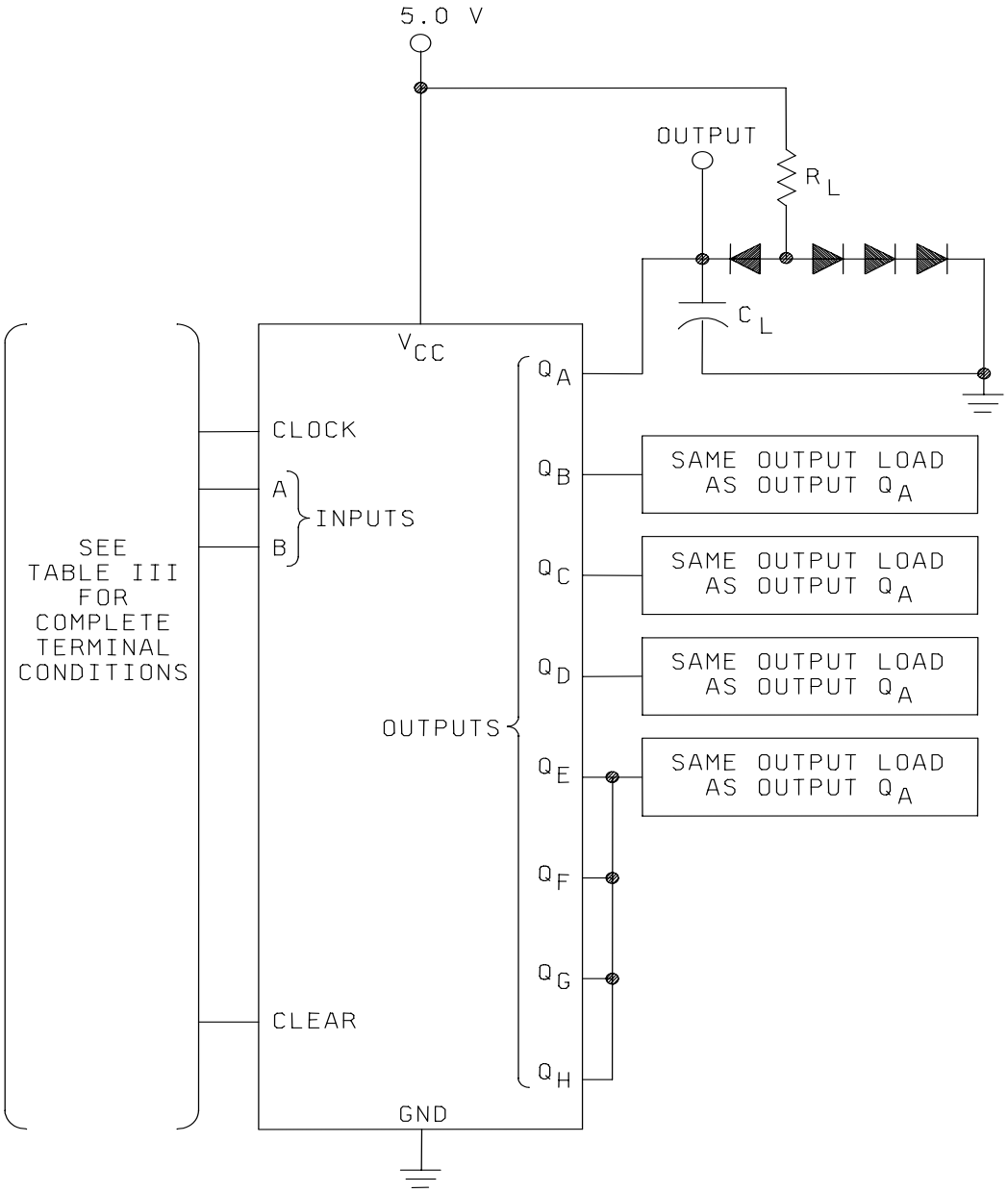


FIGURE 8. Switching test circuit and waveforms for device type 05.

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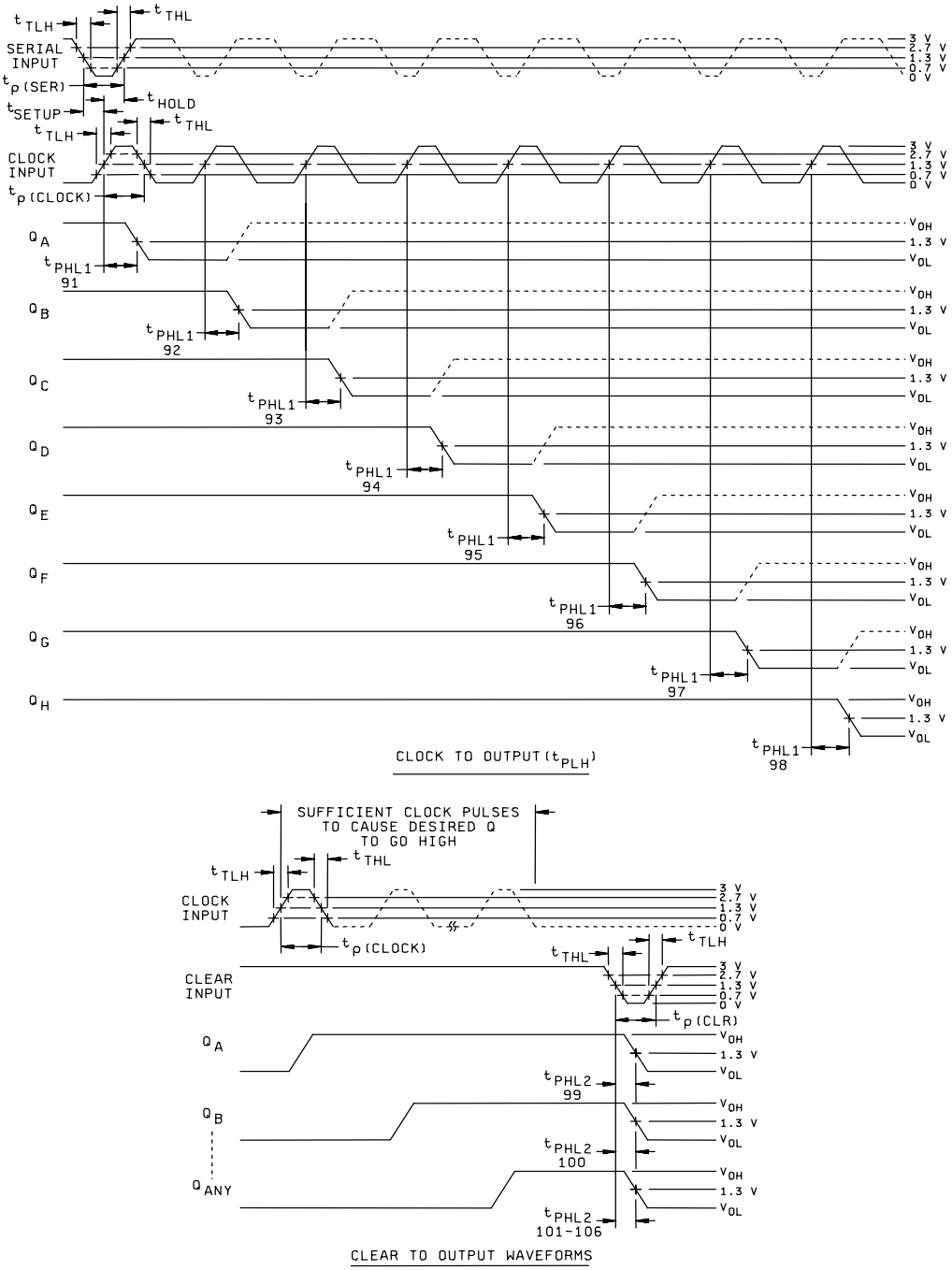
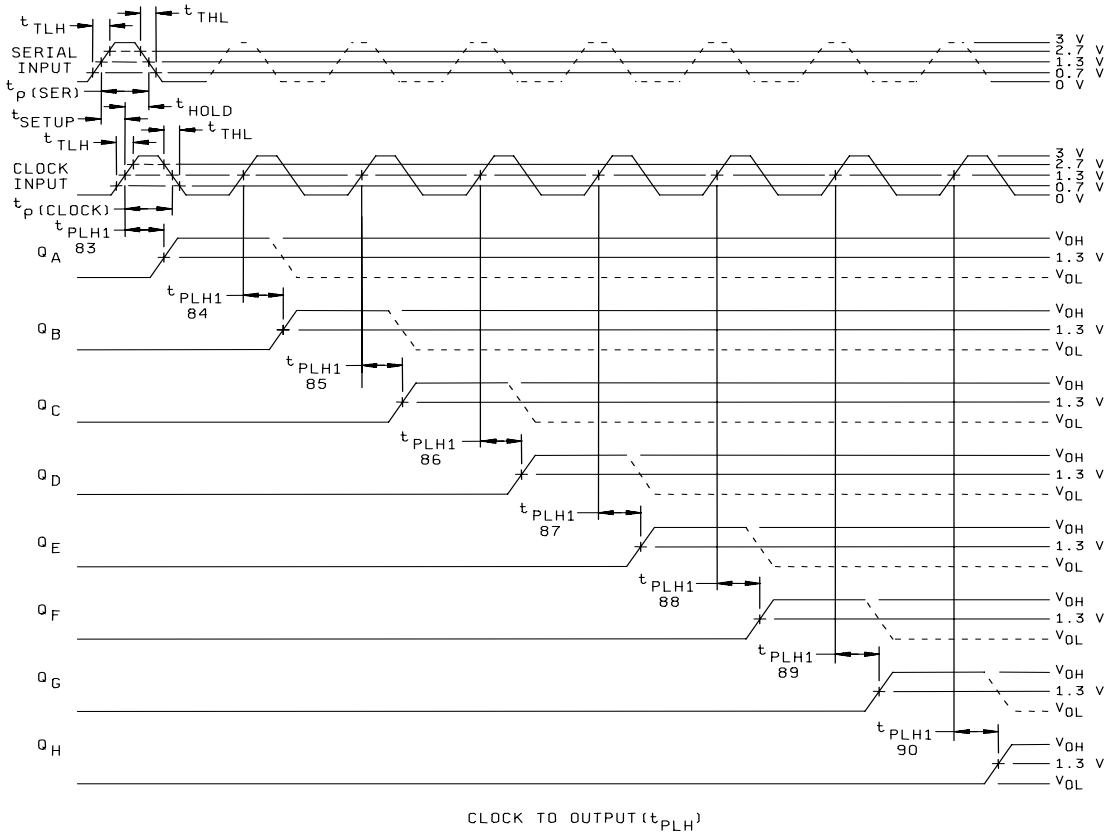


FIGURE 8. Switching test circuit and waveforms for device type 05 - Continued.

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NOTES:

1. Clock pulse characteristics: PRR ≤ 1.0 MHz, t_{TLH} ≤ 15 ns, t_{THL} ≤ 6 ns, t_p (clock) ≥ 20 ns.
2. Clear pulse characteristics: t_{TLH} ≤ 15 ns, t_{THL} ≤ 6 ns, t_p (clear) = 30 ns.
3. Serial pulse characteristics: t_{TLH} ≤ 15 ns, t_{THL} ≤ 6 ns, t_p (serial) = 30 ns, t_{SETUP} = 20 ns, t_{HOLD} = 10 ns.
4. C_L = 50 pF ±10 percent including scope, probe, wiring and stray capacitance without package in test fixture.
5. R_L = 2.0 kΩ ±5%.
6. All diodes are 1N3064, 1N916 or equivalent.
7. Prior to initiating tests, the output shall be placed in the proper state.

FIGURE 8. Switching test circuit and waveforms for device type 05 - Continued.

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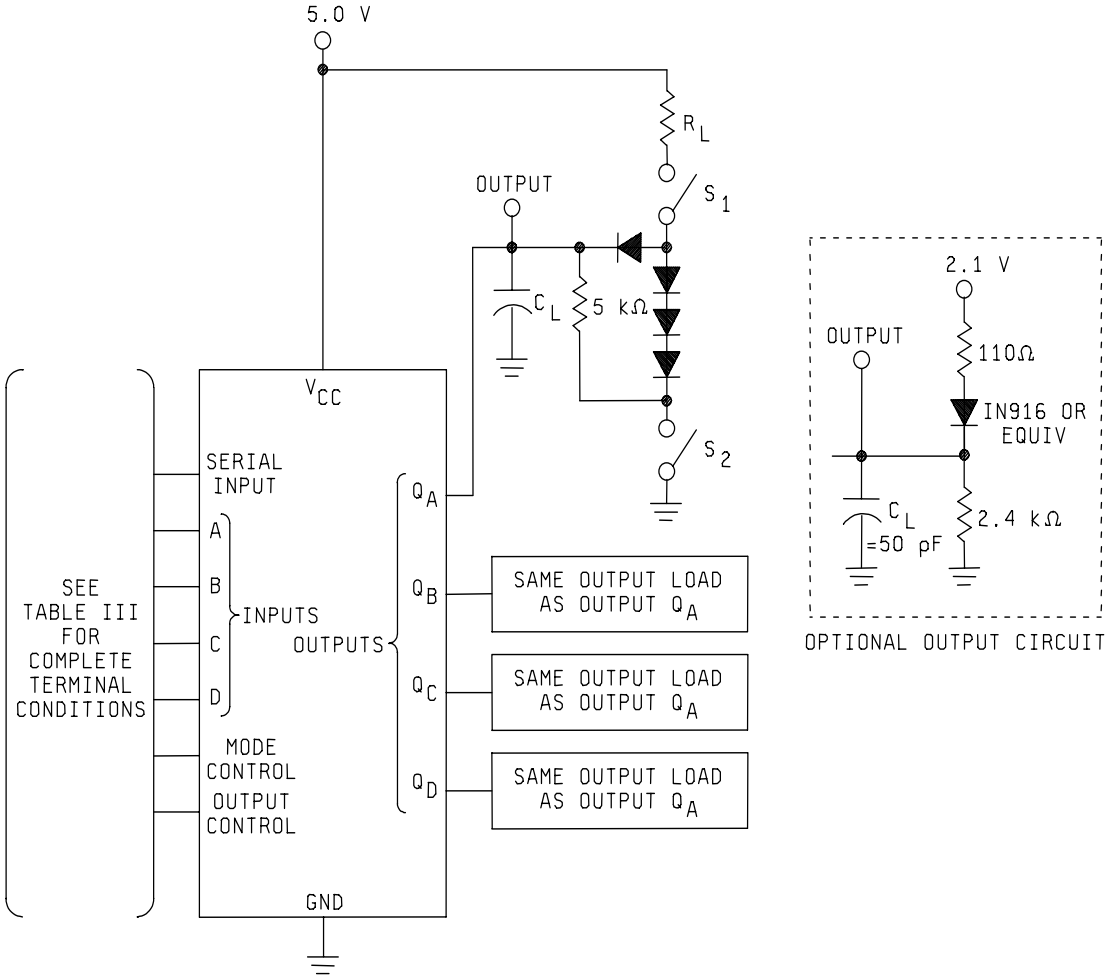


FIGURE 9. Switching test circuit and waveforms for device type 06.

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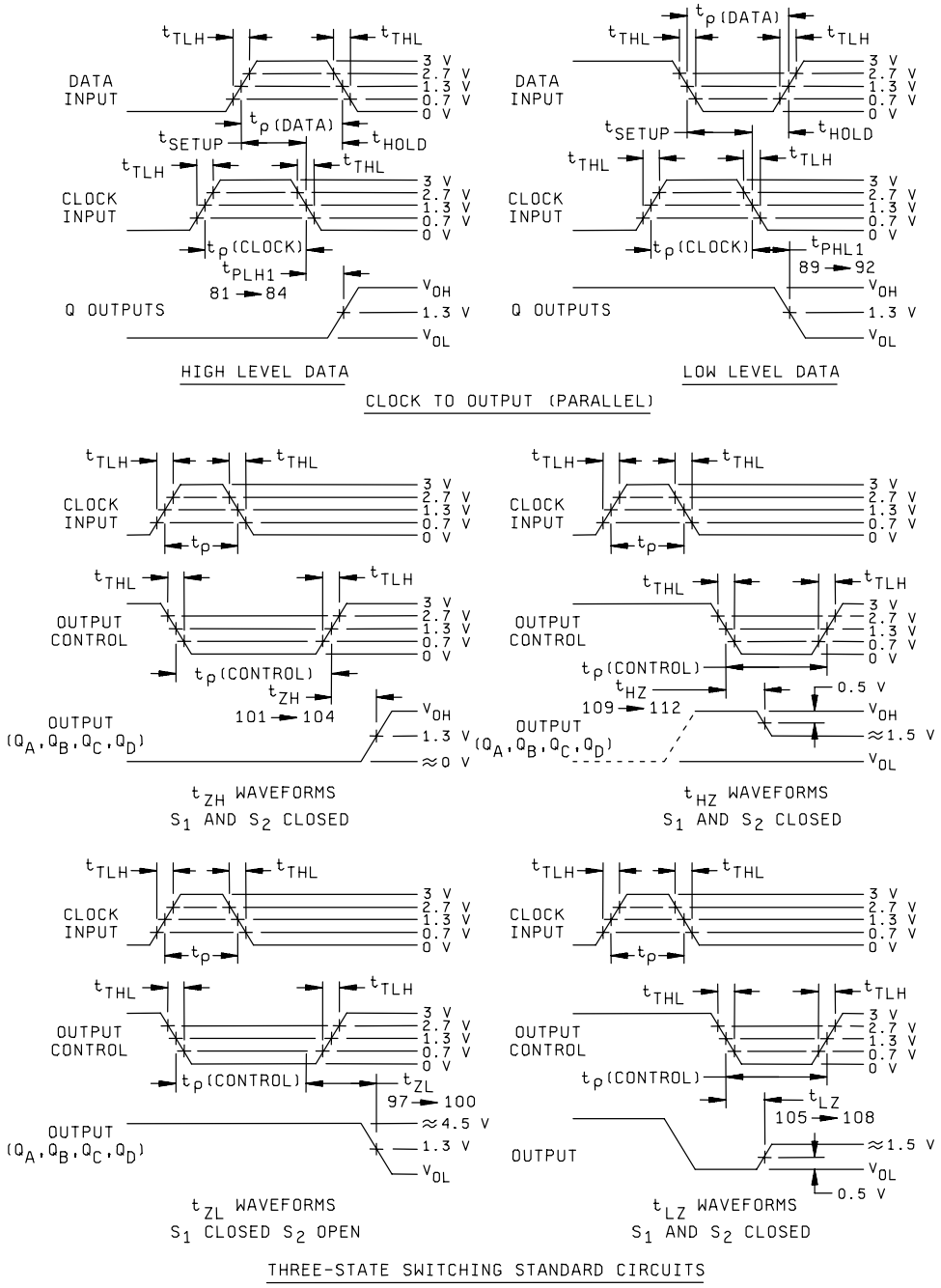
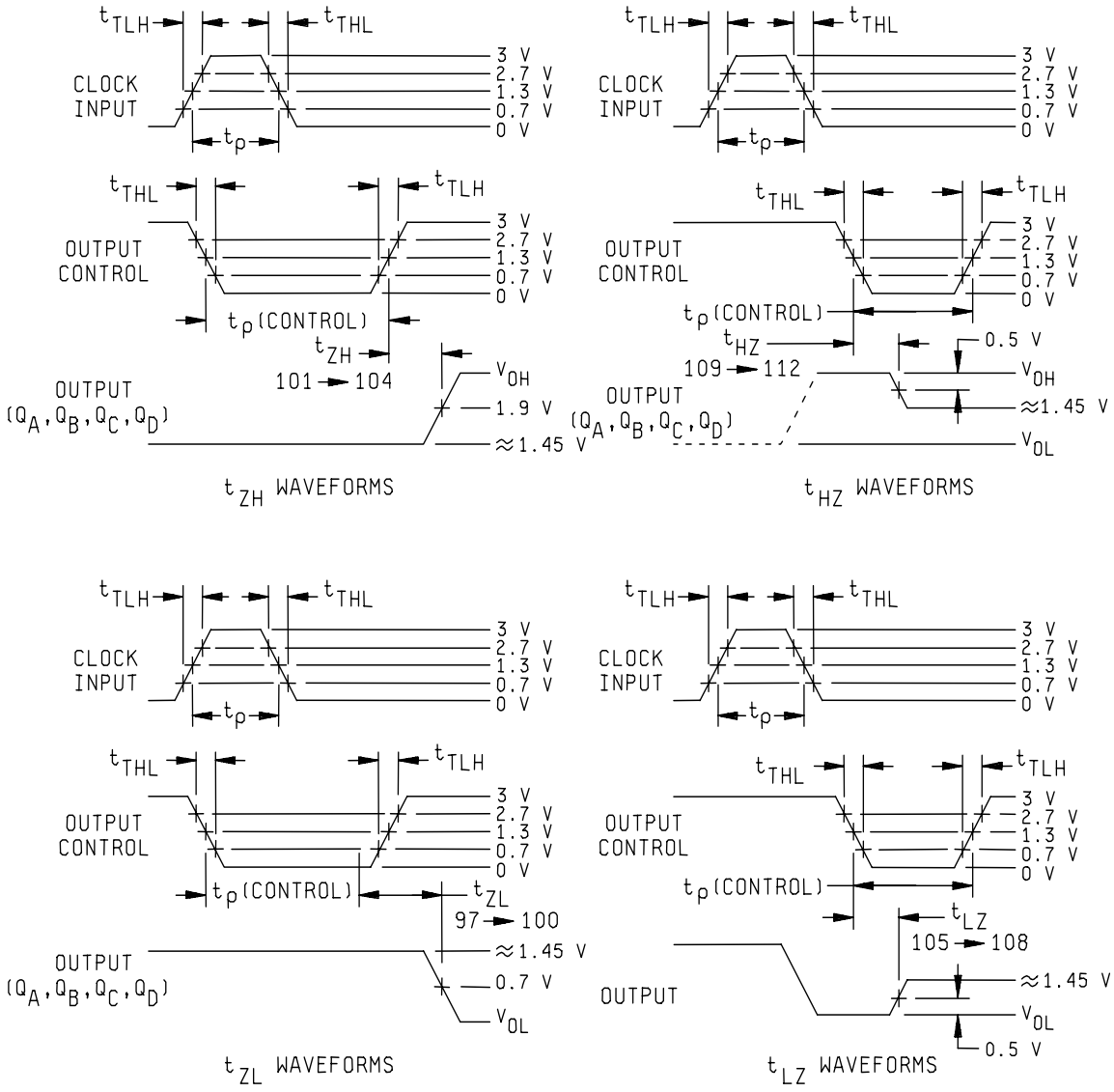


FIGURE 9. Switching test circuit and waveforms for device type 06 - Continued.

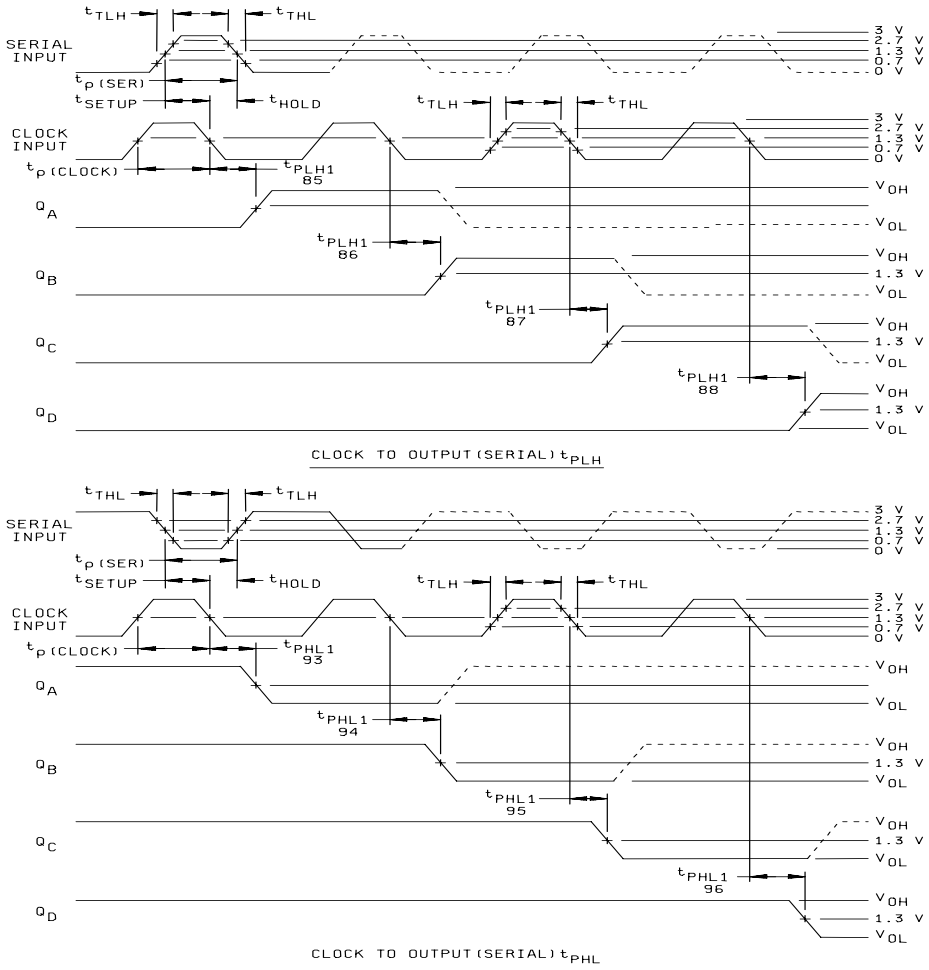
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THREE-STATE SWITCHING OPTIONAL CIRCUITS

FIGURE 9. Switching test circuit and waveforms for device type 06 - Continued.

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NOTES:
1. Clock pulse characteristics: PRR ≤ 1.0 MHz, t_{TLH} ≤ 15 ns, t_{THL} ≤ 6 ns, t_p (clock) ≥ 25 ns.
2. Data or serial pulse characteristics: t_{TLH} ≤ 15 ns, t_{THL} ≤ 6 ns, t_p (serial) or t_p (data) = 40 ns, t_{SETUP} = 20 ns, t_{HOLD} = 20 ns.
3. Output control characteristics: t_{TLH} ≤ 15 ns, t_{THL} ≤ 6 ns, t_p (control) ≥ 100 ns, except when optional load is used, C_L = 50 pF ± 10% for all tests.
4. C_L = 50 pF ±10% for propagation delay, t_{ZL}, t_{ZH}, and C_L = 15pF minimum for t_{HZ}, t_{LZ} except when optional load is used, C_L = pF ±10% for all tests.
   C_L includes scope probe, wiring, and stray capacitance without package in test fixture.
   All diodes are 1N3064, 1N916, or equivalent.
6. R_L = 680 Ω ± 5%.
7. Prior to initiating tests, the output shall be placed in the proper state.

FIGURE 9. Switching test circuit and waveforms for device type 06 - Continued.

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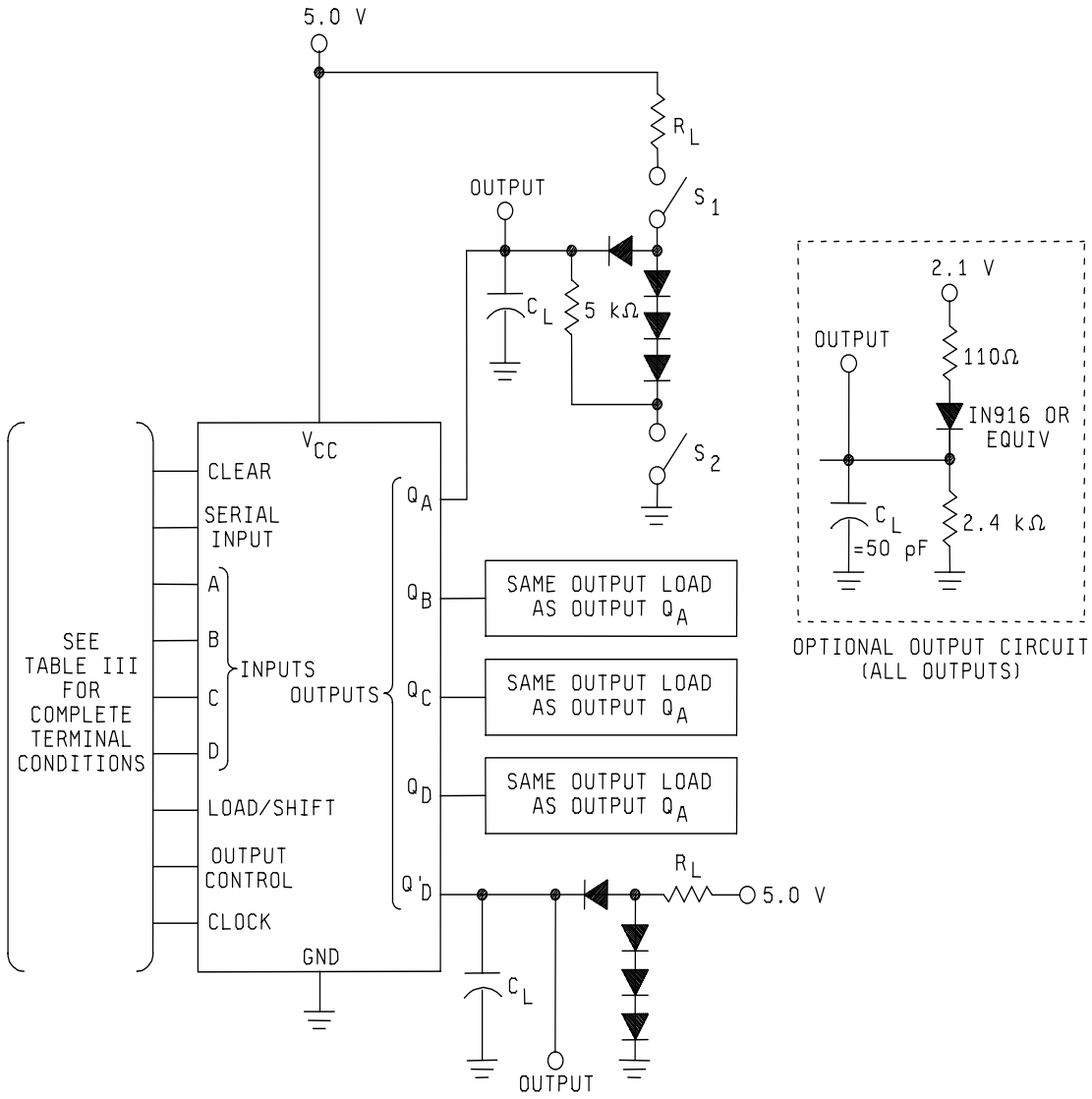


FIGURE 10. Switching test circuit and waveforms for device type 07.

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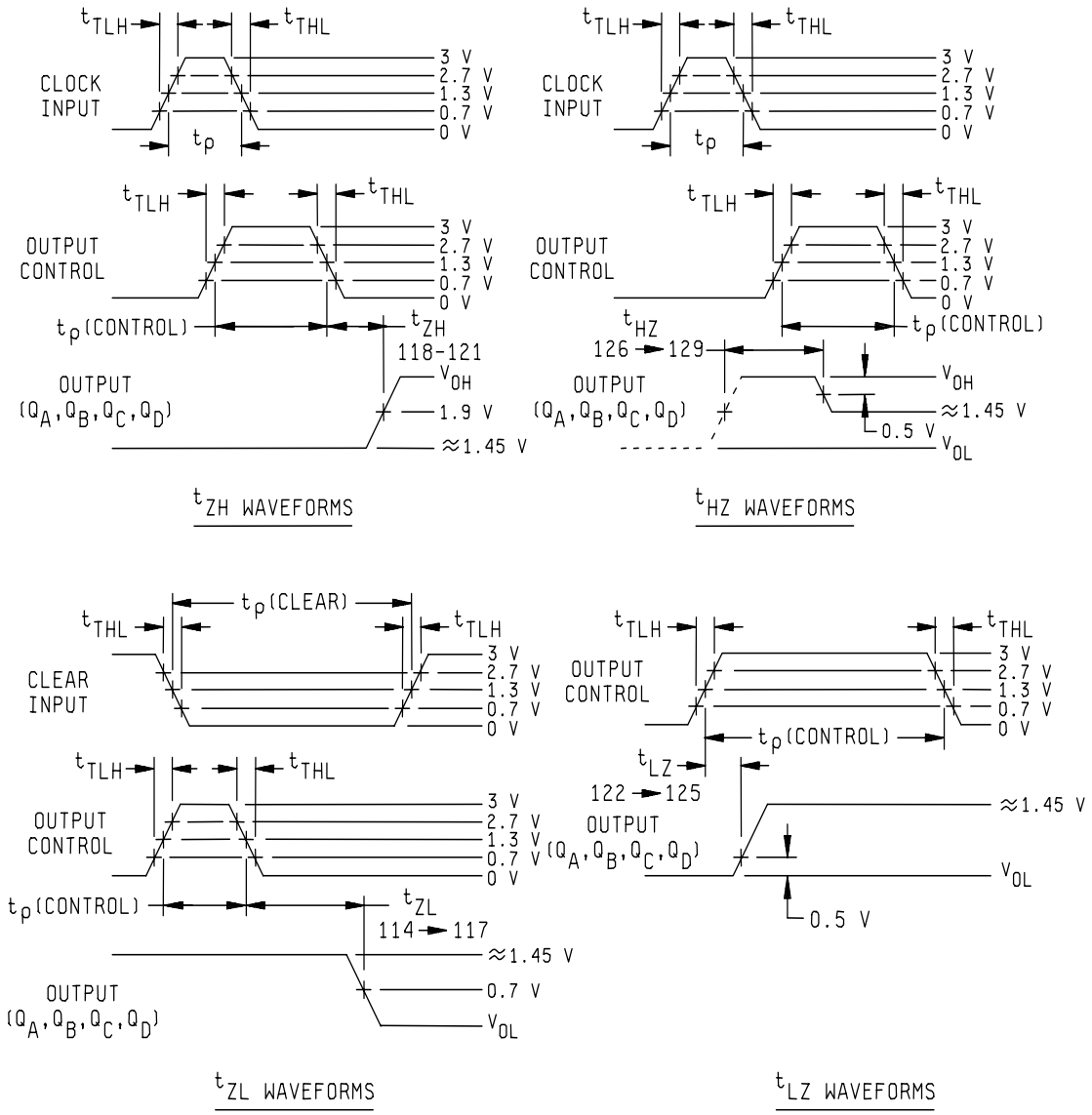


FIGURE 10. Switching test circuit and waveforms for device type 07 - Continued.

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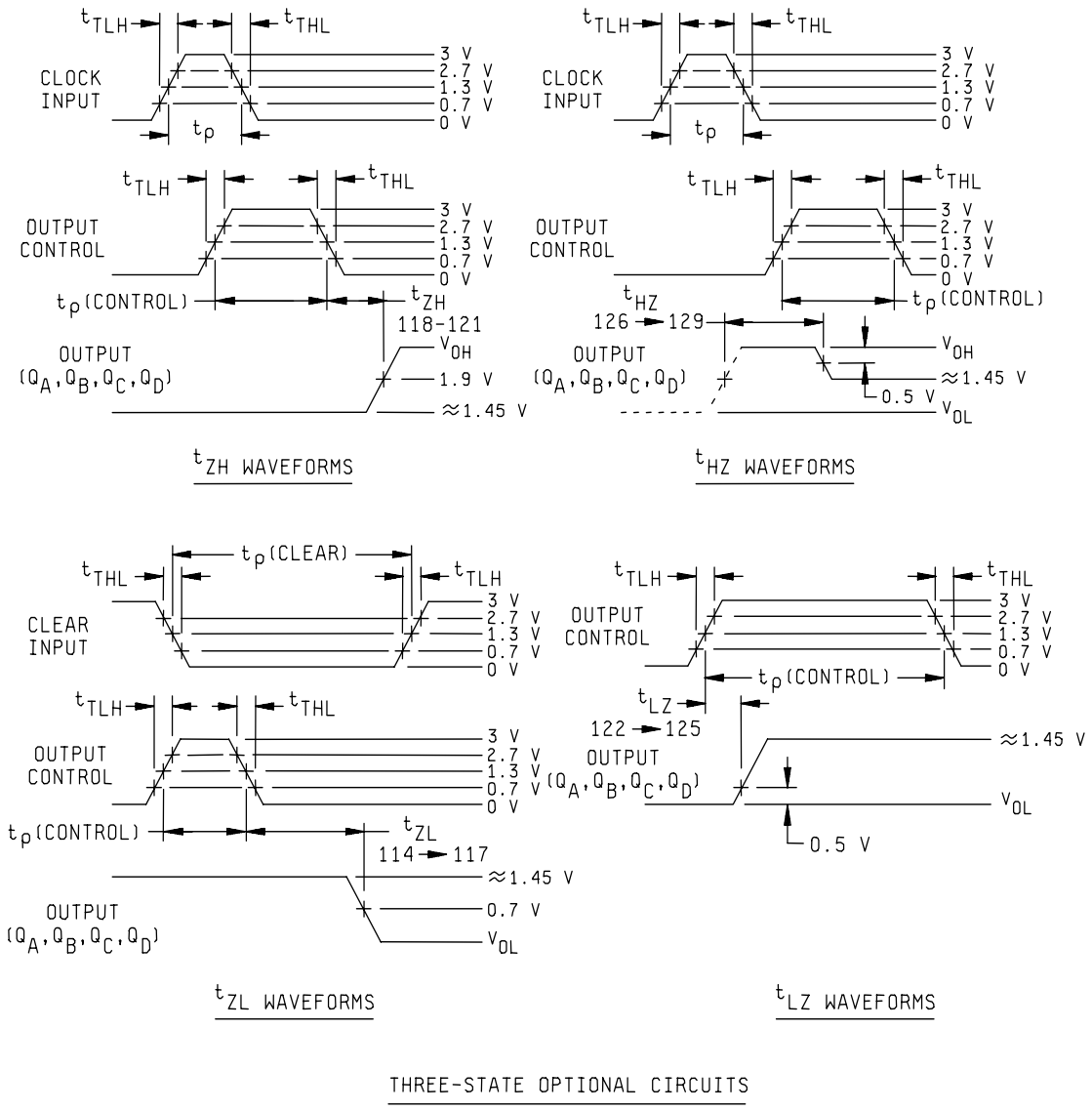
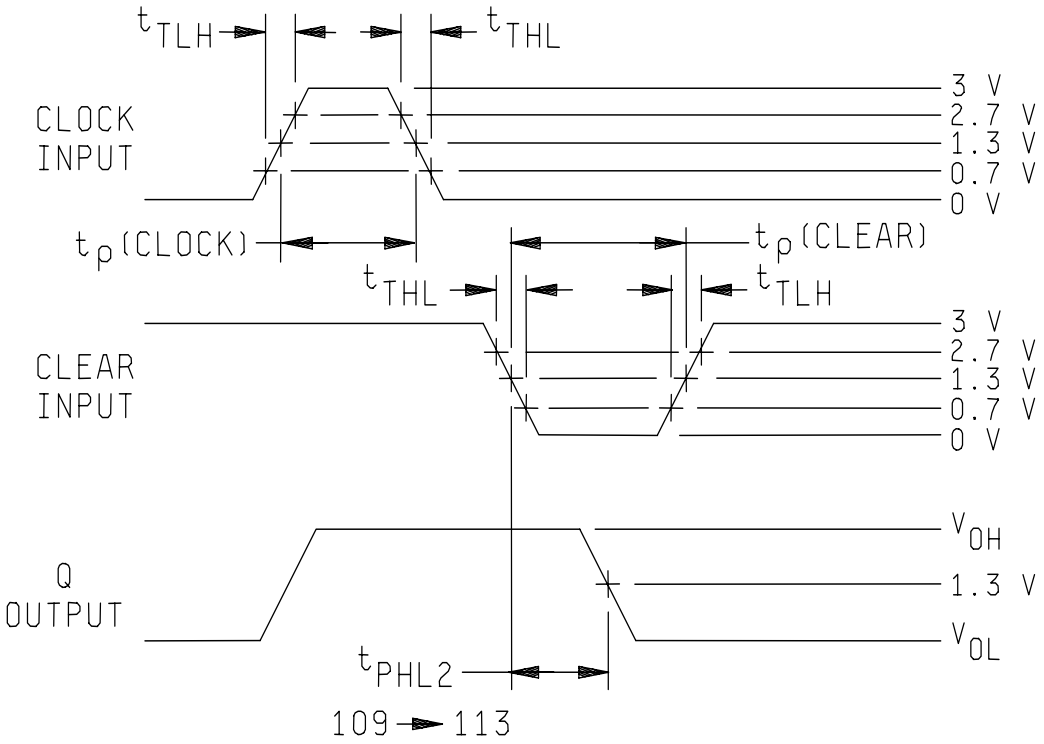


FIGURE 10. Switching test circuit and waveforms for device type 07 - Continued.

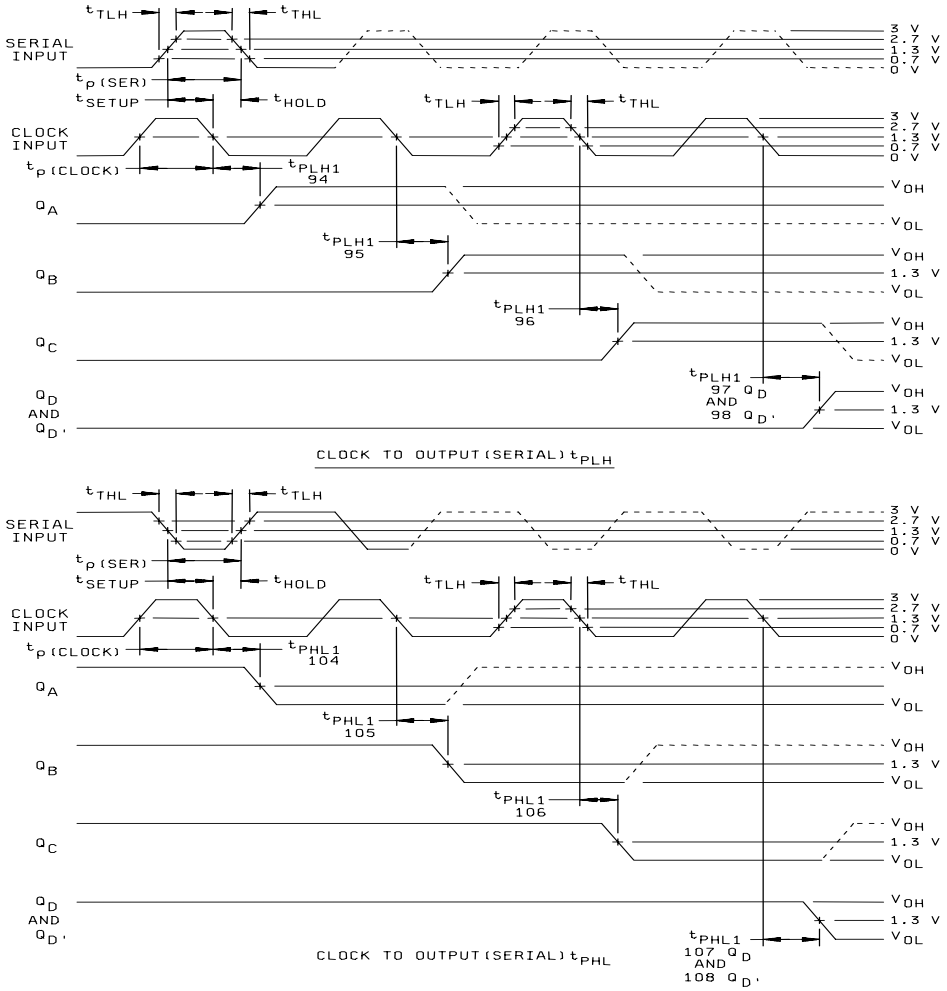
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CLEAR TO OUTPUT

FIGURE 10. Switching test circuit and waveforms for device type 07 - Continued.

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NOTES:
1. Clock pulse characteristics: PRR ≤ 1.0 MHz, t_{TLH} ≤ 15 ns, t_{THL} ≤ 6 ns, t_p (clock) ≥ 20 ns.
2. Data or serial pulse characteristics: t_{TLH} ≤ 15 ns, t_{THL} ≤ 6 ns, t_p (serial) or t_p (data) = 30 ns, t_{SETUP} = 20 ns, t_{HOLD} = 10 ns.
3. Clear pulse characteristics: t_{TLH} ≤ 15 ns, t_{THL} ≤ 6 ns, t_p (clear) = 25 ns, except ≥ 200 ns for t_{ZL} test.
4. Output control pulse characteristics: t_{TLH} ≤ 15 ns, t_{THL} ≤ 6 ns, t_p (control) ≥ 100 ns.
5. C_L = 50 pF ±10% for propagation delay, t_{ZH}, t_{ZL} test, and C_L = 15 pF minimum (all except Q_D,) for t_{HZ}, t_{LZ} tests except when optional load is used, C_L = 50 pF ±10% for all tests. C_L includes scope probe, wiring, and stray capacitance without package in test fixture.
6. All diodes are 1N3064, 1N916, or equivalent.
7. R_L = 680 Ω ± 5% except for $Q_{D'}$, R_L = 2 kΩ ±5%.
8. Prior to initiating tests, the output shall be placed in the proper state.

FIGURE 10. Switching test circuit and waveforms for device type 07 - Continued.

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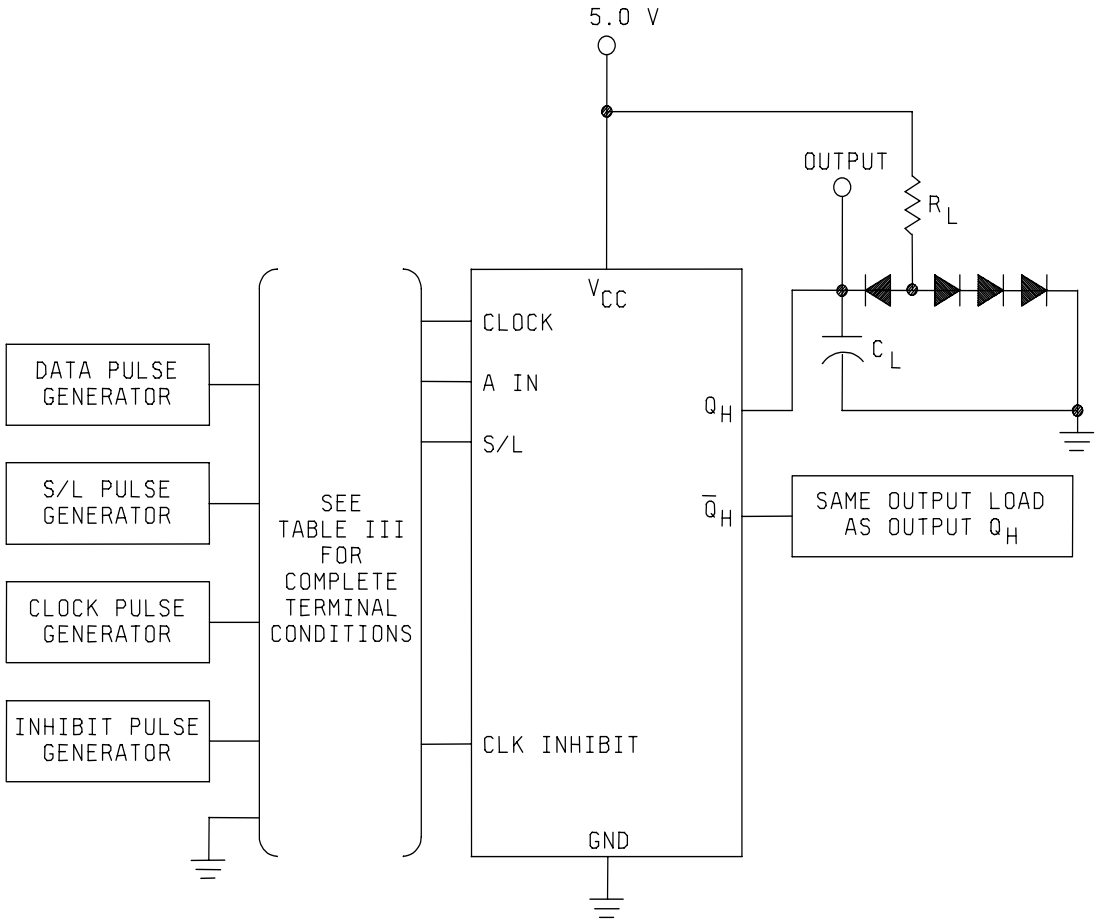


FIGURE 11. Switching test circuit and waveforms for device type 08.

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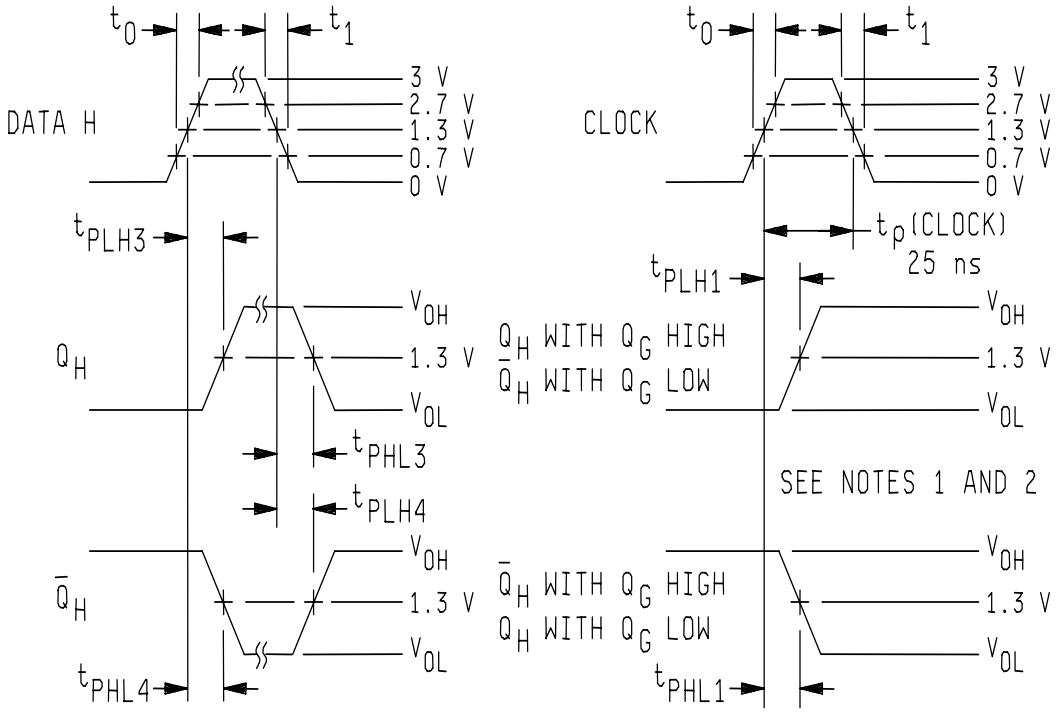
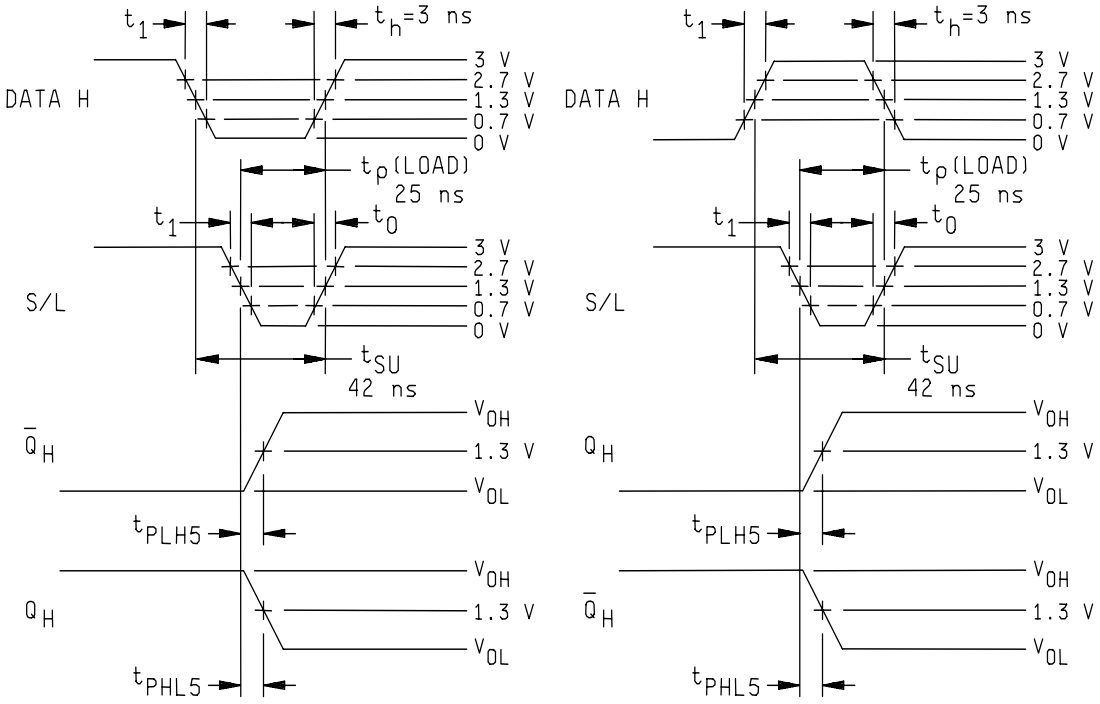


FIGURE 11. Switching test circuit and waveforms for device type 08 - Continued.

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NOTES:
1. For t_{PHL2} measurements, internal output G must be set to a low and Q_H to a high prior to tests.
2. For t_{PHL2} measurements, internal output G must be set to a high and Q_H to a low prior to test.
3. R_L = 2.0 kΩ ±5%.
4. C_L = 50 pF ±10%, which includes probe, and jug capacitance.
5. All pulse generators have the following characteristics: $Z_{OUT} \approx$ 50Ω, $t_0 \leq$ 15 ns, $t_1 \leq$ 6 ns and PRR ≤ 1 MHz.
6. All diodes 1N3064 or equivalent.

FIGURE 11. Switching test circuit and waveforms for device type 08 - Continued.

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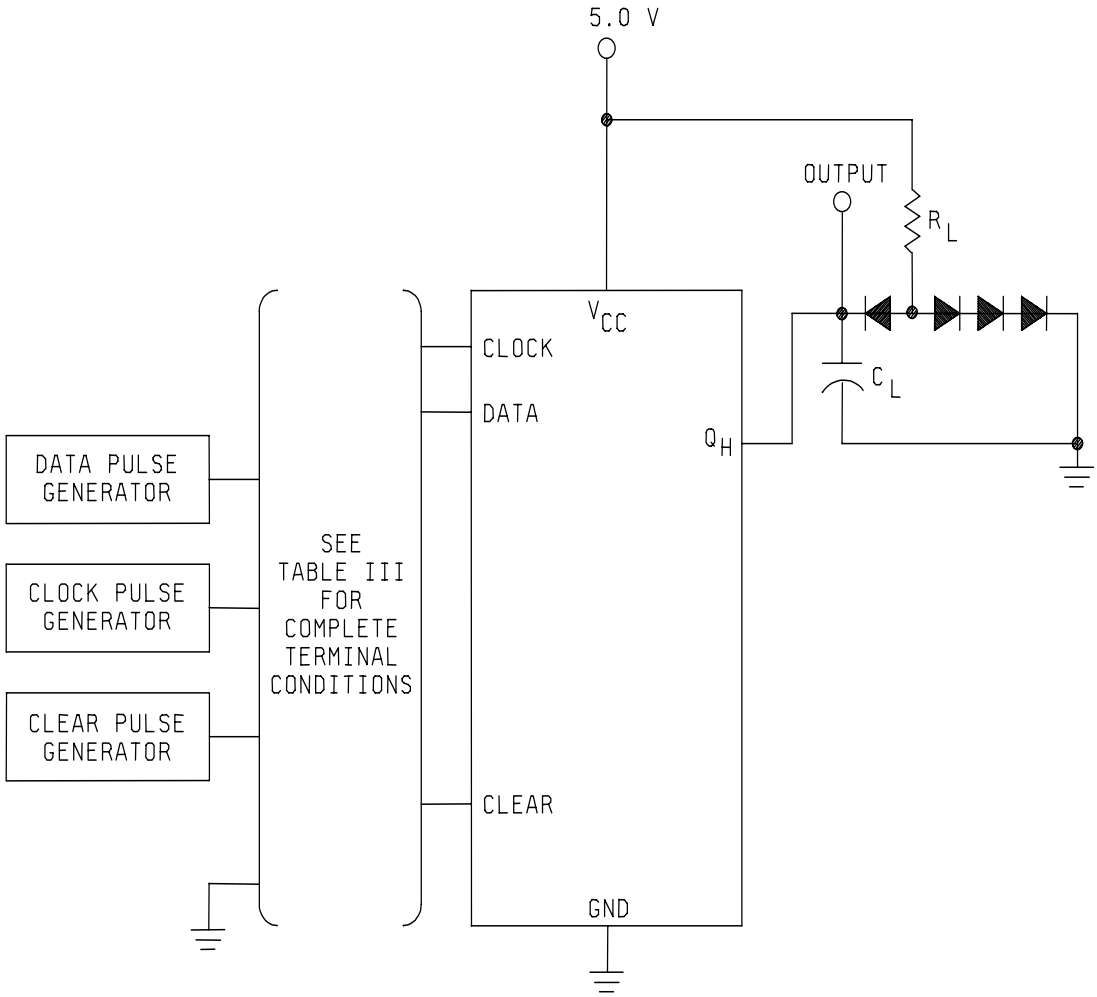
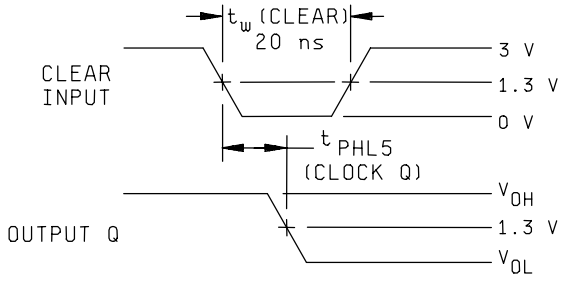
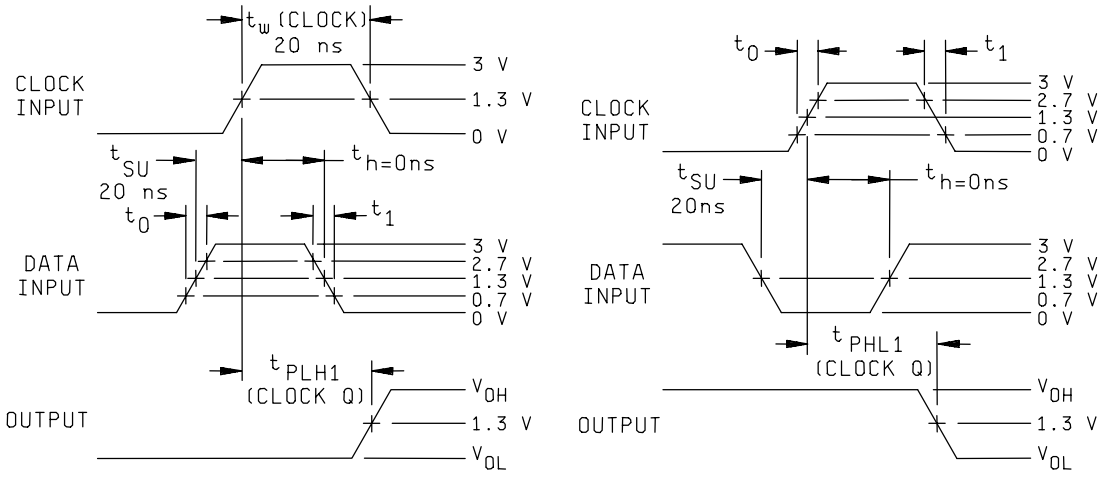


FIGURE 12. Switching test circuit and waveforms for device type 09.

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CLEAR TO OUTPUT WAVEFORM



CLOCK TO OUTPUT WAVEFORMS

NOTES:
1. R_L = 2.0 kΩ ±5%.
2. C_L = 50 pF ±10%, which includes probe, and jug capacitance.
3. All pulse generators have the following characteristics: $Z_{OUT} \approx$ 50Ω, $t_0 \leq$ 15 ns, $t_1 \leq$ 6 ns and PRR ≤ 1 MHz.
4. All diodes 1N3064 or equivalent.

FIGURE 12. Switching test circuit and waveforms for device type 09 - Continued.

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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be high ≥ 2.0; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test Limits		Unit
			Cases 2,X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	CLR	S/R Serial	A_{IN}	B_{IN}	C_{IN}	D_{IN}	S/L Serial	GND	S_0	S_1	CLK	QD	QC	QB	QA	V_{CC}				
1 Tc = 25°C	V_{OH}	3006	1	A	GND	2.0 V	GND	GND	GND	GND	GND	4.5 V	4.5 V	A 1/				-.4 mA	4.5 V	QA	2.5		V
			2	"	"	GND	2.0 V	GND	GND	"	"	"	"	"			-.4 mA		"	QB	"		"
			3	"	"	"	GND	2.0 V	GND	"	"	"	"	"		-.4 mA			"	QC	"		"
			4	"	"	"	"	GND	2.0 V	"	"	"	"	"	-.4 mA				"	QD	"		"
			5	"	2.0 V	"	"	GND	GND	"	"	"	0.7 V	"				-.4 mA	"	QA	"		"
			6	"	GND	"	"	GND	GND	2.0 V	"	0.7 V	4.5 V	"	-.4 mA				"	QD	"		"
			7	"	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	"	4.5 V	GND	"				-.4 mA	"	QA	"		"
	V_{OL}	3007	8	"	4.5 V	0.7 V	4.5 V	4.5 V	4.5 V	4.5 V	"	"	4.5 V	"				4 mA	"	QA		0.4	"
			9	"	"	4.5 V	0.7 V	4.5 V	4.5 V	"	"	"	"	"			4 mA		"	QB		"	"
			10	"	"	"	4.5 V	0.7 V	4.5 V	"	"	"	"	"		4mA			"	QC		"	"
			11	"	"	"	"	4.5 V	0.7 V	"	"	"	"	"	4 mA				"	QD		"	"
			12	"	0.7 V	"	"	4.5 V	4.5 V	"	"	"	0.7 V	"				4 mA	"	QA		"	"
			13	"	4.5 V	"	"	4.5 V	4.5 V	0.7 V	"	0.7 V	4.5 V	"	4 mA				"	QD		"	"
	V_{IC}		14	-18 mA							"								"	CLR		-1.5	"
			15		-18 mA						"								"	S/R		"	"
			16			-18 mA					"								"	A_{IN}		"	"
			17				-18 mA				"								"	B_{IN}		"	"
			18					-18 mA			"								"	C_{IN}		"	"
			19						-18 mA		"								"	D_{IN}		"	"
			20							-18 mA	"								"	S/L		"	"
			21								"	-18 mA							"	S_0		"	"
			22								"		-18 mA						"	S_1		"	"
			23								"			-18 mA					"	CLK		"	"
	I_{IH1}	3010	24	2.7 V							"								5.5 V	CLR		20	µA
			25		2.7 V						"	GND	4.5 V						"	S/R		"	"
			26			2.7 V					"	"	GND						"	A_{IN}		"	"
			27				2.7 V				"	"	"						"	B_{IN}		"	"
			28					2.7 V			"	"	"						"	C_{IN}		"	"
			29						2.7 V		"	"	"						"	D_{IN}		"	"
			30							2.7 V	"	4.5 V	"						"	S/L		"	"
			31								"	2.7 V							"	S_0		"	"
			32								"		2.7 V						"	S_1		"	"
			33	GND							"			2.7 V					"	CLK		"	"
	I_{IH2}		34	5.5 V							"		5.5 V						"	CLR		100	"
			35		5.5 V						"	GND	4.5 V						"	S/R		"	"
			36			5.5 V					"	GND	GND						"	A_{IN}		"	"
			37				5.5 V				"	"	"						"	B_{IN}		"	"
			38					5.5 V			"	"	"						"	C_{IN}		"	"
			39						5.5 V		"	:"	"						"	D_{IN}		"	"
			40							5.5 V	"	4.5 V	"						"	S/L		"	"
			41								"	5.5 V	"						"	S_0		"	"
			42								"		5.5 V						"	S_1		"	"
			43	GND							"			5.5 V					"	CLK		"	"

See footnotes at end of device types 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test Limits		Unit
			Cases 2, X	* 2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	CLR	S/R Serial	A_{IN}	B_{IN}	C_{IN}	D_{IN}	S/L Serial	GND	S_0	S_1	CLK	QD	QC	QB	QA	V_{CC}		Min	Max	
1	I_{IL1}	3009	44	0.4 V							GND								5.5 V	CLR	2/	2/	mA
Tc = 25°C	I_{IL2}		45		0.4 V						"	4.5 V	GND						"	S/R	"	"	"
			46			0.4 V					"	"	4.5 V						"	A_{IN}	"	"	"
			47				0.4 V				"	"	"						"	B_{IN}	"	"	"
			48					0.4 V			"	"	"						"	C_{IN}	"	"	"
			49						0.4 V		"	"	"						"	D_{IN}	"	"	"
			50							0.4 V	"	GND	"						"	S/L	"	"	"
	I_{IL3}		51								"	0.4 V							"	S_0	"	"	"
	I_{IL3}		52								"		0.4 V						"	S_1	"	"	"
	I_{IL4}		53	4.5 V							"			0.4 V					"	CLK	"	"	"
	I_{0S}	3011	54	A		4.5 V	GND	GND	GND		"	4.5 V	4.5 V	A 1/				GND	"	QA	-15	-100	"
			55	"		GND	4.5 V	GND	GND		"	"	"	"			GND		"	QB	"	"	"
			56	"		"	GND	4.5 V	GND		"	"	"	"		GND			"	QC	"	"	"
			57	"		"	GND	GND	4.5 V		"	"	"	"	GND				"	QD	"	"	"
	I_{CC}	3005	58	5.5 V	5.5 V	"	GND	GND	GND	5.5 V	"	5.5 V	5.5 V	"					"	V_{CC}		23	"
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = 125°C, and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C, and V_{IC} tests are omitted.																						

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured	Test Limits		Unit
			Cases 2,X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	CLR	S/R Serial	A_{IN}	B_{IN}	C_{IN}	D_{IN}	S/L Serial	GND	S_0	S_1	CLK	QD	QC	QB	QA	V_{CC}	terminal	Min	Max	
7	Truth	3014	59	C	B	B	B	B	B	B	GND	C	C	C	L	L	L	L	5.0V	All			
T_C = 25°C	table		60	"	B	"	"	"	"	B	"	C	C	B	"	"	"	"	"	outputs			
	test		61	"	B	"	"	"	"	B	"	C	C	C	"	"	"	"	"	"			
			62	"	C	"	"	"	"	C	"	B	B	C	"	"	"	"	"	"			
			63	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"			
			64	"	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"			
			65	B	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"			
			66	"	"	"	"	"	"	"	"	"	"	B	H	H	H	H	"	"			
			67	"	"	"	"	"	"	"	"	"	"	C	H	H	H	H	"	"			
			68	"	B	C	C	C	C	B	"	"	"	C	H	H	H	H	"	"			
			69	"	"	"	"	"	"	"	"	"	"	B	L	L	L	L	"	"			
			70	"	"	"	"	"	"	"	"	"	"	C	"	"	"	L	"	"			
			71	"	"	"	"	"	"	"	"	"	C	C	"	"	"	L	"	"			
			72	"	"	"	"	"	"	"	"	"	"	B	"	"	"	H	"	"			
			73	"	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"			
			74	"	"	"	"	"	"	"	"	"	"	B	"	"	H	"	"	"			
			75	"	"	"	"	"	"	"	"	"	"	C	"	"	H	"	"	"			
			76	"	"	"	"	"	"	"	"	"	"	B	"	H	H	"	"	"			
			77	"	"	"	"	"	"	"	"	"	"	C	"	H	"	"	"	"			
			78	"	"	"	"	"	"	"	"	"	"	B	H	"	"	"	"	"			
			79	"	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"			
			80	"	C	B	B	B	B	C	"	"	"	C	"	"	"	"	"	"			
			81	"	"	"	"	"	"	"	"	"	"	B	"	"	"	L	"	"			
			82	"	"	"	"	"	"	"	"	"	"	C	"	"	"		"	"			
			83	"	"	"	"	"	"	"	"	"	"	B	"	"	L		"	"			
			84	"	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"			
			85	"	"	"	"	"	"	"	"	"	"	B	"	L	"	"	"	"			
			86	"	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"			
			87	"	"	"	"	"	"	"	"	"	"	B	L	"	"	"	"	"			
			88	"	"	"	"	"	"	"	"	"	"	C	L	"	"	"	"	"			
			89	"	B	C	C	C	C	B	"	C	B	C	L	"	"	"	"	"			
			90	"	"	"	"	"	"	"	"	"	"	B	H	"	"	"	"	"			
			91	"	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"			
			92	"	"	"	"	"	"	"	"	"	"	B	"	H	"	"	"	"			
			93	"	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"			
			94	"	"	"	"	"	"	"	"	"	"	B	"	"	H	"	"	"			
			95	"	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"			
			96	"	"	"	"	"	"	"	"	"	"	B	"	"	"	H	"	"			
			97	"	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"			
			97A	"	"	"	"	"	"	"	"	"	C	C	"	"	"	"	"	"			
			97B	"	"	"	"	"	"	"	"	"	C	B	"	"	"	"	"	"			
			97C	"	"	"	"	"	"	"	"	"	C	C	"	"	"	"	"	"			
			98	"	C	B	B	B	B	C	"	"	B	C	"	"	"	"	"	"			
			99	"	"	"	"	"	"	"	"	"	"	B	L	"	"	"	"	"			
			100	"	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"			
			101	"	"	"	"	"	"	"	"	"	"	B	"	L	"	"	"	"			
			102	"	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"			
			103	"	"	"	"	"	"	"	"	"	"	B	"	"	L	"	"	"			
			104	"	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"			
			105	"	"	"	"	"	"	"	"	"	"	B	"	"	"	L	"	"			
			106	"	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"			
			107	"	"	"	"	"	"	"	"	"	C	C	"	"	"	"	"	"			
			108	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"			
			109	"	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"			
			110	"	"	C	C	C	C	"	"	"	"	C	"	"	"	"	"	"			
			111	"	"	C	C	C	C	"	"	"	"	B	"	"	"	"	"	"			

See B, C, D, and E

See footnotes at end of device type 01.

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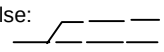
TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Cases 2, X Test no.	1 2 CLR	2 3 S/R Serial	3 4 A_{IN}	4 5 B_{IN}	5 7 C_{IN}	6 8 D_{IN}	7 9 S/L Serial	8 10 GND	9 12 S_0	10 13 S_1	11 14 CLK	12 15 QD	13 17 QC	14 18 QB	15 19 QA	16 20 V_{CC}	Measured terminal	Test Limits Min	 Max	Unit
8	Same tests, terminal conditions, and limits as subgroup 7, except T_C = 125°C and -55°C																						
9 T_C = 25°C	f_{MAX} See F and J	(Fig. 4)	112	G		IN					GND	G	G	IN				OUT	5.0 V	CLK to QA	22		MHz
	t_{PHL1}	3003 (Fig. 4)	113	"		IN					"	"	"	"				OUT	"	CLK TO QA	5	27	ns
			114	"			IN				"	"	"	"			OUT		"	CLK TO QB	"	"	"
			115	"				IN			"	"	"	"		OUT			"	CLK TO QC	"	"	"
			116	"					IN		"	"	"	"	OUT				"	CLK TO QD	"	"	"
			117	"	IN						"	"	GND	"				OUT	"	CLK TO QA	"	"	"
			118	"						IN	"	GND	G	"	OUT				"	CLK TO QD	"	"	"
	t_{PHL1}		119	"		IN					"	G	"	"				OUT	"	CLK TO QA	"	31	"
			120	"			IN				"	"	"	"			OUT		"	CLK TO QB	"	"	"
			121	"				IN			"	"	"	"		OUT			"	CLK TO QC	"	"	"
			122	"					IN		"	"	"	"	OUT				"	CLK TO QD	"	"	"
			123	"	IN						"	"	GND	"				OUT	"	CLK TO QA	"	"	"
			124	"						IN	"	GND	G	"	OUT				"	CLK TO QD	"	"	"
	t_{PHL2}		125	IN		G					"	G	"	"				OUT	"	CLK TO QA	"	35	"
			126	"			G				"	"	"	"			OUT		"	CLK TO QB	"	"	"
			127	"				G			"	"	"	"		OUT			"	CLK TO QC	"	"	"
			128	"					G		"	"	"	"	OUT				"	CLK TO QD	"	"	"
10 T_C = 25°C	f_{MAX} See F and J	3003 (Fig. 4)	129	Same tests and terminal conditions as for subgroup 9																	20		MHz
	t_{PHL1}		130 to 135																		5	41	ns
	t_{PHL1}		136 to 141																		5	47	ns
	t_{PHL2}		142 to 145																		5	53	ns
11	Same tests, terminal conditions, and limits as subgroups 10, except T_C = -55°C.																						

See footnotes at end of device type 01.

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FOOTNOTES:

A. Apply input pulse:  2.5 V minimum/5.5 V maximum
0 V

B. V_{IN} = 2.5 V.

C. V_{IN} = 0.4 V.

D. Test numbers 59 through 111 shall be run in sequence.

E. Output voltages shall be either: (1) H ≥2.5 V minimum and L ≤0.4 V maximum when using a high speed checker double comparator: (2) H ≥1.5 V and L ≤1.5 V when using a high speed checker single comparator.

F. f_{MAX} minimum limit specified is the frequency of the clock input pulse. The output frequency shall be one-half of the clock input frequency. The input frequency on the A_{IN} data shall be one-half of the clock input frequency and the A_{IN} shall be shifted such that the A_{IN} ↑ and ↓ are coincident with the clock ↓. Rise and fall times ≤ 6 ns. Input peak voltage 3 to 5 volts.

G. 3.0 V minimum/5.0 V maximum.

J. At the manufacturer's option, the following alternate procedure may be used to guarantee f_{MAX}. Serial mode - f_{MAX} for the serial mode shall be guaranteed by clocking the device four times (after reset) at f_{MAX} and looking for the Q_D output to toggle within three periods (3 x 1/ f_{MAX}) plus allowed propagation delay. Two tests are performed, depending on the state of the data input, to guarantee both LH and HL transition of the output pulse.

1/ This pulse must occur after the clear pulse.

2/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Terminal	A	B	C	D	E	F	G
I_{IL1}	CLR	-.16/-.4	-.11/-.35	-.16/-.4	-.12/-.35	-.12/-.36	-.12/-.36	-.16/-.4
I_{IL2}	S/R, A_{IN}, B_{IN} C_{IN}, D_{IN}, S/L	"	-.11/-.35	"	-.16/-.4	-.105/-.345	"	"
I_{IL3}	S_0, S_1	"	-.03/-.3	"	-.12/-.36	-.12/-.36	"	"
I_{IL4}	CLK	"	-.03/-.3	-.20/-.44	-.12/-.36	-.12/-.36	"	-.15/-.38

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TABLE III. Group A inspection for device type 02.

Terminal conditions (pins not designated may be high ≥ 2.0; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test Limits		Unit
			Cases 2,X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	CLR	J	$\overline{X}$	A_{IN}	B_{IN}	C_{IN}	D_{IN}	GND	Shift Load	CLK	$\overline{Q}$ D	QD	QC	QB	QA	V_{CC}				
1 Tc = 25°C	V_{OH}	3005	1	B	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	0.7 V	GND	0.7 V	B 1/	-.4 mA					4.5 V	$\overline{Q}$ D	2.5		V
			2	"	"	"	"	"	"	2.0 V	"	"	"		-.4 mA				"	QD	"		"
			3	"	"	"	"	"	"	"	"	"	"			-.4 mA			"	QC	"		"
			4	"	"	"	"	"	"	"	"	"	"				-.4 mA		"	QB	"		"
			5	"	"	"	"	"	"	"	"	"	"					-.4 mA	"	QA	"		"
	V_{OL}	3007	6	"	"	"	0.7	0.7	0.7	"	"	"	"	4 mA					"	$\overline{Q}$ D		0.4	"
			7	"	"	"	"	"	"	0.7	"	"	"		4 mA				"	QD		"	"
			8	"	"	"	"	"	"	"	"	"	"			4 mA			"	QC		"	"
			9	"	"	"	"	"	"	"	"	"	"				4 mA		"	QB		"	"
			10	"	"	"	"	"	"	"	"	"	"					4 mA	"	QA		"	"
	V_{IC}		11	-18 mA							"								"	CLR		-1.5	"
			12		-18 mA						"								"	J		"	"
			13			-18 mA					"								"	$\overline{K}$		"	"
			14				-18 mA				"								"	A_{IN}		"	"
			15					-18 mA			"								"	B_{IN}		"	"
			16						-18 mA		"								"	C_{IN}		"	"
			17							-18 mA	"								"	D_{IN}		"	"
			18								"	-18 mA							"	Shift load		"	"
			19								"		-18 mA						"	CLK		"	"
	I_{IH1}	3010	20	2.7 V							"								5.5 V	CLR		20	µA
			21		2.7 V						"	GND	A						"	J		"	"
			22			2.7 V					"	GND							"	$\overline{K}$		"	"
			23				2.7 V				"	4.5 V							"	A_{IN}		"	"
			24					2.7 V			"	"							"	B_{IN}		"	"
			25						2.7 V		"	"							"	C_{IN}		"	"
			26							2.7 V	"	"							"	D_{IN}		"	"
			27								"	2.7 V							"	Shift load		"	"
			28								"		2.7 V						"	CLK		"	"
	I_{IH2}		29	5.5 V							"								"	CLR		100	"
			30		5.5 V						"	GND	A						"	J		"	"
			31			5.5 V					"	GND							"	$\overline{K}$		"	"
			32				5.5 V				"	4.5 V							"	A_{IN}		"	"
			33					5.5 V			"	"							"	B_{IN}		"	"
			34						5.5 V		"	"							"	C_{IN}		"	"
			35							5.5 V	"	"							"	D_{IN}		"	"
			36								"	5.5 V							"	Shift load		"	"
			37								"		5.5 V						"	CLK		"	"

See footnotes at end of device types 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test Limits		Unit
			Cases 2, X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	CLR	J	$\overline{K}$	A_{IN}	B_{IN}	C_{IN}	D_{IN}	GND	Shift Load	CLK	$\overline{Q}$ D	QD	QC	QB	QA	V_{CC}		Min	Max	
1 Tc = 25°C	I_{IL1}	3009	38	0.4 V							GND								5.5 V	CLR	2/	2/	mA
			39	B	0.4 V						"	4.5 V							"	J	"	"	"
			40	B		0.4 V					"	4.5 V	1/ A or B						"	$\overline{K}$	"	"	"
			41				0.4 V				"	GND							"	A_{IN}	"	"	"
			42					0.4 V			"	"							"	B_{IN}	"	"	"
			43						0.4 V		"	"							"	C_{IN}	"	"	"
			44							0.4 V	"	"							"	D_{IN}	"	"	"
			45								"	0.4 V							"	Shift load	"	"	"
			46								"		0.4 V						"	CLK	"	"	"
	I_{0S}	3011	47	GND							"	4.5 V	4.5 V	GND					"	$\overline{Q}$ D	-15	-100	mA
			48	4.5 V			4.5 V	4.5 V	4.5 V	4.5 V	"	GND	B		GND				"	QD	"	"	"
			49	"			"	"	"	"	"	"	"			GND			"	QC	"	"	"
			50	"			"	"	"	"	"	"	"				GND		"	QB	"	"	"
			51	"			"	"	"	"	"	"	"					GND	"	QA	"	"	"
	I_{CC}	3005	52	B			"	"	"	"	"	"	4.5 V						"	V_{CC}		21	"
		3005	53	4.5 V			"	"	"	"	"	"	B						"	V_{CC}		21	"
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = 125°C, and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C, and V_{IC} tests are omitted.																						

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test Limits		Unit
			Cases 2,X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	CLR	J	$\overline{K}$	A_{IN}	B_{IN}	C_{IN}	D_{IN}	GND	Shift Load	CLK	$\overline{Q}_D$	QD	QC	QB	QA	V_{CC}				
7 T_C = 25°C	Truth table test	3014	54	D	C	C	C	C	C	C	GND	D	D	H	L	L	L	L	5.0V	All outputs	See C, D, E, and F		
			55	"	D	C	"	"	"	"	"	D	"	"	"	"	"	"	"	"				
			56	"	D	D	"	"	"	"	"	D	"	"	"	"	"	"	"	"	-			
			57	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"	"	"			
			58	C	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"	"	"			
			59	"	"	"	"	"	"	"	"	D	"	"	"	"	"	"	"	"	"			
			60	"	"	"	"	"	"	"	"	"	C	L	H	H	H	H	"	"			
			61	"	"	"	"	"	"	"	"	"	D	L	H	H	H	H	"	"			
			62	"	C	C	D	D	D	D	"	"	D	L	H	H	H	H	"	"			
			63	"	"	"	"	"	"	"	"	"	C	H	L	L	L	L	"	"			
			64	"	"	"	"	"	"	"	"	"	D	"	"	"	"	L	"	"			
			65	"	"	"	"	"	"	"	"	C	D	"	"	"	"	L	"	"			
			66	"	"	"	"	"	"	"	"	"	C	"	"	"	"	H	"	"			
			67	"	"	"	"	"	"	"	"	"	D	"	"	"	"	"	"	"			
			68	"	"	"	"	"	"	"	"	"	C	"	"	"	H	"	"	"			
			69	"	"	"	"	"	"	"	"	"	D	"	"	"	"	"	"	"			
			70	"	"	"	"	"	"	"	"	"	C	"	"	H	"	"	"	"			
			71	"	"	"	"	"	"	"	"	"	D	"	"	"	"	"	"	"			
			72	"	"	"	"	"	"	"	"	"	C	L	H	"	"	"	"	"			
			73	"	"	"	"	"	"	"	"	"	D	"	"	"	"	"	"	"			
			74	"	D	D	C	C	C	C	"	"	D	"	"	"	"	"	"	"			
			75	"	"	"	"	"	"	"	"	"	C	"	"	"	"	L	"	"			
			76	"	"	"	"	"	"	"	"	"	D	"	"	"	"	"	"	"			
			77	"	"	"	"	"	"	"	"	"	C	"	"	"	L	"	"	"			
			78	"	"	"	"	"	"	"	"	"	D	"	"	"	"	"	"	"			
			79	"	"	"	"	"	"	"	"	"	C	"	"	L	"	"	"	"			
			80	"	"	"	"	"	"	"	"	"	D	"	"	"	"	"	"	"			
			81	"	"	"	"	"	"	"	"	"	C	H	L	"	"	"	"	"			
			82	"	"	"	"	"	"	"	"	"	D	H	"	"	"	"	"	"			
			83	"	C	"	D	D	D	D	"	"	D	"	"	"	"	"	"	"			
			84	"	"	"	"	"	"	"	"	"	C	"	"	"	"	H	"	"			
			85	"	"	"	"	"	"	"	"	"	D	"	"	"	"	H	"	"			
			86	"	"	"	"	"	"	"	"	"	C	"	"	"	H	L	"	"			
			87	"	"	"	"	"	"	"	"	"	D	"	"	"	H	L	"	"			
			88	"	"	"	"	"	"	"	"	"	C	"	"	H	L	H	"	"			
			89	"	"	"	"	"	"	"	"	"	D	"	"	H	L	H	"	"			
			90	"	"	"	"	"	"	"	"	"	C	L	H	L	H	L	"	"			
			91	"	"	"	"	"	"	"	"	"	D	L	H	L	H	L	"	"			
			92	"	"	"	"	"	"	"	"	"	C	H	L	H	L	H	"	"			
			93	"	"	"	"	"	"	"	"	"	D	H	L	H	L	H	"	"			
			94	"	D	C	C	C	C	C	"	"	D	H	L	H	L	"	"	"			
			95	"	"	"	"	"	"	"	"	"	C	L	H	L	H	"	"	"			
			96	"	"	"	"	"	"	"	"	"	D	L	H	L	"	"	"	"			
			97	"	"	"	"	"	"	"	"	"	C	H	L	H	"	"	"	"			
			98	"	"	"	"	"	"	"	"	"	D	H	L	H	"	"	"	"			
			99	"	"	"	"	"	"	"	"	"	C	L	H	H	"	"	"	"			
8	Same tests, terminal conditions, and limits as subgroup 7 except T_C = 125°C and -55°C.																						
9 T_C = 25°C	f_{MAX} See G	(Fig. 5)	100	J			IN				GND	GND	IN					OUT	5.0 V	QA	27		MHz
			101	"				IN			"	"	"				OUT		"	QB	"		"
			102	"					IN		"	"	"			OUT			"	QC	"		"
			103	"						IN	"	"	"		OUT				"	QD	"		"
			104	"						IN	"	"	"	OUT					"	$\overline{Q}_D$	"		"

See footnotes at end of device type 02.

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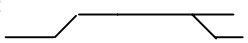
TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.7 V; or open).

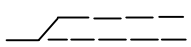
Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Cases 2, X / Test no.	1 / CLR	2 / J	3 / $\overline{K}$	4 / A_{IN}	5 / B_{IN}	6 / C_{IN}	7 / D_{IN}	8 / GND	9 / Shift load	10 / CLK	11 / $\overline{Q}$ D	12 / QD	13 / QC	14 / QB	15 / QA	16 / V_{CC}	Measured terminal	Test Limits Min	Test Limits Max	Unit
9 T_C = 25°C	t_{PHL1}	3003 (Fig. 5)	105	J	J	J					GND	J	IN					OUT	5.0 V	CLK to QA	5	27	ns
			106	"	See Fig. 5	See Fig. 5					"	"	"				OUT		"	CLK TO QB	"	"	"
			107	"	"	"					"	"	"			OUT			"	CLK TO QC	"	"	"
			108	"	"	"					"	"	"		OUT				"	CLK TO QD	"	"	"
			109	"	"	"					"	"	"	OUT					"	CLK TO $\overline{Q}$ D	"	"	"
			110	"			IN				"	GND	"					OUT	"	CLK TO QA	"	"	"
			111	"				IN			"	"	"				OUT		"	CLK TO QB	"	"	"
			112	"					IN		"	"	"			OUT			"	CLK TO QC	"	"	"
			113	"						IN	"	"	"		OUT				"	CLK TO QD	"	"	"
			114	"						IN	"	"	"	OUT						CLK TO QD	"	"	"
	t_{PHL1}		115	"	GND	GND					"	J	"					OUT	"	CLK TO QA	"	31	"
			116	"	See Fig. 5	See Fig. 5					"	"	"				OUT		"	CLK TO QB	"	"	"
			117	"	"	"					"	"	"			OUT			"	CLK TO QC	"	"	"
			118	"	"	"					"	"	"		OUT				"	CLK TO QD	"	"	"
			119	"	"	"					"	"	"	OUT					"	CLK TO $\overline{Q}$ D	"	"	"
			120	"			IN				"	GND	"					OUT	"	CLK TO QA	"	"	"
			121	"				IN			"	"	"				OUT		"	CLK TO QB	"	"	"
			122	"					IN		"	"	"			OUT			"	CLK TO QC	"	"	"
			123	"						IN	"	"	"		OUT				"	CLK TO QD	"	"	"
			124	"						IN	"	"	"	OUT					"	CLK TO $\overline{Q}$ D	"	"	"
	t_{PHL2}		125	IN			J				"	"	"					OUT	"	CLK TO QA	"	35	"
			126	"				J			"	"	"				OUT		"	CLK TO QB	"	"	"
			127	"					J		"	"	"			OUT			"	CLK TO QC	"	"	"
			128	"						J	"	"	"		OUT				"	CLK TO QD	"	"	"
	t_{PHL2}		129	"						J	"	"	"	OUT					"	CLR TO $\overline{Q}$ D	"	"	"
10 T_C = 125°C	f_{MAX} See G		130 to 134	Same tests and terminal conditions as for subgroup 9																	25		MHz
	t_{PLH1}		135 to 144																		5	41	ns
	t_{PHL1}		145 to 154																		"	47	"
	t_{PHL2}		155 to 158																		"	53	"
	t_{PHL2}		159																		"	53	"
11	Same tests, terminal conditions, and limits as subgroups 10, except T_C = -55°C.																						

See footnotes at end of device type 02.

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FOOTNOTES:

A. Apply input pulse:  2.5 V minimum/5.5 V maximum
0 V

B. Apply input pulse:  2.5 V minimum/5.5 V maximum.
0 V

C. V_{IN} = 2.5 V.

D. V_{IN} = 0.4 V.

E. Test numbers 54 through 99 shall be run in sequence.

F. Output voltages shall be either: (1) H ≥2.5 V minimum and L ≤0.4 V maximum when using a high speed checker double comparator; (2) H ≥1.5 V and L ≤1.5 V when using a high speed checker single comparator.

G. f_{MAX} minimum limit specified is the frequency of the clock input pulse. The output frequency shall be one-half of the input clock frequency. The input frequency on the parallel input shall be one-half of the clock input frequency and the parallel input shall be shifted such that the parallel input ↑ and ↓ are coincident with the clock ↓. Rise and fall times ≤ 6 ns. Input peak voltage 3 to 5 volts.

J. 3.0 V minimum/5.0 V maximum.

1/ This pulse must occur after the clear pulse.

2/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Terminal	A	B	C	D	E	F	G
I_{IL1}	CLR	-.16/-.4	-.11/-.35	-.16/-.4	-.12/-.35	-.12/-.36	-.12/-.36	-.16/-.4
	J, K, A_{IN}, B_{IN}, C_{IN}, D_{IN}	"	-.16/-.4	"	-.16/-.4	-.105/-.345	"	"
	Shift load	"	-.08/-.3	"	-.12/-.36	-.12/-.36	"	"
	CLK	"	-.03/-.3	-.20/-.44	-.12/-.36	-.12/-.36	"	-.15/-.38

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Cases 2,X	2	3	4	5	8	9	10	12	13	14	16	18	19	20		Min	Max	
			Test no.	Serial	A_{IN}	B_{IN}	C_{IN}	D_{IN}	Mode	GND	CLK_2	CLK_1	QD	QC	QB	QA	V_{CC}				
1 Tc = 25°C	V_{OH}	3006	1	2.0 V	GND	GND	GND	GND	0.7 V	GND	GND	A				-.4 mA	4.5 V	QA	2.5		V
			2		2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	"	A	GND				-.4 mA	"	QA	"		"
			3		"	"	"	"	"	"	"	"			-.4 mA		"	QB	"		"
			4		"	"	"	"	"	"	"	"		-.4 mA			"	QC	"		"
			5		"	"	"	"	"	"	"	"	-.4 mA				"	QD	"		"
	V_{OL}	3007	6	0.7 V	4.5 V	4.5 V	4.5 V	4.5 V	0.7 V	"	GND	A				4 mA	"	QA	"	0.4	"
			7		0.7 V	0.7 V	0.7 V	0.7	2.0 V	"	A	GND				4 mA	"	QA		"	"
			8		"	"	"	"	"	"	"	"			4 mA		"	QB		"	"
			9		"	"	"	"	"	"	"	"		4 mA			"	QC		"	"
			10		"	"	"	"	"	"	"	"	4 mA				"	QD		"	"
	V_{IC}		11	-18 mA						"							"	Serial		-1.5	"
			12		-18 mA					"							"	A_{IN}		"	"
			13			-18 mA				"							"	B_{IN}		"	"
			14				-18 mA			"							"	C_{IN}		"	"
			15					-18 mA		"							"	D_{IN}		"	"
			16						-18 mA	"							"	Mode		"	"
			17							"	-18 mA						"	CLK_2		"	"
			18							"		-18 mA					"	CLK_1		"	"
	I_{IH3}	3010	19	2.7 V					4.5 V	"							5.5 V	Serial		20	µA
			20		2.7 V				GND	"							"	A_{IN}		"	"
			21			2.7 V			"	"							"	B_{IN}		"	"
			22				2.7 V		"	"							"	C_{IN}		"	"
			23					2.7 V	"	"							"	D_{IN}		"	"
			24						4.5 V	"	2.7 V						"	CLK_2		"	"
			25						4.5 V	"		2.7 V					"	CLK_1		"	"
	I_{IH4}		26	5.5 V					4.5 V	"							"	Serial		100	"
			27		5.5 V				GND	"							"	A_{IN}		"	"
			28			5.5 V			"	"							"	B_{IN}		"	"
			29				5.5 V		"	"							"	C_{IN}		"	"
			30					5.5 V	"	"							"	D_{IN}		"	"
			31						4.5 V	"	5.5 V						"	CLK_2		"	"
			32						4.5 V	"		5.5 V					"	CLK_1		"	"
	I_{IH7}		33						2.7 V	"	GND						"	Mode		40	"
	I_{IH8}		34						5.5 V	"	GND						"	Mode		200	"

See footnotes at end of device types 03.

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TABLE III. Group A inspection for device type 03 - Continued

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test Limits		Unit
			Cases 2, X	2	3	4	5	8	9	10	12	13	14	16	18	19	20				
			Test no.	Serial	A_{IN}	B_{IN}	C_{IN}	D_{IN}	Mode	GND	CLK_2	CLK_1	QD	QC	QB	QA	V_{CC}		Min	Max	
1 T_C = 25°C	I_{IL2}	3009	35	0.4 V					GND	GND							5.5 V	Serial	1/	1/	mA
			36		0.4 V				4.5 V	"							"	A	"	"	"
			37			0.4 V			"	"							"	B	"	"	"
			38				0.4 V		"	"							"	C	"	"	"
			39					0.4 V	"	"							"	D	"	"	"
			40						0.4 V	"	4.5 V						"	Mode	"	"	"
	I_{IL4}		41						4.5 V	"	0.4 V						"	CLK_2	"	"	"
			42						GND	"		0.4 V					"	CLK_1	"	"	"
	I_{0S}	3011	43		4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	A	GND				GND	"	QA	-15	-100	"
			44		"	"	"	"	"	"	"	"			GND		"	QB	"	"	"
			45		"	"	"	"	"	"	"	"		GND			"	QC	"	"	"
			46		"	"	"	"	"	"	"	"	GND				"	QD	"	"	"
	I_{CC}	3005	47		GND	GND	GND	GND	5.5 V	"	"	A					"	V_{CC}		21	"
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
7 T_C = 25°C	Truth table test	3014	48	B	B	B	B	B	B	GND	B	C	X	X	X	X	5.0 V	All outputs	See B,C,D, and E		
			49	B	B	B	B	B	"	"	C	"	H	H	H	H	"				
			50	"	B	B	B	B	"	"	B	"	H	H	H	H	"	"			
			51	"	C	C	C	C	"	"	B	"	H	H	H	H	"	"			
			52	"	C	C	C	C	"	"	C	"	L	L	L	L	"	"			
			53	"	B	B	B	B	C	"	"	B	"	"	"	L	"	"			
			54	"	C	C	C	C	"	"	"	B	"	"	"	L	"	"			
			55	"	"	"	"	"	"	"	"	C	"	"	"	H	"	"			
			56	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"			
			57	"	"	"	"	"	"	"	"	C	"	"	H	"	"	"			
			58	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"			
			59	"	"	"	"	"	"	"	"	C	"	H	"	"	"	"			
			60	"	"	"	"	"	"	"	"	B	"	H	"	"	"	"			
			61	"	"	"	"	"	"	"	"	C	H	H	"	"	"	"			

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Cases 2, X Test no.	1 2 Serial	2 3 A_{IN}	3 4 B_{IN}	4 5 C_{IN}	5 8 D_{IN}	6 9 Mode	7 10 GND	8 12 CLK_2	9 13 CLK_1	10 14 QD	11 16 QC	12 18 QB	13 19 QA	14 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
7 Tc = 25°C	Truth table tests	3014	62	B	C	C	C	C	C	GND	C	B	H	H	H	H	5.0 V	All outputs	See B,C,D, and E		
			63	C	B	B	B	B	"	"	B	B	"	"	"	H	"				
			64	"	"	"	"	"	"	"	"	C	"	"	"	L	"	"			
			65	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"			
			66	"	"	"	"	"	"	"	"	C	"	"	L	"	"	"			
			67	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"			
			68	"	"	"	"	"	"	"	"	C	"	L	"	"	"	"			
			69	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"			
			70	"	"	"	"	"	"	"	"	C	L	"	"	"	"	"			
			71	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			72	"	"	"	"	"	B	"	C	"	"	"	"	"	"	"			
			73	"	"	"	"	"	C	"	C	"	"	"	"	"	"	"			
			74	"	"	"	"	"	C	"	B	"	"	"	"	"	"	"			
			75	"	"	"	"	"	B	"	B	"	"	"	"	"	"	"			
			76	"	"	"	"	"	B	"	C	B	H	H	H	H	"	"			
			77	"	"	"	"	"	C	"	C	"	"	"	"	"	"	"			
			78	"	"	"	"	"	C	"	B	"	"	"	"	"	"	"			
			79	"	"	"	"	"	B	"	B	"	"	"	"	"	"	"			
8	Same tests, terminal conditions, and limits as subgroup 7 except T_C = 125°C and -55°C.																				
9 T_C = 25°C	f_{MAX} See F,J	(Fig. 6)	80		IN				G	GND	IN					OUT	5.0 V	QA	22		MHz
			81			IN			"	"	"				OUT		"	QB	"		"
			82				IN		"	"	"			OUT			"	QC	"		"
			83					IN	"	"	"		OUT				"	QD	"		"
			84	IN					GND	"		IN				OUT	"	QA	"		"
	t_{PLH1}	3003 (Fig. 6)	85		IN				G	"	IN					OUT	"	CLK to QA	5	32	ns
			86			IN			"	"	"				OUT		"	CLK to QB	"	"	"
			87				IN		"	"	"			OUT			"	CLK to QC	"	"	"
			88					IN	"	"	"		OUT				"	CLK to QD	"	"	"
			89	IN					GND	"		IN				OUT	"	CLK to QA	"	"	"
			90	"					"	"		"			OUT		"	CLK to QB	"	"	"
			91	"					"	"		"		OUT				CLK to QC	"	"	"
			92	"					"	"		"	OUT				"	CLK to QD	"	"	"
	t_{PLH1}		93		IN				G	"	IN					OUT	"	CLK to QA	"	37	"
			94			IN			"	"	"				OUT		"	CLK to QB	"	"	"
			95				IN		"	"	"			OUT			"	CLK to QC	"	"	"
			96					IN	"	"	"		OUT				"	CLK to QD	"	"	"
			97	IN					GND	"		IN				OUT	"	CLK to QA	"	"	"
			98	"					"	"		"			OUT		"	CLK to QB	"	"	"
			99	"					"	"		"		OUT			"	CLK to QC	"	"	"
			100	"					"	"		"	OUT				"	CLK to QD	"	"	"

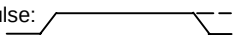
See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Cases 2, X	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	Serial	A_{IN}	B_{IN}	C_{IN}	D_{IN}	Mode	GND	CLK_2	CLK_1	QD	QC	QB	QA	V_{CC}		Min	Max	
10 Tc = 25°C	f_{MAX} See F,J	3003 (Fig. 6)	101 to 105	Same tests and terminal conditions as for subgroup 9.															20	---	MHz
	t_{PLH1}	3003 (Fig. 6)	106 to 113																5	48	ns
	t_{PHL1}	3003 (Fig. 6)	114 to 121																5	56	ns
11	Same tests, terminal conditions as subgroup 10 except T_C = -55°C.																				

Notes:

A. Apply input pulse:  2.5 V minimum/5.5 V maximum
0 V

B. V_{IN} = 2.5 V.

C. V_{IN} = 0.4 V.

D. Tests numbers 48 through 79 shall be run in sequence.

E. Output voltages shall be either: (1) H ≥2.5 minimum and L ≤0.4 V maximum when using a high speed checker double comparator; (2) H ≥1.5 V and L ≤1.5 V when using a high speed checker single comparator.

F. f_{MAX} minimum limit specified is the frequency of the clock input pulse. The output frequency shall be one-half of the input clock frequency. The input frequency on the serial input shall be one-half of the clock input frequency and the input shall be shifted such that the input ↑ and ↓ are coincident with the clock ↑. Rise and fall times ≤ 6 ns. Input peak voltage 3 to 5 volts.

G. 3.0 V minimum/5.0 V maximum.

J. At the manufacturer's option, the following alternate procedures may be used to guarantee f_{MAX}:

a. Parallel mode. f_{MAX} for the parallel mode shall be guaranteed by performing propagation delay measurements with the clock pulse width at 1/2 x $1/f_{MAX}$. In addition to the constraints on the clock pulse, the inputs are set to the worst-case condition for the $t_{set\text{-}up}$ and t_{hold} requirements. Both positive and negative clock pulse widths shall be tested. The five tests to justify each JAN f_{MAX} requirement shall be used to test all possible input/output combinations. A failing limit or nontoggle will indicate that the device fails to function at f_{MAX} and/or the propagation delay from input to output has exceeded the allowed limit .

b. Serial mode. f_{MAX} for the serial mode shall be guaranteed by clocking the device four times (after reset) at f_{MAX} and looking for the Q_D output to toggle within three periods (3 x $1/f_{MAX}$) plus allowed propagation delay. Two tests are performed, depending on the state of data input, to guarantee both LH and HL transition of the output pulse.

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1/ I_L limits (mA) min/max values for circuits shown:

Parameter	Terminal	A	B	C	D	E
I_{IL2}	Serial A, B, C, D	-.16/-.4	-.11/-.35	-.16/-.4	-.105/-.345	-.12/-.35
I_{IL4}	Mode	"	-.06/-.6	-.30/-.75	-.24/-.72	"
	CLK_2, CLK_1	"	-.03/-.3	-.20/-.44	-.12/-.36	"

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TABLE III. Group A inspection for device type 04.

Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2,X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	CLK	A_{IN}	B_{IN}	C_{IN}	V_{CC}	D_{IN}	E_{IN}	Enable	Serial	QE	QD	GND	QC	QB	QA	CLR		Min	Max	
1 Tc = 25°C	V_{OH}	3006	1	2.0 V	2.0 V	2.0 V	2.0 V	4.5 V	2.0 V	2.0 V	2.0 V	2.0 V			GND			-.4 mA	2.0 V	QA	2.5		V
			2	"	"	"	"	"	"	"	"	"			"		-.4 mA		"	QB	"		"
			3	"	"	"	"	"	"	"	"	"			"	-.4 mA			"	QC	"		"
			4	"	"	"	"	"	"	"	"	"		-.4 mA	"				"	QD	"		"
			5	"	"	"	"	"	"	"	"	"	-.4 mA		"				"	QE	"		"
	V_{OL}	3007	6	"	"	"	"	"	"	"	0.7 V				"			4 mA	0.7 V	QA		0.4	"
			7	"	"	"	"	"	"	"	"				"		4 mA		"	QB		"	"
			8	"	"	"	"	"	"	"	"				"	4 mA			"	QC		"	"
			9	"	"	"	"	"	"	"	"			4 mA	"				"	QD		"	"
			10	"	"	"	"	"	"	"	"		4 mA		"				"	QE		"	"
	V_{IC}		11	-18 mA				"							"					CLK		-1.5	"
			12		-18 mA			"							"					A_{IN}		"	"
			13			-18 mA		"							"					B_{IN}		"	"
			14				-18 mA	"							"					C_{IN}		"	"
			15					"	-18 mA						"					D_{IN}		"	"
			16					"		-18 mA					"					E_{IN}		"	"
			17					"			-18 mA				"					Enable		"	"
			18					"				-18 mA			"					Serial		"	"
			19					"							"				-18 mA	CLR		"	"
	I_{IH5}		20	2.7 V	GND	GND	GND	5.5 V	GND	GND	GND	GND			"				GND	CLK		20	µA
			21	GND	2.7 V	GND	GND	"	"	"	"	"			"				"	A_{IN}		"	"
			22	"	GND	2.7 V	GND	"	"	"	"	"			"				"	B_{IN}		"	"
			23	"		GND	2.7 V	"	"	"	"	"			"				"	C_{IN}		"	"
			24	"	"	"	GND	"	2.7 V	"	"	"			"				"	D_{IN}		"	"
			25	"	"	"	"	"	GND	2.7 V	"	"			"				"	E_{IN}		"	"
			26	"	"	"	"	"	"	GND	"	2.7 V			"				"	Serial		"	"
			27	"	"	"	"	"	"	"	"	GND			"				2.7 V	CLR		"	"
	I_{IH6}		28	5.5 V	"	"	"	"	"	"	"	"			"				GND	CLK		100	"
			29	GND	5.5 V	"	"	"	"	"	"	"			"				"	A_{IN}		"	"
			30	"	GND	5.5 V	"	"	"	"	"	"			"				"	B_{IN}		"	"
			31	"	"	GND	5.5 V	"	"	"	"	"			"				"	C_{IN}		"	"
			32	"	"	"	GND	"	5.5 V	"	"	"			"				"	D_{IN}		"	"
			33	"	"	"	"	"	GND	5.5 V	"	"			"				"	E_{IN}		"	"
			34	"	"	"	"	"	"	GND	"	5.5 V			"				"	Serial		"	"
			35	"	"	"	"	"	"	"	"	GND			"				5.5 V	CLR		"	"
	I_{IH9}		36	"	"	"	"	"	"	"	2.7 V	GND			"				GND	Enable		"	"
	I_{IH10}		37	"	"	"	"	"	"	"	5.5 V	GND			"				GND	Enable		500	"
	I_{IL3}	3009	38	0.4 V	4.5 V	4.5 V	4.5 V	"	4.5 V	4.5 V	4.5 V	4.5 V			"				4.5 V	CLK	1/	1/	mA
			39	4.5 V	0.4 V	4.5 V	"	"	"	"	"	"			"				"	A_{IN}	"	"	"
			40	"	4.5 V	0.4 V	"	"	"	"	"	"			"				"	B_{IN}	"	"	"
			41	"	"	4.5 V	0.4 V	"	"	"	"	"			"				"	C_{IN}	"	"	"
			42	"	"	4.5 V	4.5 V	"	0.4 V	"	"	"			"				"	D_{IN}	"	"	"

See footnotes at end of device types 04.

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TABLE III. Group A inspection for device type 04 - Continued

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Cases 2, X Test no.	1 CLK	2 A_{IN}	3 B_{IN}	4 C_{IN}	5 V_{CC}	6 D_{IN}	7 E_{IN}	8 Enable	9 Serial	10 QE	11 QD	12 GND	13 QC	14 QB	15 QA	16 CLR	Measured terminal	Test Limits Min	 Max	Unit
1 T_C = 25°C	I_{IL3}	3009	43	4.5 V	4.5 V	4.5 V	4.5 V	5.5 V	4.5 V	0.4 V	4.5 V	4.5 V			GND				4.5 V	E_{IN}	1/	1/	mA
			44	"	"	"	"	"	"	4.5 V	"	0.4 V			"				4.5 V	Serial	"	"	"
			45	"	"	"	"	"	"	"	"	4.5 V			"				0.4 V	CLR	"	"	"
	I_{IL5}		46	"	"	"	"	"	"	"	0.4 V	"			"				4.5 V	Enable	"	"	"
	I_{OS}	3011	47	"	"	"	"	"	"	"	4.5 V	"			"			GND	"	QA	-15	-100	"
			48	"	"	"	"	"	"	"	"	"			"		GND		"	QB	"	"	"
			49	"	"	"	"	"	"	"	"	"			"	GND			"	QC	"	"	"
			50	"	"	"	"	"	"	"	"	"		GND	"				"	QD	"	"	"
			51	"	"	"	"	"	"	"	"	"	GND		"				"	QE	"	"	"
	I_{CC}	3005	52					"							"				GND	V_{CC}		20	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = 25°C	Truth table test	3014	53	B	A	A	A	5.0 V	A	A	B	B	L	L	GND	L	L	L	B	All outputs	See A,B,C, and D		
			54	A	"	"	"	"	"	"	B	A	L	L	"	L	L	L	B				
			55	B	"	"	"	"	"	"	B	B	L	L	"	L	L	L	B	"			
			56	"	"	"	"	"	"	"	A	"	H	H	"	H	H	H	A	"			
			57	"	B	B	B	"	B	B	"	"	H	H	"	H	H	H	A	"			
			58	"	"	"	"	"	"	"	"	"	L	L	"	L	L	L	B	"			
			59	"	"	"	"	"	"	"	B	"	"	"	"		"	L	A	"			
			60	"	"	"	"	"	"	"	"	A	"	"	"	"	"	L	"	"			
			61	A	"	"	"	"	"	"	"	"	"	"	"	"	"	H	"	"			
			62	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			63	A	"	"	"	"	"	"	"	"	"	"	"	"	H	"	"	"			
			64	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			65	A	"	"	"	"	"	"	"	"	"	"	"	H	"	"	"	"			
			66	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			67	A	"	"	"	"	"	"	"	"	"	H	"	"	"	"	"	"			
			68	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			69	A	"	"	"	"	"	"	"	"	H	"	"	"	"	"	"	"			
			70	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			71	B	A	A	A	"	A	A	"	B	"	"	"	"	"	"	"	"			
			72	A	"	"	"	"	"	"	"	"	"	"	"	"	"	L	"	"			
			73	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			74	A	"	"	"	"	"	"	"	"	"	"	"	"	L	"	"	"			
			75	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			76	A	"	"	"	"	"	"	"	"	"	"	"	L	"	"	"	"			
			77	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			78	A	"	"	"	"	"	"	"	"	"	L	"	"	"	"	"	"			
			79	B	"	"	"	"	"	"	"	"	"	L	"	"	"	"	"	"			
			80	A	"	"	"	"	"	"	"	"	L	L	"	"	"	"	"	"			

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Cases 2, X Test no.	1 2 CLK	2 3 A_{IN}	3 4 B_{IN}	4 5 C_{IN}	5 7 V_{CC}	6 8 D_{IN}	7 9 E_{IN}	8 10 Enable	9 12 Serial	10 13 QE	11 14 QD	12 15 GND	13 17 QC	14 18 QB	15 19 QA	16 20 CLR	Measured terminal	Test Limits Min	 Max	Unit
8	Same tests, terminal conditions, and limits as subgroup 7, except T_C = 125°C and -55°C																						
9 T_C = 25°C	f_{MAX} see note E	(Fig. 7)	81	IN				5.0 V			GND	IN			GND			OUT	F	QA	20		MHz
	t_{PLH1}	3003 (Fig. 7)	82	"				"			"	IN			"			OUT	"	CLK TO QA	5	45	ns
			83	"				"			"	See figure 7			"		OUT		"	CLK TO QB	"	"	"
			84	"				"			"				"	OUT			"	CLK TO QC	"	"	"
			85	"				"			"			OUT	"				"	CLK TO QD	"	"	"
			86	"				"			"		OUT						"	CLK TO QE	"	"	"
	t_{PLH2}		87	GND	IN			"			F				"			OUT	IN	A_{IN} TO QA	"	40	"
			88	"		IN		"			"				"		OUT		"	B_{IN} TO QB	"	"	"
			89	"			IN	"			"				"	OUT			"	C_{IN} TO QC	"	"	"
			90	"				"	IN		"			OUT	"				"	D_{IN} TO QD	"	"	"
			91	"				"		IN	"		OUT		"				"	E_{IN} TO QE	"	"	"
			92	"	F			"			IN				"			OUT	"	Enable to QA	"	"	"
			93	"		F		"			"				"		OUT		"	Enable to QB	"	"	"
			94	"			F	"			"				"	OUT			"	Enable to QC	"	"	"
			95	"				"	F		"			OUT	"				"	Enable to QD	"	"	"
			96	"				"	"	F	"		OUT		"				"	Enable to QE	"	"	"
	t_{PHL1}		97	IN				"			GND	IN			"			OUT	F	CLK TO QA	"	45	
			98	"				"			"				"		OUT		"	CLK TO QB	"	"	
			99	"				"			"				"	OUT			"	CLK TO QC	"	"	
			100	"				"			"			OUT	"				"	CLK TO QD	"	"	
			101	"				"			"		OUT		"				"	CLK to QE	"	"	
	t_{PHL2}		102	GND	F			"			IN				"			OUT	IN	CLR to QA	"	60	
			103	"		F		"			"				"		OUT		"	CLR to QB	"	"	
			104	"			F	"			"				"	OUT			"	CLR to QC	"	"	
			105	"				"	F		"			OUT	"				"	CLR to QD	"	"	
			106	"				"		F	"		OUT		"				"	CLR to QE	"	"	

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F / Cases 2, X / Test no.	1 / 2 / CLK	2 / 3 / A_{IN}	3 / 4 / B_{IN}	4 / 5 / C_{IN}	5 / 7 / V_{CC}	6 / 8 / D_{IN}	7 / 9 / E_{IN}	8 / 10 / Enable	9 / 12 / Serial	10 / 13 / QE	11 / 14 / QD	12 / 15 / GND	13 / 17 / QC	14 / 18 / QB	15 / 19 / QA	16 / 20 / CLR	Measured terminal	Test Limits Min	Test Limits Max	Unit
10 T_C = 25°C	f_{max} See E	(Fig. 7)	107																		17	---	MHz
	t_{PLH1}	3003 (Fig, 7)	108 to 112	Same tests and terminal conditions as for subgroup 9.																	5	68	ns
	t_{PLH2}		113 to 122																		"	60	"
	t_{PHL1}		123 to 127																		"	68	"
	t_{PHL2}		128 to 132																		"	90	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

Notes:

A. V_{IN} = 2.5 V.

B. V_{IN} = 0.4 V.

C. Tests numbers 53 through 80 shall be run in sequence.

D. Output voltages shall be either: (1) H ≥2.5 V minimum and L ≤0.4 V maximum when using a high speed checker double double comparator; (2) H ≥1.5 V and L ≤1.5 V when using a high speed checker single comparator.

E. f_{MAX} minimum limit specified is the frequency of the clock input pulse. The output frequency shall be one-half of the input clock frequency. The input frequency on the serial data shall be one-half of the clock input frequency and the serial shall be shifted such that the serial ↑ and ↓ are coincident with the clock ↓. Rise and fall times ≤ 6 ns. Input peak voltage 3 to 5 volts.

1/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Terminal	A	B
I_{IL3}	CLK	-.16/-.40	-.16/-.40
	A_{IN}, B_{IN}, C_{IN} D_{IN}, E_{IN}, CLR	-.16/-.40	-.12/-.36
	Serial	-.10/-.34	-.10/-.34
I_{IL5}	Enable	-.8/-2.0	-.6/-1.8

F. 3.0 V minimum/5.0 V maximum.

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TABLE III. Group A inspection for device type 05.

Terminal conditions (pins not designated may be high ≥ 2.0; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test Limits		Unit
			Cases 2,X	2	3	4	5	8	9	10	12	13	14	16	18	19	20				
			Test no.	A_{IN}	B_{IN}	QA	QB	QC	QD	GND	CLK	CLR	QE	QF	QG	QH	V_{CC}		Min	Max	
1 Tc = 25°C	V_{OH}	3006	1	2.0 V	2.0 V	-.4 mA				GND	J 1/	2.0 V					4.5 V	QA	2.5		V
			2	"	"		-.4 mA			"	" 2/	"					"	QB	"		"
			3	"	"			-.4 mA		"	" 3/	"					"	QC	"		"
			4	"	"				-.4 mA	"	" 4/	"					"	QD	"		"
			5	"	"					"	" 5/	"	-.4 mA				"	QE	"		"
			6	"	"					"	" 6/	"		-.4 mA			"	QF	"		"
			7	"	"					"	" 7/	"			-.4 mA		"	QG	"		"
			8	"	"					"	" 8/	"				-.4 mA	"	QH	"		"
	V_{OL}	3007	9			4 mA				"		0.7 V					"	QA		0.4	"
			10				4 mA			"		"					"	QB		"	"
			11					4 mA		"		"					"	QC		"	"
			12						4 mA	"		"					"	QD		"	"
			13							"		"	4 mA				"	QE		"	"
			14							"		"		4 mA			"	QF		"	"
			15							"		"			4 mA		"	QG		"	"
			16							"		"				4 mA	"	QH		"	"
	V_{IC}		17	-18 mA						"							"	A_{IN}		-1.5	"
			18		-18 mA					"							"	B_{IN}		"	"
			19							"	-18 mA						"	CLK		"	"
			20							"		-18 mA					"	CLR		"	"
	I_{IH1}	3010	21	2.7 V	GND					"							5.5 V	A_{IN}		20	µA
			22	GND	2.7 V					"							"	B_{IN}		"	"
			23							"	2.7 V						"	CLK		"	"
			24							"		2.7 V					"	CLR		"	"
	I_{IH2}		25	5.5 V	GND					"							"	A_{IN}		100	"
			26	GND	5.5 V					"							"	B_{IN}		"	"
			27							"	5.5 V						"	CLK		"	"
			28							"		5.5 V					"	CLR		"	"
	I_{IL1}	3009	29	0.4 V	4.5 V					"							"	A_{IN}	10/	10/	mA
			30	4.5 V	0.4 V					"							"	B_{IN}	"	"	"
			31							"	0.4 V						"	CLK	"	"	"
			32							"		0.4 V					"	CLR	"	"	"

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test Limits		Unit
			Cases 2, X	2	3	4	5	8	9	10	12	13	14	16	18	19	20				
			Test no.	A_{IN}	B_{IN}	QA	QB	QC	QD	GND	CLK	CLR	QE	QF	QG	QH	V_{CC}		Min	Max	
1	I_{OS}	3011	33 9/	4.5 V	4.5 V	GND				"	A 1/	4.5 V					5.5 V	QA	-15	-100	mA
T_C = 25°C			34 "	"	"		GND			"	" 2/	"					"	QB	"	"	"
			35 "	"	"			GND		"	" 3/	"					"	QC	"	"	"
			36 "	"	"				GND	"	" 4/	"					"	QD	"	"	"
			37 "	"	"					"	" 5/	"	GND				"	QE	"	"	"
			38 "	"	"					"	" 6/	"		GND			"	QF	"	"	"
			39 "	"	"					"	" 7/	"			GND		"	QG	"	"	"
			40 "	"	"					"	" 8/	"				GND	"	QH	"	"	"
	I_{CC}	3005	41	GND	GND					"	5.5 V	J					"	V_{CC}		27	"
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = 125°C and V_{IC} tests are omitted																				
7	Truth	3014	42	B	B	L	L	L	L	GND	C	C	L	L	L	L	5.0 V	All	See B,C,D and E		
T_C = 25°C	table		43	"	"	"	"	"	"	"	B	C	"	"	"	"	"	outputs			
	test		44	"	"	"	"	"	"	"	C	C	"	"	"	"	"	"			
			45	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"			
			46	C	C	"	"	"	"	"	"	"	"	"	"	"	"	"			
			47	B	B	"	"	"	"	"	"	"	"	"	"	"	"	"			
			48	"	"	H	"	"	"	"	B	"	"	"	"	"	"	"			
			49	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			50	"	"	"	H	"	"	"	B	"	"	"	"	"	"	"			
			51	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			52	"	"	"	"	H	"	"	B	"	"	"	"	"	"	"			
			53	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			54	"	"	"	"	"	H	"	B	"	"	"	"	"	"	"			
			55	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			56	"	"	"	"	"	"	"	B	"	H	"	"	"	"	"			
			57	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			58	"	"	"	"	"	"	"	B	"	"	H	"	"	"	"			
			59	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			60	"	"	"	"	"	"	"	B	"	"	"	H	"	"	"			
			61	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			62	"	"	"	"	"	"	"	B	"	"	"	"	H	"	"			
			63	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			64	C	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			65	"	"	L	"	"	"	"	B	"	"	"	"	"	"	"			
			66	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			67	"	"	"	L	"	"	"	B	"	"	"	"	"	"	"			
			68	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			69	"	C	"	"	"	"	"	C	"	"	"	"	"	"	"			
			70	"	"	"	"	L	"	"	B	"	"	"	"	"	"	"			
			71	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			72	"	"	"	"	"	L	"	B	"	"	"	"	"	"	"			
			73	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			74	B	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			75	"	"	"	"	"	"	"	B	"	L	"	"	"	"	"			
			76	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			77	"	"	"	"	"	"	"	B	"	"	L	"	"	"	"			
			78	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			79	"	"	"	"	"	"	"	B	"	"	"	L	"	"	"			
			80	"	"	"	"	"	"	"	C	"	"	"	L	"	"	"			
			81	"	"	"	"	"	"	"	B	"	"	"	L	L	"	"			

See footnotes at end of device type 05.

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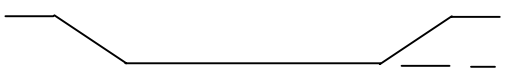
TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Cases 2, X	2	3	4	6	8	9	10	12	13	14	16	18	19	20				
			Test no.	A_{IN}	B_{IN}	QA	QB	QC	QD	GND	CLK	CLR	QE	QF	QG	QH	V_{CC}		Min	Max	
8	Same tests, terminal conditions, and limits as subgroup 7 except T_C = 125°C and -55°C.																				
9 T_C = 25°C	f_{MAX} See note F	(Fig. 8)	82	IN	G	OUT				GND	IN	G					5.0 V	QA	22		MHz
	t_{PLH1}	3003 (Fig. 8)	83	IN	G	OUT				"	"	"					"	CLK TO QA	5	32	ns
			84	See fig. 8	See fig. 8		OUT				"	"					"	CLK TO QB	"	"	"
			85	"	"			OUT		"	"	"					"	CLK TO QC	"	"	"
			86	"	"				OUT	"	"	"					"	CLK TO QD	"	"	"
			87	"	"					"	"	"	OUT				"	CLK TO QE	"	"	"
			88	"	"					"	"	"		OUT			"	CLK TO QF	"	"	"
			89	"	"					"	"	"			OUT		"	CLK TO QG	"	"	"
			90	"	"					"	"	"				OUT	"	CLK TO QH	"	"	"
	t_{PHL1}		91	"	"	OUT				"	"	"					"	CLK TO QA	"	37	ns
			92	"	"		OUT			"	"	"					"	CLK TO QB	"	"	"
			93	"	"			OUT		"	"	"					"	CLK TO QC	"	"	"
			94	"	"				OUT	"	"	"					"	CLK TO QD	"	"	"
			95	"	"					"	"	"	OUT				"	CLK TO QE	"	"	"
			96	"	"					"	"	"		OUT			"	CLK TO QF	"	"	"
			97	"	"					"	"	"			OUT		"	CLK TO QG	"	"	"
			98	"	"					"	"	"				OUT	"	CLK TO QH	"	"	"
	t_{PHL2}		99	G	G	OUT				"	"	IN					"	CLK TO QA	"	41	"
			100	"	"		OUT			"	"	"					"	CLK TO QB	"	"	"
			101	"	"			OUT		"	"	"					"	CLK TO QC	"	"	"
			102	"	"				OUT	"	"	"					"	CLK TO QD	"	"	"
			103	"	"					"	"	"	OUT				"	CLK TO QE	"	"	"
			104	"	"					"	"	"		OUT			"	CLK TO QF	"	"	"
			105	"	"					"	"	"			OUT		"	CLK TO QG	"	"	"
			106	"	"					"	"	"				OUT	"	CLK TO QH	"	"	"
10 T_C = 125°C	f_{MAX} See F	(Fig. 8)	107																20		MHz
	t_{PLH1}	3003 (Fig. 8)	108 to 115																5	48	ns
	t_{PHL1}	3003 (Fig. 8)	116 to 123																5	66	ns
	t_{PHL2}	3003 (Fig. 8)	124 to 131																5	62	ns
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																				

See footnotes at end of device type 05.

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FOOTNOTES:

A. Apply input pulse:  2.5 V minimum/5.5 V maximum. 0 V

B. V_{IN} = 2.5 V.

C. V_{IN} = 0.4 V.

D. Test numbers 42 through 81 shall be run in sequence.

E. Output voltages shall be either: (1) H ≥2.5 V minimum and L ≤0.4 V maximum when using a high speed checker double comparator; (2) H ≥1.5 V and L ≤1.5 V when using a high speed checker single comparator.

F. f_{MAX} minimum limit specified is the frequency of the clock input pulse. The output frequency shall be one-half of the input clock frequency. The input frequency on the A_{IN} data shall be one-half of the clock input frequency and the A_{IN} shall be shifted such that the A_{IN} and are coincident with the clock. Rise and fall times ≤ 6 ns. Input peak voltage 3 to 5 volts.

G. 3.0 V minimum/5.0 V maximum.

J. Apply input pulse:  2.5 V minimum/5.5 V maximum 0 V

1/ One pulse minimum.

2/ Two pulses minimum.

3/ Three pulses minimum.

4/ Four pulses minimum.

5/ Five pulses minimum.

6/ Six pulses minimum.

7/ Seven pulses minimum.

8/ Eight pulses minimum.

9/ At the manufacturer's option, I_{OS} tests 33 through 40, the following alternate procedure may be used; apply 2.75 volts @; test 33, QA , test 34, QB, test 35, QC, test 36, QD, test 37, QE, test 38, QF, test 39, QG, test 40, QH, and min/max limits of -7.5/-50 mA.

10/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Terminal	A	B	C	D	E	F	G
I_{IL1}	A_{IN}, B_{IN}	0/-.34	-.10/-.34	-.16/-.40	-.16/-.40	-.135/-.370	-.12/-.36	-.16/-.40
	CLK	0/-.4	-.16/-.4	-.12/-.36	-.20/-.44	"	"	"
	CLR	0/-.4	-.16/-.4	-.12/-.36	-.16/-.40	"	"	"

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TABLE III. Group A inspection for device type 06.

Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Cases 2,X	2	3	4	5	8	9	10	12	13	14	16	18	19	20		Min	Max	
			Test no.	Serial	A_{IN}	B_{IN}	C_{IN}	D_{IN}	Mode	GND	CONT	CLK	QD	QC	QB	QA	V_{CC}				
1 Tc = 25°C	V_{OH}	3006	1	2.0 V					0.7 V	GND	4.5 V	A				-1.0 mA	4.5 V	QA	2.4		V
			2		2.0 V				2.0 V	"	"	"				-1.0 mA	"	QA	"		"
			3			2.0 V			"	"	"	"			-1.0 mA		"	QB	"		"
			4				2.0 V		"	"	"	"		-1.0 mA			"	QC	"		"
			5					2.0 V	"	"	"	"	-1.0 mA				"	QD	"		"
	V_{OL}	3007	6	0.7 V					0.7 V	"	"	"				12 mA	"	QA		0.4	"
			7		0.7 V				2.0 V	"	"	"				12 mA	"	QA		"	"
			8			0.7 V			"	"	"	"			12 mA		"	QB		"	"
			9				0.7 V		"	"	"	"		12 mA			"	QC		"	"
			10					0.7 V	"	"	"	"	12 mA				"	QD		"	"
	V_{IC}		11	-18 mA						"							"	Serial		-1.5	"
			12		-18 mA					"							"	A_{IN}		"	"
			13			-18 mA				"							"	B_{IN}		"	"
			14				-18 mA			"							"	C_{IN}		"	"
			15					-18 mA		"							"	D_{IN}		"	"
			16						-18 mA	"							"	Mode		"	"
			17							"	-18 mA						"	CONT		"	"
			18							"		-18 mA					"	CLK		"	"
	I_{IH1}	3010	19	2.7 V					4.5 V	"							5.5 V	Serial		20	µA
			20		2.7 V				GND	"							"	A_{IN}		"	"
			21			2.7 V			"	"							"	B_{IN}		"	"
			22				2.7 V		"	"							"	C_{IN}		"	"
			23					2.7 V	"	"							"	D_{IN}		"	"
			24						2.7 V	"							"	Mode		"	"
			25							"	2.7 V						"	CONT		"	"
			26							"		2.7 V					"	CLK		"	"
	I_{IH2}		27	5.5 V					4.5 V	"							"	Serial		100	"
			28		5.5 V				GND	"							"	A_{IN}		"	"
			29			5.5 V			"	"							"	B_{IN}		"	"
			30				5.5 V		"	"							"	C_{IN}		"	"
			31					5.5 V	"	"							"	D_{IN}		"	"
			32						5.5 V	"							"	Mode		"	"
			33							"	5.5 V						"	CONT		"	"
			34							"		5.5 V					"	CLK		"	"
	I_{OZH}		35		0.7 V				4.5 V	"	0.7 V	A				2.7 V	"	QA		20	"
			36			0.7 V			"	"	"	"			2.7 V		"	QB		"	"
			37				0.7 V		"	"	"	"		2.7 V			"	QC		"	"
			38					0.7 V	"	"	"	"	2.7 V				"	QD		"	"
	I_{OZL}		39		2.0 V				"	"	"	"				0.4 V	"	QA		-20	"
			40			2.0 V			"	"	"	"			0.4 V		"	QB		"	"
			41				2.0 V		"	"	"	"		0.4 V			"	QC		"	"
			42					2.0 V	"	"	"	"	0.4 V				"	QD		"	"

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test Limits		Unit
			Cases 2, X	2	3	4	5	8	9	10	12	13	14	16	18	19	20				
			Test no.	Serial	A_{IN}	B_{IN}	C_{IN}	D_{IN}	Mode	GND	CONT	CLK	QD	QC	QB	QA	V_{CC}		Min	Max	
1 Tc = 25°C	I_{IL1}	3009	43	0.4 V					GND	"							5.5 V	Serial	1/	1/	mA
			44		0.4 V				4.5 V	"							"	A_{IN}	"	"	"
			45			0.4 V			"	"							"	B_{IN}	"	"	"
			46				0.4 V		"	"							"	C_{IN}	"	"	"
			47					0.4 V	"	"							"	D_{IN}	"	"	"
			48						0.4 V	"							"	Mode	"	"	"
			49							"	0.4 V						"	CONT	"	"	"
			50							"		0.4 V					"	CLK	"	"	"
	I_{0S}	3011	51		4.5 V				4.5 V	"	4.5 V	A				GND	"	QA	2/	2/	"
			52			4.5 V			"	"	"	"			GND		"	QB	"	"	"
			53				4.5 V		"	"	"	"		GND			"	QC	"	"	"
			54					4.5 V	"	"	"	"	GND				"	QD	"	"	"
	I_{CC}	3005	55	5.5 V	GND	GND	GND	GND	5.5 V	"	5.5 V	"					"	V_{CC}		27	"
	I_{CC}	3005	56	5.5 V	GND	GND	GND	GND	5.5 V	"	GND	GND					"	V_{CC}		29	"
2	Same tests, terminal conditions and limits as subgroup 1 except T_C = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as subgroup 1 except T_C = -55°C and V_{IC} tests are omitted.																				

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Cases 2, X	2	3	4	5	8	9	10	12	13	14	16	18	19	20				
			Test no.	Serial	A_{IN}	B_{IN}	C_{IN}	D_{IN}	Mode	GND	CONT	CLK	QD	QC	QB	QA	V_{CC}		Min	Max	
7 Tc = 25°C	Truth table tests	3014	57	B	B	B	B	B	B	GND	B	B	X	X	X	X	5.0 V	All outputs	See B,C,D, and E		
			58	"	B	B	B	B	"	"	"	C	H	H	H	H	"				
			59	"	B	B	B	B	"	"	"	B	H	H	H	H	"	"			
			60	"	C	C	C	C	"	"	"	B	H	H	H	H	"	"			
			61	"	"	"	"	"	"	"	"	C	L	L	L	L	"	"			
			62	"	"	"	"	"	"	"	"	B	"	"	"	L	"	"			
			63	"	"	"	"	"	C	"	"	B	"	L	"	L	"	"			
			64	"	"	"	"	"	"	"	"	C	"	"	"	H	"	"			
			65	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"			
			66	"	"	"	"	"	"	"	"	C	"	"	H	"	"	"			
			67	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"			
			68	"	"	"	"	"	"	"	"	C	"	H	"	"	"	"			
			69	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"			
			70	"	"	"	"	"	"	"	"	C	H		"	"	"	"			
			71	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"			
			72	C	B	B	B	B	"	"	"	B	"	"	"	"	"	"			
			73	"	"	"	"	"	"	"	"	C	"	"	"	L	"	"			
			74	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"			
			75	"	"	"	"	"	"	"	"	C	"	"	L	"	"	"			
			76	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"			
			77	"	"	"	"	"	"	"	"	C	"	L	"	"	"	"			
			78	"	"	"	"	"	"	"	"	B	"	L	"	"	"	"			
			79	"	"	"	"	"	"	"	"	C	L	L	"	"	"	"			
8	Same tests, terminal conditions, and limits as subgroup 7 except T_C = 125°C and -55°C.																				
9 T_C = 25°C	f_{MAX} See note F	(Fig. 9)	80	IN					GND	GND	G	IN				OUT	5.0 V	QA	20		MHz
	t_{PLH1}	3003 (Fig. 9)	81		IN				G	"	"	"				OUT	"	CLK to QA	6	35	ns
			82			IN			"	"	"	"			OUT		"	CLK to QB	"	"	"
			83				IN		"	"	"	"		OUT			"	CLK to QC	"	"	"
			84					IN	"	"	"	"	OUT				"	CLK to QD	"	"	"
			85	IN					GND	"	"	"				OUT	"	CLK to QA	"	"	'
			86	See fig. 9					"	"	"	"			OUT		"	CLK to QB	"	"	"
			87	See fig. 9					"	"	"	"		OUT			"	CLK to QC	"	"	"
			88	See fig. 9					"	"	"	"	OUT				"	CLK to QD	"	"	"
	t_{PHL1}		89		IN				G	"	"					OUT	"	CLK to QA	"	40	"
			90			IN			"	"	"	"			OUT		"	CLK to QB	"	"	"
			91				IN		"	"	"	"		OUT				CLK to QC	"	"	"
			92					IN	"	"	"	"	OUT				"	CLK to QD	"	"	"
			93	IN					GND	"	"					OUT	"	CLK to QA	"	"	"
			94	See fig. 9					"	"	"	"			OUT		"	CLK to QB	"	"	"
			95	See fig. 9					"	"	"	"		OUT			"	CLK to QC	"	"	"
			96	See fig. 9					"	"	"	"	OUT				"	CLK to QD	"	"	"

See footnotes at end of device type 06.

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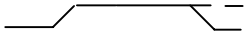
TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Cases 2, X	2	3	4	5	8	9	10	12	13	14	16	18	19	20				
			Test no.	Serial	A_{IN}	B_{IN}	C_N	D_{IN}	Mode	GND	CONT	CLK	QD	QC	QB	QA	V_{CC}		Min	Max	
9 T_C = 25°C	t_{ZL}	(Fig. 9)	97		GND				G	GND	IN	IN				OUT	5.0 V	CONT to QA	5	35	ns
			98			GND			"	"	"	"			OUT		"	CONT TO QB	"	"	"
			99				GND		"	"	"	"		OUT			"	CONT TO QC	"	"	"
			100					GND	"	"	"	"	OUT				"	CONT TO QD	"	"	"
	t_{ZH}		101		G				"	"	"	"				OUT	"	CLK TO QA	"	30	"
			102			G			"	"	"	"			OUT		"	CLK TO QB	"	"	"
			103				G		"	"	"	"		OUT			"	CLK TO QC	"	"	"
			104					G	"	"	"	"	OUT				"	CLK TO QD	"	"	"
	t_{LZ}		105		GND				"	"	"	"				OUT	"	CLK TO QA	"	55	"
			106			GND			"	"	"	"			OUT		"	CLK TO QB	"	"	"
			107				GND		"	"	"	"		OUT			"	CLK TO QC	"	"	"
			108					GND	"	"	"	"	OUT				"	CLK TO QD	"	"	"
	t_{HZ}		109		G				"	"	"	"				OUT	"	CLK TO QA	"	65	"
			110			G			"	"	"	"			OUT		"	CLK TO QB	"	"	"
			111				G		"	"	"	"		OUT			"	CLK TO QC	"	"	"
			112					G	"	"	"	"	OUT				"	CLK TO QD	"	"	"
10 T_C =125°C	f_{MAX} See F		113	Same test and terminal conditions as subgroup 9.															18		MHz
	t_{PLH1}	3003 (Fig. 9)	114 to 121																5	46	ns
	t_{PHL1}		122 to 129																"	52	"
	t_{ZL}		130 to 133																"	45	"
	t_{ZH}		134 to 137																"	39	"
	t_{LZ}		138 to 141																"	71	"
	t_{HZ}		142 to 145																"	84	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																				

See footnotes at end of device type 06.

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FOOTNOTES:

A. Apply input pulse:  2.5 V minimum/5.5 V maximum
0 V

B. V_{IN} = 2.4 V.

C. V_{IN} = 0.4 V.

D. Test numbers 57 through 79 shall be run in sequence.

E. Output voltages shall be either: (1) H ≥2.5 V minimum and L ≤0.4 V maximum when using a high speed checker double comparator; (2) H ≥1.5 V and L ≤1.5 V when using a high speed checker single comparator.

F. f_{MAX} minimum limit specified is the frequency of the clock input pulse. The output frequency shall be one-half of the input clock frequency. The input frequency on the serial shall be one-half of the clock input frequency and the serial shall be shifted such that the serial ↑ and ↓ are coincident with the clock ↑. Rise and fall times ≤ 6 ns. Input peak voltage 3 to 5 volts.

G. 3.0 V minimum/5.0 V maximum.

1/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Terminal	A	B	C	D	E
I_{IL1}	Serial	-.075/-.250	-.16/-.40	-.16/-.40	-.105/-.345	-.12/-.36
	A_{IN}, B_{IN}, C_{IN}, D_{IN}	-.12/-.36	-.16/-.40	-.16/-.40	-.105/-.345	-.12/-.36
	Mode	-.16/-.40	-.15/-.38	-.03/-.3	-.12/-.36	-.12/-.36
	CONT	-.16/-.40	-.16/-.40	-.03/-.3	-.12/-.36	-.12/-.36
	CLK	-.16/-.40	-.20/-.44	-.03/-.3	-.12/-.36	-.12/-.36

2/ I_{OS} limits (mA) min/max values for circuit A: -30/-130.
for circuits B, C, D, E: -15/-100.

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TABLE III. Group A inspection for device type 07 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test Limits		Unit
			Cases 2,X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	CLR	Serial	A_{IN}	B_{IN}	C_{IN}	D_{IN}	Load	GND	CONT	CLK	QD'	QD	QC	QB	QA	V_{CC}				
1 Tc = 25°C	V_{OH}	3006	1	2.0 V		2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	GND	0.7 V	A					-.1 mA	4.5 V	QA	2.4		V
			2	"		"	"	"	"	"	"	"	"				-.1 mA		"	QB	"		"
			3	"		"	"	"	"	"	"	"	"			-.1 mA			"	QC	"		"
			4	"		"	"	"	"	"	"	"	"		-.1 mA				"	QD	"		"
			5	"		"	"	"	"	"	"	"	"	-.4 mA					"	QD'	2.5		"
	V_{OL}	3007	6	"		0.7 V	0.7 V	0.7 V	0.7 V	"	"	"	"					12 mA	"	QA		0.4	"
			7	"		"	"	"	"	"	"	"	"				12 mA		"	QB		"	"
			8	"		"	"	"	"	"	"	"	"			12 mA			"	QC		"	"
			9	"		"	"	"	"	"	"	"	"		12 mA				"	QD		"	"
			10	"		"	"	"	"	"	"	"	"	4 mA					"	QD'		"	"
	V_{IC}		11	-18 mA							"								"	CLR		-1.5	"
			12		-18 mA						"								"	Serial		"	"
			13			-18 mA					"								"	A_{IN}		"	"
			14				-18 mA				"								"	B_{IN}		"	"
			15					-18 mA			"								"	C_{IN}		"	"
			16						-18 mA		"								"	D_{IN}		"	"
			17							-18 mA	"								"	Load		"	"
			18								"	-18 mA							"	CONT		"	"
			19								"		-18 mA						"	CLK		"	"
	I_{IH1}	3010	20	2.7 V							"								5.5 V	CLR		20	µA
			21		2.7 V					4.5 V	"								"	Serial		"	"
			22			2.7 V				GND	"								"	A_{IN}		"	"
			23				2.7 V			"	"								"	B_{IN}		"	"
			24					2.7 V		"	"								"	C_{IN}		"	"
			25						2.7 V	"	"								"	D_{IN}		"	"
			26							2.7 V	"								"	Load		"	"
			27							GND	"	2.7 V							"	CONT		"	"
			28							GND	"		2.7 V						"	CLK		"	"
	I_{IH2}		29	5.5 V							"								"	CLR		100	"
			30		5.5 V					4.5 V	"								"	Serial		"	"
			31			5.5 V				GND	"								"	A_{IN}		"	"
			32				5.5 V			"	"								"	B_{IN}		"	"
			33					5.5 V		"	"								"	C_{IN}		"	"
			34						5.5 V	"	"								"	D_{IN}		"	"
			35							5.5 V	'								"	Load		"	"
			36							GND	"	5.5 V							"	CONT		"	"
			37							GND	"		5.5 V						"	CLK		"	"
	I_{OZH}		38	2.0 V		0.7 V	0.7 V	0.7 V	0.7 V	2.0 V	"	2.0 V	A					2.7 V	"	QA		20	"
			39	"		"	"	"	"	"	"	"	"				2.7 V		"	QB		"	"
			40	"		"	"	"	"	"	"	"	"			2.7 V			"	QC		"	"
			41	"		"	"	"	"	"	"	"	"		2.7 V				"	QD		"	"
	I_{OZL}		42	"		2.0 V	2.0 V	2.0 V	2.0 V	"	"	"	"					0.4 V	"	QA		-20	"
			43	"		"	"	"	"	"	"	"	"				0.4 V		"	QB		"	"
			44	"		"	"	"	"	"	"	"	"			0.4 V			"	QC		"	"
			45	"		"	"	"	"	"	"	"	"		0.4 V				"	QD		"	"

See footnotes at end of device types 07.

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TABLE III. Group A inspection for device type 07 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test Limits		Unit
			Cases 2,X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	CLR	Serial	A_{IN}	B_{IN}	C_{IN}	D_{IN}	Load	GND	CONT	CLK	QD'	QD	QC	QB	QA	V_{CC}		Min	Max	
1	I_{IL1}	3009	46	0.4 V							GND								5.5 V	CLR	1/	1/	mA
T_C = 25°C			47	B	0.4 V					GND	"		A						"	Serial	"	"	"
			48			0.4 V				4.5 V	"								"	A_{IN}	"	"	"
			49				0.4 V			"	"								"	B_{IN}	"	"	"
			50					0.4 V		"	"								"	C_{IN}	"	"	"
			51						0.4 V	"	"								"	D_{IN}	"	"	"
			52							0.4 V	"								"	Load	"	"	"
			53							4.5 V	"		0.4 V						"	CONT	"	"	"
			54							"	"			0.4 V					"	CLK	"	"	"
	I_{OS}	3011	55	4.5 V	GND	4.5 V	4.5 V	4.5 V	4.5 V	"	"	GND	A					GND	"	Q_A	2/	2/	"
			56	"	"	"	"	"	"	"	"	"	"				GND		"	Q_B	"	"	"
			57	"	"	"	"	"	"	"	"	"	"			GND			"	Q_C	"	"	"
			58	"	"	"	"	"	"	"	"	"	"		GND				"	Q_D	"	"	"
			59	"	"	"	"	"	"	"	"	"	"	GND					"	$Q_{D'}$	"	"	"
	I_{CC}	3005	60	5.5 V	5.5 V	GND	GND	GND	GND	5.5 V	"	5.5 V	"						"	V_{CC}		34	"
	I_{CC}	3005	61	GND	5.5 V	GND	GND	GND	GND	5.5 V	"	GND	GND						"	V_{CC}		31	"
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7	Truth	3014	62	D	C	C	C	C	C	C	GND	D	C	L	L	L	L	L	5.0v	All	See C,D,E, and F		
T_C = 25° C	table		63	D	D	D	D	D	D	D	"	"	D	"	"	"	"	"	"	outputs			
	test		64	D	C	C	C	C	C	C	"	"	C	"	"	"	"	"	"	"			
			65	C	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			66	C	"	"	"	"	"	"	"	"	D	H	H	H	H	H	"	"			
			67	C	"	"	"	"	"	"	"	"	C	H	H	H	H	H	"	"			

See footnotes at end of device types 07.

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TABLE III. Group A inspection for device type 07 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Cases 2, X / Test no.	1 / 2 / CLK	2 / 3 / Serial	3 / 4 / A_{IN}	4 / 5 / B_{IN}	5 / 7 / C_{IN}	6 / 8 / D_{IN}	7 / 9 / Load	8 / 10 / GND	9 / 12 / CONT	10 / 13 / CLK	11 / 14 / QD'	12 / 15 / QD	13 / 17 / QC	14 / 18 / QB	15 / 19 / QA	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
7 T_C = 25°C	Truth table tests	3014	68	C	C	D	D	D	D	C	GND	D	C	H	H	H	H	H	5.0 V	All outputs	See C,D,E, and F		
			69	"	"	"	"	"	"	C	"	"	D	L	L	L	L	L	"				
			70	"	"	"	"	"	"	C	"	"	C	"	"	"	"	L	"	"			
			71	"	"	"	"	"	"	D	"	"	C	"	"	"	"	L	"	"			
			72	"	"	"	"	"	"	"	"	"	D	"	"	"	"	H	"	"			
			73	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			74	"	"	"	"	"	C	"	"	"	D	"	"	"	H	"	"	"			
			75	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			76	"	"	"	"	"	"	"	"	"	D	"	"	H	"	"	"	"			
			77	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			78	"	"	"	"	"	"	"	"	"	D	H	H	"	"	"	"	"			
			79	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			80	"	D	C	C	C	C	"	"	"	C	"	"	"	"	"	"	"			
			81	"	"	"	"	"	"	"	"	"	D	"	"	"	"	L	"	"			
			82	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			83	"	"	"	"	"	"	"	"	"	D	"	"	"	L	"	"	"			
			84	"	"	"	"	"	"	"	"	"	C	"	"	"	"	"	"	"			
			85	"	"	"	"	"	"	"	"	"	D	"	"	L	"	"	"	"			
			86	"	"	"	"	"	"	"	"	"	C	"	"	L	"	"	"	"			
			87	"	"	"	"	"	"	"	"	"	D	L	L	L	"	"	"	"			
8	Same tests, terminal conditions, as subgroup 7 except T_C = 125°C and -55°C.																						
9 T_C = 25°C	f_{MAX} See G	(Fig. 10)	88	J	IN					GND	GND	GND	IN					OUT	5.0 V	QA	22		MHz
	t_{PLH1}	3003 (Fig. 10)	89	"		IN				J	"	"	"					OUT	"	CLK to QA	5	37	ns
			90	"			IN			"	"	"	"				OUT		"	CLK to QB	"	"	"
			91	"				IN		"	"	"	"			OUT			"	CLK to QC	"	"	"
			92	"					IN	"	"	"	"		OUT				"	CLK to QD	"	"	"
			93	"					IN	"	"	"	"	OUT					"	CLK to QD'	"	"	'
			94	"	IN					GND	"	"	"					OUT	"	CLK to QA	"	"	"
			95	"	See fig. 10					"	"	"	"				OUT		"	CLK to QB	"	"	"
			96	"	"					"	"	"	"			OUT			"	CLK to QC	"	"	"
			97	"	"					"	"		"		OUT				"	CLK to QD	"	"	"
			98	"	"					"	"	"	"	OUT					"	CLK to QD'	"	"	"
	t_{PHL1}		99	"		IN				J	"	"	"					OUT		CLK to QA	"	"	"
			100	"			IN			"	"	"	"				OUT		"	CLK to QB	"	"	"
			101	"				IN		"	"		"			OUT			"	CLK to QC	"	"	"
			102	"					IN	"	"	"	"		OUT				"	CLK to QD	"	"	"
			103	"					IN	"	"	"	"	OUT					"	CLK to QD'	"	"	"
			104	"	IN					GND	"	"	"					OUT	"	CLK to QA	"	"	"
			105	"	See (fig. 10)					"	"	"	"				OUT		"	CLK to QB	"	"	"
			106	"	"					"	"	"	"			OUT			"	CLK to QC	"	"	"
			107	"	"					"	"	"	"		OUT				"	CLK to QD	"	"	"
			108	"	"					"	"	"	"	OUT					"	CLK to QD'	"	"	"

See footnotes at end of device type 07.

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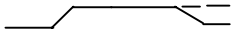
TABLE III. Group A inspection for device type 07 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.7 V; or open).

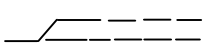
Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Cases 2, X / Test no.	1 / 2 / CLK	2 / 3 / Serial	3 / 4 / A_{IN}	4 / 5 / B_N	5 / 7 / C_{IN}	6 / 8 / D_{IN}	7 / 9 / Load	8 / 10 / GND	9 / 12 / CONT	10 / 13 / CLK	11 / 14 / QD'	12 / 15 / QD	13 / 17 / QC	14 / 18 / QB	15 / 19 / QA	14 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
9 T_C = 25°C	t_{PHL2}	3003 (Fig. 10)	109	IN		J				J	GND	GND	IN					OUT	5.0 V	CLR to QA	5	37	ns
			110	"			J			"	"	"	"				OUT		"	CLR to QB	"	"	"
			111	"				J		"	"	"	"			OUT			"	CLR to QC	"	"	"
			112	"					J	"	"	"	"		OUT				"	CLR to QD	"	"	"
			113	"					J	"	"	"	"	OUT					"	CLR to QD'	"	"	"
	t_{ZL}		114	"							"	IN						OUT	"	CONT to QA	"	35	"
			115	"							"	"					OUT		"	CONT TO QB	"	"	"
			116	"							"	"				OUT			"	CONT TO QC	"	"	"
			117	"							"	"			OUT				"	CONT TO QD	"	"	"
	t_{ZH}		118	J		J				J	"	"	IN					OUT	"	CONT TO QA	"	"	"
			119	"			J			"	"	"	"				OUT		"	CONT TO QB	"	"	"
			120	"				J		"	"	"	"			OUT			"	CONT TO QC	"	"	"
			121	"					J	"	"	"	"		OUT				"	CONT TO QD	"	"	"
	t_{LZ}		122	GND							"	"						OUT	"	CONT TO QA	"	"	"
			123	"							"	"					OUT		"	CONT TO QB	"	"	"
			124	"							"	"				OUT			"	CONT TO QC	"	"	"
			125	"							"	"			OUT				"	CONT TO QD	"	"	"
	t_{HZ}		126	J		J				J	"	"	IN					OUT	"	CONT TO QA	"	"	"
			127	"			J			"	"	"	"				OUT		"	CONT TO QB	"	"	"
			128	"				J		"	"	"	"			OUT			"	CONT TO QC	"	"	"
			129	"					J	"	"	"	"		OUT				"	CONT TO QD	"	"	"
10 T_C = 125°C	f_{MAX} See G	(Fig. 10)	130	Same tests and terminal conditions as for subgroup 9.																	20		MHz
	t_{PLH1}	3003 (Fig. 10)	131 to 140																		5	56	ns
	t_{PHL1}		141 to 150																		"	56	"
	t_{PHL2}		151 to 155																		"	56	"
	t_{ZL}		156 to 159																		"	53	"
	t_{ZH}		160 to 163																		"	"	"
	t_{LZ}		164 to 167																		"	"	"
	t_{HZ}		168 to 171																		"	"	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C																						

See footnotes at end of device type 07.

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FOOTNOTES:

A. Apply input pulse:  2.5 V minimum/5.5 V maximum
0 V-

B. Apply input pulse:  2.5 V minimum/5.5 V maximum.
0 V

C. V_{IN} = 2.4V.

D. V_{IN} = 0.4 V.

E. Test numbers 62 through 87 shall be run in sequence.

F. Output voltages shall be either: (1) H ≥2.5 V minimum and L ≤0.4 V maximum when using a high speed checker double comparator: (2) H ≥1.5 V and L ≤1.5 V when using a high speed checker single comparator.

G. f_{MAX} minimum limit specified is the frequency of the clock input pulse. The output frequency shall be one-half of the input clock frequency. The input frequency on the parallel input shall be one-half of the clock input frequency and the parallel input shall be shifted such that the parallel input ↑ and ↓ are coincident with the clock ↑. Rise and fall times ≤ 6 ns. Input peak voltage 3 to 5 volts.

J. 3.0 V minimum/5.0 V maximum.

1/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Terminal	A	B	C	D
I_{IL1}	Serial	-.075/-.250	-.16/-.40	-.105/-.345	-.12/-.36
	A_{IN}, B_{IN}, C_{IN}	-.12/-.36	"	-.105/-.345	-.12/-.36
	D_{IN}	-.16/-.40	"	-.16/-.40	-.105/-.345
	CLR, Load, CONT, CLK	-.16/-.40	-.03/-.30	-.12/-.36	-.12/-.36

2/ I_{OS} limits for circuit A for QA through QD are -30 to -130 mA, for QD' is -20 to -100 mA, and for circuits B, C, and D are -15 to -100 mA.

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TABLE III. Group A inspection for device type 08

Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test Limits		Unit
			Cases 2,X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	Shift Load	CLK	E	F	G	H	$\overline{Q}_H$	GND	Q_H	Serial INP	A	B	C	D	CLK INHB	V_{CC}				
1 Tc = 25°C	V_{OH}	3006	1	0.7 V					2.0 V		GND	-.4 mA							4.5 V	Q_H	2.5		V
		3006	2	"					0.7 V	-.4 mA	"								"	$\overline{Q}_H$	2.5		"
	V_{OL}	3007	3	"					0.7 V		"	4 mA							"	Q_H		0.4	"
		3007	4	"					2.0 V	4 mA	"								"	$\overline{Q}_H$		0.4	"
	V_{IC}		5	-18 mA							"								"	S/L		-1.5 V	"
			6		-18 mA						"								"	CLK		"	"
			7			-18 mA					"								"	E		"	"
			8				-18 mA				"								"	F		"	"
			9					-18 mA			"								"	G		"	"
			10						-18 mA		"								"	H		"	"
			11								"		-18 mA						"	S/INP		"	"
			12								"			-18 mA					"	A		"	"
			13								"				-18 mA				"	B		"	"
			14								"					-18 mA			"	C		"	"
			15								"						-18 mA		"	D		"	"
			16								"							-18 mA	"	CLK/INHB		"	"
	I_{IL1}	3009	17		0.4 V						"								5.5 V	CLK	1/	1/	mA
	I_{IL6}		18	GND		0.4 V					"								"	E	"	"	"
			19	"			0.4 V				"								"	F	"	"	"
			20	"				0.4 V			"								"	G	"	"	"
			21	"					0.4 V		"								"	H	"	"	"
			22								"		0.4 V						"	S/INP	"	"	"
			23	GND							"			0.4 V					"	A	"	"	"
			24								"				0.4 V				"	B	"	"	"
			25								"					0.4 V			"	C	"	"	"
			26								"						0.4 V		"	D	"	"	"

See footnotes at end of device types 08.

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TABLE III. Group A inspection for device type 08 - Continued

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Cases 2, X Test no.	1 2 Shift Load	2 3 CLK	3 4 E	4 5 F	5 7 G	6 8 H	7 9 $\overline{Q}_H$	8 10 GND	9 12 Q_H	10 13 Serial INP	11 14 A	12 15 B	13 17 C	14 18 D	15 19 CLK INHB	16 20 V_{CC}	Measured terminal	Test Limits Min	 Max	Unit
1	I_{IL1}	3009	27								GND							0.4 V	5.5 V	CLK/INHB	1/	1/	mA
Tc = 25°C	I_{IL7}		28	0.4 V							"								"		1/	1/	mA
	I_{IH1}	3010	29		2.7 V						"								"	CLK		20	µA
			30			2.7 V					"								"	E		"	"
			31				2.7 V				"								"	F		"	"
			32					2.7 V			"								"	G		"	"
			33						2.7 V		"								"	H		"	"
			34								"		2.7 V						"	S/INP		"	"
			35								"			2.7 V					"	A		"	"
			36								"				2.7 V				"	B		"	"
			37								"					2.7 V			"	C		"	"
			38								"						2.7 V		"	D		"	"
			39								"							2.7 V	"	CLK/INHB		"	"
	I_{IH11}		40	2.7 V							"								"	S/L		60	"
	I_{IH2}		41		5.5 V						"								"	CLK		0.1	mA
			42			5.5 V					"								"	E		"	"
			43				5.5 V				"								"	F		"	"
			44					5.5 V			"								"	G		"	"
			45						5.5 V		"								"	H		"	"
			46								"		5.5 V						"	S/INP		"	"
			47								"			5.5 V					"	A		"	"
			48								"				5.5 V				"	B		"	"
			49								"					5.5 V			"	C		"	"
			50								"						5.5 V		"	D		"	"
			51								"							5.5 V	"	CLK/INHB		"	"
	I_{IH12}		52	5.5 V							"								"	S/L		0.3	"
	I_{OS}	3011	53	GND					5.5 V		"	GND							"	Q_H	-15	-100	"
	I_{OS}	3011	54	"					GND	GND	"								"	$\overline{Q}_H$	-15	-100	"
	I_{CC}	3005	55	"	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V		"		4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	V_{CC}		36	"
	I_{CC}	3005	56	"	4.5 V	GND	GND	GND	GND		"		GND	GND	GND	GND	GND	4.5 V	"	V_{CC}		36	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

See footnotes at end of device type 08.

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TABLE III. Group A inspection for device type 08 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2, X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	Shift Load	CLK	E	F	G	H	$\overline{Q}_H$	GND	Q_H	Serial INP	A	B	C	D	CLK INHB	V_{CC}		Min	Max	
7 2/ Tc = 25°C	Truth table tests	3014	57	B	B	B	A	B	A	L	GND	H	A	B	A	B	A	B	5.0 V		3/		
			58	A	B	"	"	"	"	L	"	H	A	"	"	"	"	"	"				
			59	"	A	"	"	"	"	H	"	L	A	"	"	"	"	"	"				
			60	"	B	"	"	"	"	H	"	L	B	"	"	"	"	"	"				
			61	"	A	"	"	"	"	L	"	H	"	"	"	"	"	"	"				
			62	"	B	"	"	"	"	L	"	H	"	"	"	"	"	"	"				
			63	"	A	"	"	"	"	H	"	L	"	"	"	"	"	"	"				
			64	"	B	"	"	"	"	H	"	L	"	"	"	"	"	"	"				
			65	"	A	"	"	"	"	L	"	H	"	"	"	"	"	"	"				
			66	"	B	"	"	"	"	L	"	H	"	"	"	"	"	"	"				
			67	"	A	"	"	"	"	H	"	L	"	"	"	"	"	"	"				
			68	"	B	"	"	"	"	H	"	L	"	"	"	"	"	"	"				
			69	"	A	"	"	"	"	L	"	H	"	"	"	"	"	"	"				
			70	"	B	"	"	"	"	L	"	H	"	"	"	"	"	"	"				
			71	"	A	"	"	"	"	H	"	L	"	"	"	"	"	"	"				
			72	"	B	"	"	"	"	H	"	L	"	"	"	"	"	"	"				
			73	"	A	"	"	"	"	L	"	H	"	"	"	"	"	"	"				
			74	"	B	"	"	"	"	L	"	H	"	"	"	"	"	"	"				
			75	"	A	"	"	"	"	H	"	L	"	"	"	"	"	"	"				
			76	"	B	"	"	"	"	H	"	L	A	"	"	"	"	A	"				
			77	"	A	"	"	"	"	H	"	L	A	"	"	"	"	A	"				
8	Same tests, terminal conditions, as subgroup 7 except T_C = 125°C and -55°C.																						
9 T_C = 25°C	f_{MAX} 4/		78	5.0 v	IN						GND	OUT	IN					GND	5.0 V	CLK to Q_H	25		MHz
	t_{PLH5}	3003	79	IN					IN		"	OUT							"	S/L to Q_H	5	40	ns
	t_{PHL5}	See fig. 11	80	"					"	OUT	"								"	S/L to $\overline{Q}_H$	"	"	"
	t_{PLH5}		81	"					"	OUT	"								"	S/L to $\overline{Q}_H$	"	"	"
	t_{PHL5}		82	"					"		"	OUT							"	S/L to $\overline{Q}_H$	"	"	"
	t_{PLH1}		83	5.0 V	IN						"	OUT						GND	"	CLK to Q_H	"	45	'
	t_{PHL1}		84	"	"					OUT	"							"		CLK to $\overline{Q}_H$	"	"	"
	t_{PLH1}		85	"	"					OUT	"							"	"	CLK to $\overline{Q}_H$	"	"	"
	t_{PHL1}		86	"	"						"	OUT						"	"	CLK to $\overline{Q}_H$	"	"	"
	t_{PLH3}		87	GND					IN		"	OUT							"	H to Q_H	"	30	"
	t_{PHL3}		88	"					"		"	OUT								H to Q_H	"	35	"
	t_{PLH4}		89	"					"	OUT	"								"	H to $\overline{Q}_H$	"	35	"
	t_{PHL4}		90	"					"	OUT	"								"	H to $\overline{Q}_H$	"	30	"

See footnotes at end of device type 08.

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TABLE III. Group A inspection for device type 08 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Cases 2, X Test no.	1 2 Shift Load	2 3 CLK	3 4 E	4 5 F	5 7 G	6 8 H	7 9 $\overline{Q}_H$	8 10 GND	9 12 Q_H	10 13 Serial INP	11 14 A	12 15 B	13 17 C	14 18 D	15 19 CLK INHB	16 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
10 T_C = 125°C	f_{MAX}		91	Same tests and terminal conditions as subgroup 9, except T_C = 125°C.																	20		MHz
	t_{PLH5}	3003 See fig. 11	92																		5	52	ns
	t_{PHL5}		93																		"	"	"
	t_{PLH5}		94																		"	"	"
	t_{PHL5}		95																		"	"	"
	t_{PLH1}		96																		"	58	'
	t_{PHL1}		97																		"	"	"
	t_{PLH1}		98																		"	"	"
	t_{PHL1}		99																		"	"	"
	t_{PLH3}		100																		"	39	"
	t_{PHL3}		101																		"	46	"
	t_{PLH4}		102																		"	46	"
	t_{PHL4}		103																		"	39	"
11 T_C = -55°C	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

NOTES:

2/ A = 2.5 V and B = 0.4 V.

3/ Output voltages shall be either:

(a) H = 2.5 V minimum and L = 0.4 V maximum when using a high speed checker double comparator or,
(b) H ≥1.5 V and L ≤1.5 V when using a high speed checker single comparator.

4/ f_{MAX} minimum limit specified is the frequency of the clock input pulse. The output frequency shall be one-half of the input clock frequency. The input frequency on the serial shall be one-half of the clock input frequency and the serial shall be shifted such that the serial ↑ and ↓ are coincident with the clock ↓, but may be offset sufficiently to assure adequate t_{SETUP} and t_{HOLD}. Rise and fall times ≤ 6 ns. Input peak voltage 3 to 5 volts.

1/ I_{IL} limits (mA) min/max values for circuit shown:

Parameter	Terminal	A	C	F
I_{IL1}	CLK, CLK/INHIB	-.001/-.150	-.12/-.38	-.005/-.72
I_{IL6}	A,B,C,D, E,F,G,H	-.120/-.360	-.12/-.38	-.12/-.38
	S/IN	-.100/-.340	-.12/-.38	-.12/-.38
I_{IL7}	S/L	-.001/-.150	-.36/-1.08	-.005/-.72

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TABLE III. Group A inspection for device type 09- Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test Limits		Unit
			Cases 2,X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	Ser. in	A	B	C	D	CLK INHB	CLK	GND	CLR	E	F	G	Q_H	H	Shift load	V_{CC}				
1 Tc = 25°C	V_{OH}	3006	1						0.7 V	1/	GND					-.4 mA	2.0 V	0.7 V	4.5 V	Q_H	2.5		V
	V_{OL}	3007	2						0.7 V	1/	"					4 mA	0.7 V	0.7 V	"	Q_H		0.4	"
	V_{IC}		3	-18 mA							"								"	S/IN		-1.5	"
			4		-18 mA						"								"	A		"	"
			5			-18 mA					"								"	B		"	"
			6				-18 mA				"								"	C		"	"
			7					-18 mA			"								"	D		"	"
			8						-18 mA		"								"	CLK INHB		"	"
			9							-18 mA	"								"	CLK		"	"
			10								"	-18 mA							"	CLR		"	"
			11								"		-18 mA						"	E		"	"
			12								"			-18 mA					"	F		"	"
			13								"				-18 mA				"	G		"	"
			14								"						-18 mA		"	H		"	"
			15								"							-18 mA	"	Shift load		"	"
	I_{IL6}	3009	16	0.4 V							"								5.5 V	S/IN	-.100	-.340	mA
			17		0.4 V						"							GND	"	A	"	"	"
			18			0.4 V					"							"	"	B	"	"	"
			19				0.4 V				"							"	"	C	"	"	"
			20					0.4 V			"							"	"	D	"	"	"
	I_{IL1}		21						0.4 V		"								"	CLK INHB	-.001	-.150	"
	I_{IL1}		22							0.4 V	"								"	CLK	-.001	-.150	"
	I_{IL1}		23								"	0.4 V							"	CLR	-.001	-.150	"
	I_{IL6}		24								"		0.4 V					GND	"	E	-.100	-.340	"
			25								"			0.4 V				"	"	F	"	"	"
			26								"				0.4 V			"	"	G	"	"	"
			27								"						0.4 V	"	"	H	"	"	"
	I_{IL7}		28								"							0.4 V	"	Shift load	-.001	-.150	"
	I_{IH1}	3010	29	2.7 V							"								"	S/IN		20	µA
			30		2.7 V						"								"	A		"	"
			31			2.7 V					"								"	B		"	"
			32				2.7 V				"								"	C		"	"
			33					2.7 V			"								"	D		"	"
			34						2.7 V		"								"	CLK INHB		"	"
			35							2.7 V	"								"	CLK		"	"
			36								"	2.7 V							"	CLR		"	"
			37								"		2.7 V						"	E		"	"
			38								"			2.7 V					"	F		"	"
			39								"				2.7 V				"	G		"	"
			40								"						2.7 V		"	H		"	"
			41								"							2.7 V	"	Shift load		"	"

See footnotes at end of device types 09

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TABLE III. Group A inspection for device type 09- Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test Limits		Unit
			Cases 2,X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	Ser. in	A	B	C	D	CLK INHB	CLK	GND	CLR	E	F	G	Q_H	H	Shift load	V_{CC}				
1 Tc = 25°C	I_{IH2}	3010	42	5.5 V							GND								5.5 V	S/IN		0.1	mA
			43		5.5 V						"								"	A		"	
			44			5.5 V					"								"	B		"	"
			45				5.5 V				"								"	C		"	"
			46					5.5 V			"								"	D		"	"
			47						5.5 V		"								"	CLK INHB		"	"
			48							5.5 V	"								"	CLK		"	"
			49								"	5.5 V							"	CLR		"	"
			50								"		5.5 V						"	E		"	"
			51								"			5.5 V					"	F		"	"
			52								"				5.5 V				"	G		"	"
			53								"						5.5 V		"	H		"	"
			54								"							5.5 V	"	Shift load		"	"
	I_{OS}	3011	55						GND	1/	"					GND	5.5 V	GND	"	Q_H	-15	-100	"
	I_{CC}	3005	56	4.5	GND	GND	GND	GND	GND	1/	"	GND	GND	GND	GND		GND	GND	"	V_{CC}		38	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = 125° C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55° C and V_{IC} tests are omitted.																						
7 2/ Tc = 25°C	Truth table test	3014	57	B	A	B	A	B	B	B	GND	B	A	B	A	L	A	B	5.0 V		3/		
			58	"	"	"	"	"	"	B	"	A	"	"	"	L	"	B	"				
			59	"	"	"	"	"	"	A	"	"	"	"	"	H	"	B	"				
			60	"	"	"	"	"	"	B	"	"	"	"	"	"	"	A	"				
			61	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"				
			62	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"				
			63	"	"	"	"	"	"	A	"	"	"	"	"	L	"	"	"				
			64	"	"	"	"	"	"	B	"	"	"	"	"	L	"	"	"				
			65	"	"	"	"	"	"	A	"	"	"	"	"	H	"	"	"				
			66	"	"	"	"	"	"	B	"	"	"	"	"	H	"	"	"				
			67	"	"	"	"	"	"	A	"	"	"	"	"	L	"	"	"				
			68	"	"	"	"	"	"	B	"	"	"	"	"	L	"	"	"				
			69	"	"	"	"	"	"	A	"	"	"	"	"	H	"	"	"				
			70	"	"	"	"	"	"	B	"	"	"	"	"	H	"	"	"				
			71	"	"	"	"	"	"	A	"	"	"	"	"	L	"	"	"				
			72	"	"	"	"	"	"	B	"	"	"	"	"	L	"	"	"				
			73	"	"	"	"	"	"	A	"	"	"	"	"	H	"	"	"				
			74	"	"	"	"	"	A	B	"	"	"	"	"	H	"	"	"				
			75	"	"	"	"	"	A	A	"	"	"	"	"	H	"	"	"				
8	Same tests, terminal conditions, and limits as subgroup 7, except T_C = 125° C and -55°C.																						

See footnotes at end of device types 09.

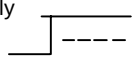
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TABLE III. Group A inspection for device type 09- Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V; or low ≤ 0.7 V; or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test Limits		Unit
			Cases 2,X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	Ser. in	A	B	C	D	CLK INHB	CLK	GND	CLR	E	F	G	Q_H	H	Shift load	V_{CC}				
9 4/ Tc = 25°C	f_{MAX}	3003	76						GND	IN	GND	5.0 V				OUT	IN	GND	5.0 V	CLK to Q_H	25		MHz
	t_{PHL5}	See fig. 12	77								"	IN				"			"	CLR to Q_H	5	40	ns
											"					"			"		"		
	t_{PLH1}		78						GND	IN	"	5.0 V				"	IN	GND	"	CLR to Q_H	"	31	ns
	t_{PHL1}		79						GND	IN	"	5.0 V				"	IN	GND	"	CLR to Q_H	"	35	ns
10	f_{MAX}		Same tests and terminal as subgroup 9, except T_C = 125°C.																		20		MHz
	t_{PHL5}																				5	52	ns
	t_{PLH1}																				5	40	ns
	t_{PHL1}																				5	46	ns
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

NOTES:

1/ Apply  2.5 V minimum, 5.5 V maximum to clock input prior to test.

2/ A = 2.5 V and B = 0.4 V.

3/ Output voltages shall be either:

a. H = 2.5 V minimum and L = 0.4 V maximum when using a high speed checker double comparator or,
b. H ≥1.5 V and L ≤1.5 V when using a high speed checker single comparator.

4/ f_{MAX} minimum limit specified is the frequency of the clock input pulse. The output frequency shall be one-half of the input clock frequency. The input frequency on the "H" shall be one-half of the clock input frequency and the "H" shall be shifted such that the "H" ↑ and ↓ are coincident with the clock ↓. Rise and fall times ≤6 ns. Input peak voltage 3 to 5 volts.

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5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's system Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Complete part number (see 1.2).

c. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

d. Requirements for certificate of compliance, if applicable.

e. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

g. Requirements for product assurance options.

h. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements should not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

j. Requirements for "JAN" marking.

6.3 Superseding information. The requirements of MIL-M-38510 have been superseded to take advantage of the available Qualified Manufacturer Listing (QML) system provided by MIL-PRF-38535. Previous references to MIL-M-38510 in this document have been replaced by appropriate references to MIL-PRF-38535. All technical requirements now consist of this specification and MIL-PRF-38535. The MIL-M-38510 specification sheet number and PIN have been retained to avoid adversely impacting existing government logistics systems and contractor's parts lists.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-38535 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DSCC-VQ, 3990 E. Broad Street, Columbus, Ohio 43123-1199.

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6.5 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-38535, MIL-HDBK-1331, and as follows:

GND ..... Ground zero voltage potential
I_{IN} ..... Current flowing into an input terminal
V_{IC} ..... Input clamp voltage
V_{IN} ..... Voltage level at an input terminal

6.6 Logistic support. Lead materials and finishes (see 3.4) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish A (see 3.4). Longer length leads and lead forming should not affect the part number.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information should not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-PRF-38535.

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Military device type	Company							Generic Industry type
	Texas Instruments	Signetics Corp.	Raytheon Company	Advanced Micro Devices	Fairchild Semi-conductor	Motorola, Inc.	National Semi-conductor	
01, circuit--	A	B	C	D	E	F	G	54LS194A
02, circuit--	A	B	C	D	E	F	G	54LS195A
03, circuit--	A	B	C	---	D	E	---	54LS95B
04, circuit--	A	B	---	---	----	---	---	54LS96
05, circuit--	A	B	G	C	E	F	D	54LS164
06, circuit--	A	C	B	---	D	E	---	54LS295B
07, circuit--	A	B	---	---	C	D	---	54LS395A
08, circuit--	A	---	---	---	C	F	---	54LS165A
09, circuit--	A	---	---	---	---	F	---	54LS166

6.6 Change from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5962-1960)

Review activities:
Army – SM
Navy - AS, CG, MC, SH, TD
Air Force - 03, 19, 99

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